

## *Trigonometry*

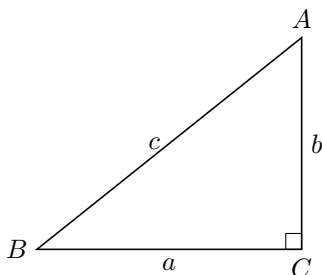
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## 4.1 Revision

EMCG9

**Trigonometric ratios**

We defined the basic trigonometric ratios using the lengths of the sides of a right-angled triangle.



$$\sin \hat{A} = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c}$$

$$\sin \hat{B} = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{b}{c}$$

$$\cos \hat{A} = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$$

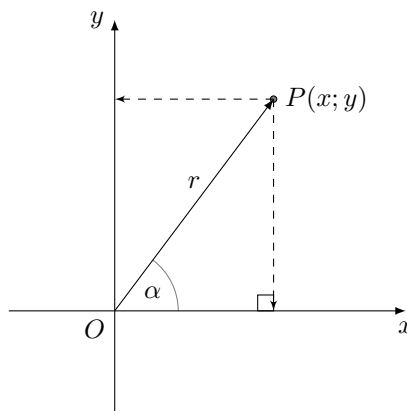
$$\cos \hat{B} = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{a}{c}$$

$$\tan \hat{A} = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b}$$

$$\tan \hat{B} = \frac{\text{opposite}}{\text{adjacent}} = \frac{b}{a}$$

**Trigonometric ratios in the Cartesian plane**

We also defined the trigonometric ratios with respect to any point in the Cartesian plane in terms of  $x$ ,  $y$  and  $r$ . Using the theorem of Pythagoras,  $r^2 = x^2 + y^2$ .



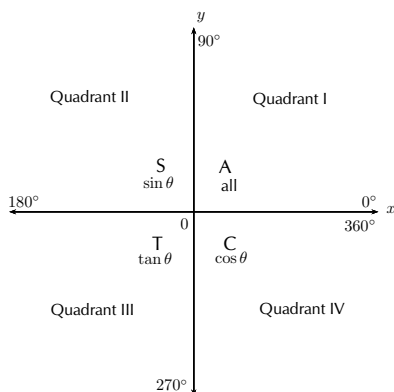
$$\sin \alpha = \frac{y}{r}$$

$$\cos \alpha = \frac{x}{r}$$

$$\tan \alpha = \frac{y}{x}$$

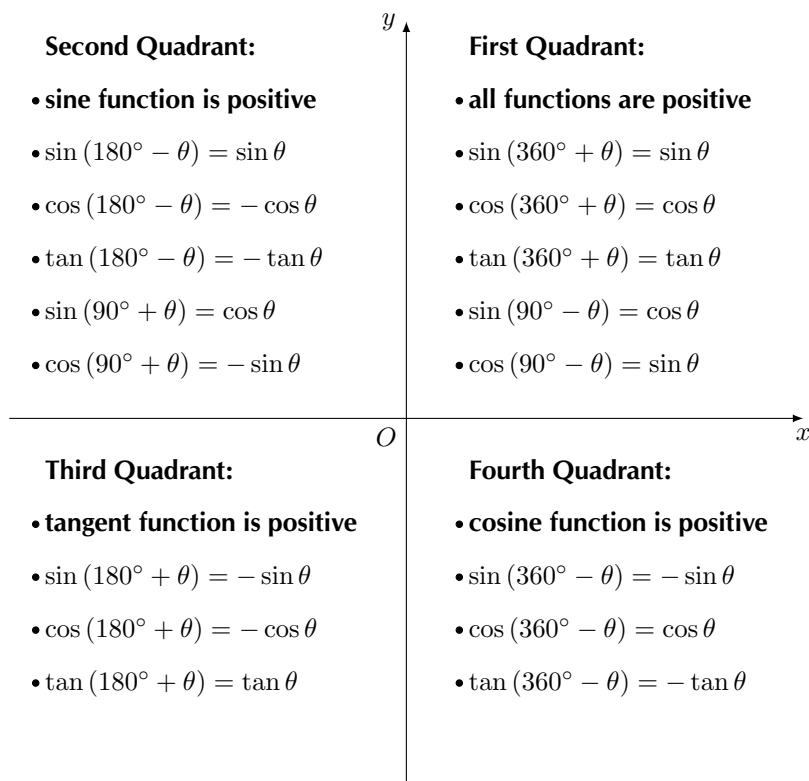
## CAST diagram

The sign of a trigonometric ratio depends on the signs of  $x$  and  $y$ :



## Reduction formulae and co-functions:

1. The reduction formulae hold for any angle  $\theta$ . For convenience, we assume  $\theta$  is an acute angle ( $0^\circ < \theta < 90^\circ$ ).
2. When determining function values of  $(180^\circ \pm \theta)$ ,  $(360^\circ \pm \theta)$  and  $(-\theta)$  the function does not change.
3. When determining function values of  $(90^\circ \pm \theta)$  and  $(\theta \pm 90^\circ)$  the function changes to its co-function.



## Negative angles

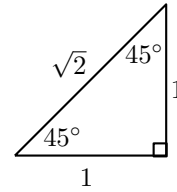
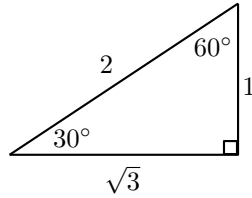
$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

## Special angle triangles

These values are useful when we need to solve a problem involving trigonometric functions without using a calculator. Remember that the lengths of the sides of a right-angled triangle obey the theorem of Pythagoras.



| $\theta$      | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|---------------|-----------|----------------------|----------------------|----------------------|------------|
| $\cos \theta$ | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0          |
| $\sin \theta$ | 0         | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| $\tan \theta$ | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | undef      |

## Trigonometric identities

Quotient identity:

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad (\cos \theta \neq 0)$$

Square identity:

$$\sin^2 \theta + \cos^2 \theta = 1$$

It also follows that:

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\sin \theta = \pm \sqrt{1 - \cos^2 \theta}$$

$$\cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

All these relationships and identities are very useful for simplifying trigonometric expressions.

### Worked example 1: Revision

#### QUESTION

Determine the value of the expression, without using a calculator:

$$\frac{\cos 420^\circ - \sin 225^\circ \cos(-45^\circ)}{\tan 315^\circ}$$

#### SOLUTION

**Step 1: Use reduction formulae to express each trigonometric ratio in terms of an acute angle**

$$\begin{aligned} & \frac{\cos 420^\circ - \sin 225^\circ \cos(-45^\circ)}{\tan 315^\circ} \\ &= \frac{\cos(360^\circ + 60^\circ) - \sin(180^\circ + 45^\circ) \cos(-45^\circ)}{\tan(360^\circ - 45^\circ)} \\ &= \frac{\cos 60^\circ - (-\sin 45^\circ)(\cos 45^\circ)}{-\tan 45^\circ} \\ &= \frac{\cos 60^\circ + \sin 45^\circ \cos 45^\circ}{-\tan 45^\circ} \end{aligned}$$

Now use special angles to evaluate the simplified expression:

$$\begin{aligned} &= \frac{\cos 60^\circ + \sin 45^\circ \cos 45^\circ}{-\tan 45^\circ} \\ &= \frac{\frac{1}{2} + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)}{-1} \\ &= -\left(\frac{1}{2} + \frac{1}{2}\right) \\ &= -1 \end{aligned}$$

### Worked example 2: Revision

#### QUESTION

Prove:

$$\sin^2 \alpha - (\tan \alpha - \cos \alpha)(\tan \alpha + \cos \alpha) = \frac{\cos^2 \alpha - \sin^2 \alpha}{\cos^2 \alpha}$$

State restrictions where applicable.

#### SOLUTION

**Step 1: Use trigonometric identities to simplify each side separately**

Simplify the left-hand side of the identity:

$$\begin{aligned} \text{LHS} &= \sin^2 \alpha - (\tan \alpha - \cos \alpha)(\tan \alpha + \cos \alpha) \\ &= \sin^2 \alpha - (\tan^2 \alpha - \cos^2 \alpha) \\ &= \sin^2 \alpha - \tan^2 \alpha + \cos^2 \alpha \\ &= (\sin^2 \alpha + \cos^2 \alpha) - \tan^2 \alpha \\ &= 1 - \tan^2 \alpha \end{aligned}$$

Simplify the right-hand side of the identity so that it equals the left-hand side:

$$\begin{aligned}\text{RHS} &= \frac{\cos^2 \alpha - \sin^2 \alpha}{\cos^2 \alpha} \\ &= \frac{\cos^2 \alpha}{\cos^2 \alpha} - \frac{\sin^2 \alpha}{\cos^2 \alpha} \\ &= 1 - \tan^2 \alpha\end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

Alternative method: we could also have started with the left-hand side of the identity and substituted  $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$  and simplified to get the right-hand side.

### Restrictions

We need to determine the values of  $\alpha$  for which any of the terms in the identity will be undefined:

$$\begin{aligned}\cos^2 \alpha &= 0 \\ \therefore \cos \alpha &= 0 \\ \therefore \alpha &= 90^\circ \text{ or } 270^\circ\end{aligned}$$

We must also consider the values of  $\alpha$  for which  $\tan \alpha$  is undefined. Therefore, the identity is undefined for  $\alpha = 90^\circ + k \cdot 180^\circ$ .

### Useful tips:

- It is sometimes useful to write  $\tan \theta$  in terms of  $\sin \theta$  and  $\cos \theta$ .
- Never write a trigonometric ratio without an angle. For example,  $\tan = \frac{\sin}{\cos}$  has no meaning.
- For proving identities, only simplify one side of the identity at a time.
- As seen in the worked example above, sometimes both sides of the identity need to be simplified.
- Remember to write down restrictions:
  - the values for which any of the trigonometric ratios are not defined;
  - the values of the variable which make any of the denominators in the identity equal to zero.

### Exercise 4 – 1: Revision - reduction formulae, co-functions and identities

1. Given:  $\sin 31^\circ = A$

Write each of the following expressions in terms of  $A$ :

- |                      |                                    |
|----------------------|------------------------------------|
| a) $\sin 149^\circ$  | d) $\tan 211^\circ \cos 211^\circ$ |
| b) $\cos(-59^\circ)$ | e) $\tan 31^\circ$                 |
| c) $\cos 329^\circ$  |                                    |

2. a) Simplify  $P$  to a single trigonometric ratio:

$$P = \sin(360^\circ + \theta) \cos(180^\circ + \theta) \tan(360^\circ + \theta)$$

- b) Simplify  $Q$  to a single trigonometric ratio:

$$Q = \frac{\cos(\theta - 360^\circ) \sin(90^\circ + \theta) \sin(-\theta)}{\sin(\theta + 180^\circ)}$$

- c) Hence, determine:

i.  $P + Q$

ii.  $\frac{Q}{P}$

3. If  $p = \sin \beta$ , express the following in terms of  $p$ :

$$\frac{\cos(\beta + 360^\circ) \tan(\beta - 360^\circ) \cos(\beta + 90^\circ)}{\sin^2(\beta + 180^\circ) \cos(\beta - 90^\circ)}$$

4. Evaluate the following without the use of a calculator:

a)  $\frac{\cos(-120^\circ)}{\tan 150^\circ} + \cos 390^\circ$

b)  $(1 - \sin 45^\circ)(1 - \sin 225^\circ)$

5. Reduce the following to one trigonometric ratio:

a)  $\tan^2 \beta - \frac{1}{\cos^2 \beta}$

b)  $\sin^2(90^\circ + \theta) \tan^2 \theta + \tan^2 \theta \cos^2(90^\circ - \theta)$

c)  $\sin \alpha \cos \alpha \tan \alpha - 1$

d)  $\tan^2 \theta + \frac{\cos^2 \theta - 1}{\cos^2 \theta}$

6. a) Use reduction formulae and special angles to show that

$$\frac{\sin(180^\circ + \theta) \tan(720^\circ + \theta) \cos(-\theta)}{\cos(90^\circ + \theta)}$$

can be simplified to  $\sin \theta$ .

- b) Without using a calculator, determine the value of  $\sin 570^\circ$ .

7. Troy's mathematics teacher asks the class to answer the following question.

**Question:**

Prove that  $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$ .

**Troy's answer:**

$$\begin{aligned} \frac{\cos \theta}{1 + \sin \theta} &= \frac{1 - \sin \theta}{\cos \theta} \\ (\cos \theta)(\cos \theta) &= (1 + \sin \theta)(1 - \sin \theta) \\ \cos^2 \theta &= 1 - \sin^2 \theta \\ \cos^2 \theta &= \cos^2 \theta \\ \therefore \text{LHS} &= \text{RHS} \end{aligned}$$

Comment on Troy's answer and show the correct method for proving this identity.

8. Prove the following identities:

(State any restricted values in the interval  $[0^\circ; 360^\circ]$ , where appropriate.)

a)  $\sin^2 \alpha + (\cos \alpha - \tan \alpha)(\cos \alpha + \tan \alpha) = 1 - \tan^2 \alpha$

b)  $\frac{1}{\cos \theta} - \frac{\cos \theta \tan^2 \theta}{1} = \cos \theta$

c)  $\frac{2 \sin \theta \cos \theta}{\sin \theta + \cos \theta} = \sin \theta + \cos \theta - \frac{1}{\sin \theta + \cos \theta}$

d)  $\left( \frac{\cos \beta}{\sin \beta} + \tan \beta \right) \cos \beta = \frac{1}{\sin \beta}$

e)  $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = \frac{2 \tan \theta}{\sin \theta \cos \theta}$

f)  $(1 + \tan^2 \alpha) \cos \alpha = \frac{1 - \tan \alpha}{\cos \alpha - \sin \alpha}$

9. Determine whether the following statements are true or false.

If the statement is false, choose a suitable value between  $0^\circ$  and  $90^\circ$  to confirm your answer.

a)  $\cos(180^\circ - \theta) = -1 - \cos \theta$

d)  $\frac{1}{3} \sin 3\alpha = \sin \alpha$

b)  $\sin(\alpha + \beta) = \sin \alpha + \sin \beta$

e)  $\cos \beta = \sqrt{1 - \sin^2 \beta}$

c)  $\sin \alpha = 2 \sin \frac{\alpha}{2} \sin \frac{\alpha}{2}$

f)  $\sin \theta = \tan \theta \cos \theta$

10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. [28M3](#) 1b. [28M4](#) 1c. [28M5](#) 1d. [28M6](#) 1e. [28M7](#) 2. [28M8](#)

3. [28M9](#) 4a. [28MB](#) 4b. [28MC](#) 5a. [28MD](#) 5b. [28MF](#) 5c. [28MG](#)

5d. [28MH](#) 6. [28MJ](#) 7. [28MK](#) 8a. [28MM](#) 8b. [28MN](#) 8c. [28MP](#)

8d. [28MQ](#) 8e. [28MR](#) 8f. [28MS](#) 9a. [28MT](#) 9b. [28MV](#) 9c. [28MW](#)

9d. [28MX](#) 9e. [28MY](#) 9f. [28MZ](#)



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## 4.2 Compound angle identities

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### Derivation of $\cos(\alpha - \beta)$

EMCGC

#### Investigation: Compound angles

Danny is studying for a trigonometry test and completes the following question:

#### Question:

Evaluate the following:  $\cos(180^\circ - 120^\circ)$

**Danny's solution:**

$$\cos(180^\circ - 120^\circ) = \cos 180^\circ - \cos 120^\circ \quad (\text{line 1})$$

$$= -1 - \cos(90^\circ + 30^\circ) \quad (\text{line 2})$$

$$= -1 + \sin 30^\circ \quad (\text{line 3})$$

$$= -1 + \frac{1}{2} \quad (\text{line 4})$$

$$= -\frac{1}{2} \quad (\text{line 5})$$

1. Consider Danny's solution and determine why it is incorrect.
2. Use a calculator to check that Danny's answer is wrong.
3. Describe in words the mistake(s) in his solution.
4. Is the following statement true or false?  
"A trigonometric ratio can be distributed to the angles that lie within the brackets."

From the investigation above, we know that  $\cos(\alpha - \beta) \neq \cos \alpha - \cos \beta$ . It is wrong to apply the distributive law to the trigonometric ratios of compound angles.

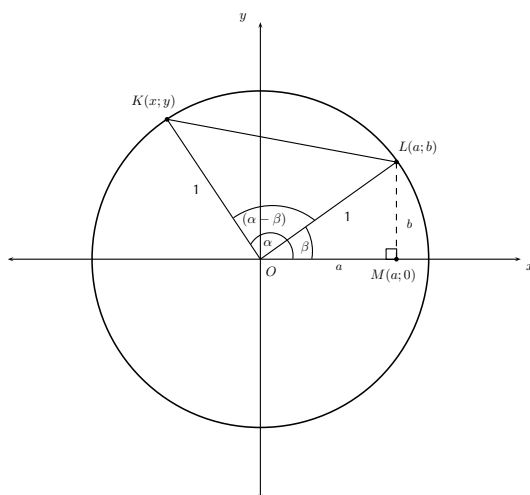
$$\text{Distance formula: } d_{AB} = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$$

$$\text{Cosine rule: } a^2 = b^2 + c^2 - 2bc \cdot \cos \hat{A}$$

Using the distance formula and the cosine rule, we can derive the following identity for compound angles:

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

Consider the unit circle ( $r = 1$ ) below. The two points  $L(a; b)$  and  $K(x; y)$  are shown on the circle.



We can express the coordinates of  $L$  and  $K$  in terms of the angles  $\alpha$  and  $\beta$ :

$$\begin{aligned}\text{In } \triangle LOM, \quad \sin \beta &= \frac{b}{1} \\ \therefore b &= \sin \beta \\ \cos \beta &= \frac{a}{1} \\ \therefore a &= \cos \beta \\ L &= (\cos \beta; \sin \beta)\end{aligned}$$

Similarly,  $K = (\cos \alpha; \sin \alpha)$

We use the distance formula to determine  $KL^2$ :

$$\begin{aligned}d^2 &= (x_K - x_L)^2 + (y_K - y_L)^2 \\ KL^2 &= (\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2 \\ &= \cos^2 \alpha - 2 \cos \alpha \cos \beta + \cos^2 \beta + \sin^2 \alpha - 2 \sin \alpha \sin \beta + \sin^2 \beta \\ &= (\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) - 2 \cos \alpha \cos \beta - 2 \sin \alpha \sin \beta \\ &= 1 + 1 - 2 (\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ &= 2 - 2 (\cos \alpha \cos \beta + \sin \alpha \sin \beta)\end{aligned}$$

Now we determine  $KL^2$  using the cosine rule for  $\triangle KOL$ :

$$\begin{aligned}KL^2 &= KO^2 + LO^2 - 2 \cdot KO \cdot LO \cdot \cos(\alpha - \beta) \\ &= 1^2 + 1^2 - 2(1)(1) \cos(\alpha - \beta) \\ &= 2 - 2 \cdot \cos(\alpha - \beta)\end{aligned}$$

Equating the two expressions for  $KL^2$ , we have

$$\begin{aligned}2 - 2 \cdot \cos(\alpha - \beta) &= 2 - 2 (\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ 2 \cdot \cos(\alpha - \beta) &= 2 (\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ \therefore \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta\end{aligned}$$

### Worked example 3: Derivation of $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$

#### QUESTION

Derive an expression for  $\cos(\alpha + \beta)$  in terms of the trigonometric ratios of  $\alpha$  and  $\beta$ .

#### SOLUTION

**Step 1: Use the compound angle formula for  $\cos(\alpha - \beta)$**

We use the compound angle formula for  $\cos(\alpha - \beta)$  and manipulate the sign of  $\beta$  in  $\cos(\alpha + \beta)$  so that it can be written as a difference of two angles:

$$\cos(\alpha + \beta) = \cos(\alpha - (-\beta))$$

And we have shown  $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

$$\therefore \cos[\alpha - (-\beta)] = \cos \alpha \cos(-\beta) + \sin \alpha \sin(-\beta)$$

$$\therefore \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

**Step 2: Write the final answer**

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

#### Worked example 4: Derivation of $\sin(\alpha - \beta)$ and $\sin(\alpha + \beta)$

##### QUESTION

Derive the expanded formulae for  $\sin(\alpha - \beta)$  and  $\sin(\alpha + \beta)$  in terms of the trigonometric ratios of  $\alpha$  and  $\beta$ .

##### SOLUTION

**Step 1: Use the compound angle formula and co-functions to expand  $\sin(\alpha - \beta)$**

Using co-functions, we know that  $\sin \hat{A} = \cos(90^\circ - \hat{A})$ , so we can write  $\sin(\alpha + \beta)$  in terms of the cosine function as:

$$\begin{aligned}\sin(\alpha - \beta) &= \cos(90^\circ - (\alpha - \beta)) \\ &= \cos(90^\circ - \alpha + \beta) \\ &= \cos[(90^\circ - \alpha) + \beta]\end{aligned}$$

Apply the compound angle formula:

$$\begin{aligned}\cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ \therefore \cos[(90^\circ - \alpha) + \beta] &= \cos(90^\circ - \alpha) \cos \beta - \sin(90^\circ - \alpha) \sin \beta \\ \therefore \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta\end{aligned}$$

To derive the formula for  $\sin(\alpha + \beta)$ , we use the compound angle formula for  $\sin(\alpha - \beta)$  and manipulate the sign of  $\beta$ :

$$\begin{aligned}\sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \text{We can write } \sin(\alpha + \beta) &= \sin[\alpha - (-\beta)] \\ \therefore \sin[\alpha - (-\beta)] &= \sin \alpha \cos(-\beta) - \cos \alpha \sin(-\beta) \\ \therefore \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta\end{aligned}$$

**Step 2: Write the final answers**

$$\begin{aligned}\sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta\end{aligned}$$

##### Compound angle formulae

- $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$
- $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
- $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$
- $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

Note: we can use the compound angle formulae to expand and simplify compound angles in trigonometric expressions (using the equations from left to right) or we can use the expanded form to determine the trigonometric ratio of a compound angle (using the equations from right to left).

### Worked example 5: Compound angle formulae

#### QUESTION

Prove that  $\sin 75^\circ = \frac{\sqrt{2}(\sqrt{3}+1)}{4}$  without using a calculator.

#### SOLUTION

##### Step 1: Consider the given identity

We know the values of the trigonometric functions for the special angles ( $30^\circ$ ,  $45^\circ$ ,  $60^\circ$ , etc.) and we can write  $75^\circ = 30^\circ + 45^\circ$ .

Therefore, we can use the compound angle formula for  $\sin(\alpha + \beta)$  to express  $\sin 75^\circ$  in terms of known trigonometric function values.

##### Step 2: Prove the left-hand side of the identity equals the right-hand side

When proving an identity is true, remember to only work with one side of the identity at a time.

$$\begin{aligned}\text{LHS} &= \sin 75^\circ \\ &= \sin (45^\circ + 30^\circ) \\ \sin (45^\circ + 30^\circ) &= \sin (45^\circ) \cos (30^\circ) + \cos (45^\circ) \sin (30^\circ) \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2}(\sqrt{3} + 1)}{4} \\ &= \text{RHS}\end{aligned}$$

Therefore, we have shown that  $\sin 75^\circ = \frac{\sqrt{2}(\sqrt{3}+1)}{4}$ .

### Worked example 6: Compound angle formulae

#### QUESTION

Determine the value of the following expression without the use of a calculator:

$$\cos 65^\circ \cos 35^\circ + \cos 25^\circ \cos 55^\circ$$

#### SOLUTION

##### Step 1: Use co-functions to simplify the expression

- We need to change two of the trigonometric functions from cosine to sine so that we can apply the compound angle formula.

- We also need to make sure that the sum (or difference) of the two angles is equal to a special angle so that we can determine the value of the expression without using a calculator. Notice that  $35^\circ + 25^\circ = 60^\circ$ .

$$\begin{aligned}\cos 65^\circ \cos 35^\circ + \cos 25^\circ \cos 55^\circ \\&= \cos(90^\circ - 25^\circ) \cos 35^\circ + \cos 25^\circ \cos(90^\circ - 35^\circ) \\&= \sin 25^\circ \cos 35^\circ + \cos 25^\circ \sin 35^\circ\end{aligned}$$

**Step 2: Apply the compound angle formula and use special angles to evaluate the expression**

$$\begin{aligned}\sin 25^\circ \cos 35^\circ + \cos 25^\circ \sin 35^\circ \\&= \sin(25^\circ + 35^\circ) \\&= \sin 60^\circ \\&= \frac{\sqrt{3}}{2}\end{aligned}$$

**Step 3: Write the final answer**

$$\cos 65^\circ \cos 35^\circ + \cos 25^\circ \cos 55^\circ = \frac{\sqrt{3}}{2}$$

**Checking answers:** It is always good to check answers. The question stated that we could not use a calculator to find the answer, but we can use a calculator to check that the answer is correct:

$$\begin{aligned}\text{LHS} &= \cos 65^\circ \cos 35^\circ + \cos 25^\circ \cos 55^\circ = 0,866 \dots \\ \text{RHS} &= \frac{\sqrt{3}}{2} = 0,866 \dots \\ \therefore \text{LHS} &= \text{RHS}\end{aligned}$$

#### Exercise 4 – 2: Compound angle formulae

1. Given:

$$13 \sin \alpha + 5 = 0 \quad (0^\circ < \alpha < 270^\circ)$$

$$13 \cos \beta - 12 = 0 \quad (90^\circ < \beta < 360^\circ)$$

Draw a sketch and determine the following, without the use of a calculator:

- $\tan \alpha - \tan \beta$
- $\sin(\beta - \alpha)$
- $\cos(\alpha + \beta)$

2. Calculate the following without the use of a calculator (leave answers in surd form):

- a)  $\sin 105^\circ$
- b)  $\cos 15^\circ$
- c)  $\sin 15^\circ$
- d)  $\tan 15^\circ$
- e)  $\cos 20^\circ \cos 40^\circ - \sin 20^\circ \sin 40^\circ$
- f)  $\sin 10^\circ \cos 80^\circ + \cos 10^\circ \sin 80^\circ$
- g)  $\cos(45^\circ - x) \cos x - \sin(45^\circ - x) \sin x$
- h)  $\cos^2 15^\circ - \sin^2 15^\circ$

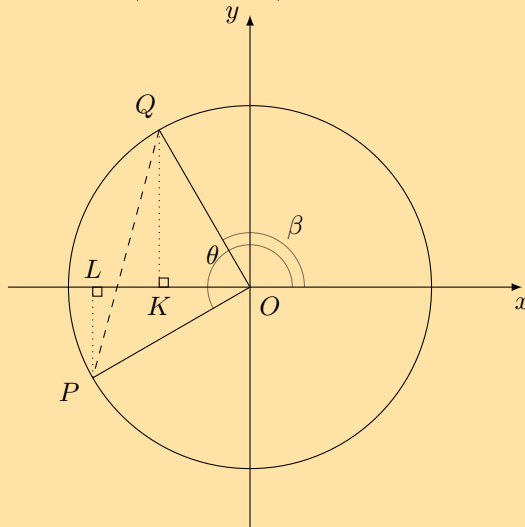
3. a) Prove:  $\sin(60^\circ - x) + \sin(60^\circ + x) = \sqrt{3} \cos x$   
 b) Hence, evaluate  $\sin 15^\circ + \sin 105^\circ$  without using a calculator.  
 c) Use a calculator to check your answer.

4. Simplify the following without using a calculator:

$$\frac{\sin p \cos(45^\circ - p) + \cos p \sin(45^\circ - p)}{\cos p \cos(60^\circ - p) - \sin p \sin(60^\circ - p)}$$

5. a) Prove:  $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$   
 b) Hence, calculate the value of  $\cos 75^\circ \sin 15^\circ$  without using a calculator.
6. In the diagram below, points  $P$  and  $Q$  lie on the circle with radius of 2 units and centre at the origin.

Prove  $\cos(\theta - \beta) = \cos \theta \cos \beta + \sin \theta \sin \beta$ .



7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28N3    2a. 28N4    2b. 28N5    2c. 28N6    2d. 28N7    2e. 28N8  
 2f. 28N9    2g. 28NB    2h. 28NC    3. 28ND    4. 28NF    5. 28NG  
 6. 28NH



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Derivation of  $\sin 2\alpha$ 

EMCGF

We have shown that  $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ . If we let  $\alpha = \beta$ , then we can write the formula as:

$$\begin{aligned}\sin(2\alpha) &= \sin(\alpha + \alpha) \\ &= \sin \alpha \cos \alpha + \cos \alpha \sin \alpha \\ \therefore \sin 2\alpha &= 2 \sin \alpha \cos \alpha\end{aligned}$$

Derivation of  $\cos 2\alpha$ 

EMCGG

Similarly, we know that  $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$ . If we let  $\alpha = \beta$ , then we have:

$$\begin{aligned}\cos(2\alpha) &= \cos(\alpha + \alpha) \\ &= \cos \alpha \cos \alpha - \sin \alpha \sin \alpha \\ \therefore \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha\end{aligned}$$

Using the square identity,  $\sin^2 \alpha + \cos^2 \alpha = 1$ , we can also derive the following formulae:

$$\begin{aligned}\cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= (1 - \sin^2 \alpha) - \sin^2 \alpha \\ \therefore \cos 2\alpha &= 1 - 2 \sin^2 \alpha\end{aligned}$$

And

$$\begin{aligned}\cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= \cos^2 \alpha - (1 - \cos^2 \alpha) \\ &= \cos^2 \alpha - 1 + \cos^2 \alpha \\ \therefore \cos 2\alpha &= 2 \cos^2 \alpha - 1\end{aligned}$$

**Double angle formulae**

- $\sin 2\alpha = 2 \sin \alpha \cos \alpha$
- $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$
- $\cos 2\alpha = 1 - 2 \sin^2 \alpha$
- $\cos 2\alpha = 2 \cos^2 \alpha - 1$

### Worked example 7: Double angle identities

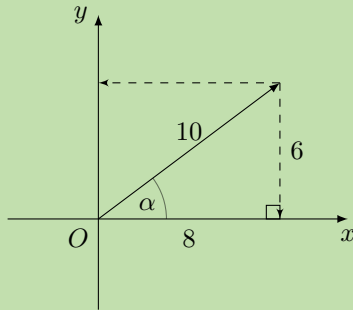
#### QUESTION

If  $\alpha$  is an acute angle and  $\sin \alpha = 0,6$ , determine the value of  $\sin 2\alpha$  without using a calculator.

#### SOLUTION

##### Step 1: Draw a sketch

We convert 0,6 to a fraction so that we can use the ratio to represent the sides of a triangle.



$$\begin{aligned}\sin \alpha &= 0,6 \\ &= \frac{6}{10}\end{aligned}$$

Using Pythagoras:

$$\begin{aligned}x^2 &= r^2 - y^2 \\ &= 10^2 - 6^2 \\ &= 100 - 36 \\ &= 64 \\ \therefore x &= 8\end{aligned}$$

##### Step 2: Use the double angle formula to determine the value of $\sin 2\alpha$

$$\begin{aligned}\sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ &= 2 \left( \frac{6}{10} \right) \left( \frac{8}{10} \right) \\ &= \frac{96}{100} \\ &= 0,96\end{aligned}$$

##### Step 3: Write the final answer

$$\sin 2\alpha = 0,96$$

Check the answer using a calculator:

$$\begin{aligned}\sin \alpha &= 0,6 \\ \therefore \alpha &\approx 36,87^\circ \\ 2\alpha &\approx 73,74^\circ \\ \therefore \sin (73,74^\circ) &\approx 0,96\end{aligned}$$

## Worked example 8: Double angle identities

### QUESTION

Prove that  $\frac{\sin \theta + \sin 2\theta}{1 + \cos \theta + \cos 2\theta} = \tan \theta$ .

For which values of  $\theta$  is the identity not valid?

### SOLUTION

#### Step 1: Consider the given expressions

The right-hand side (RHS) of the identity cannot be simplified, so we simplify the left-hand side (LHS). We also notice that the trigonometric function on the RHS does not have a  $2\theta$  dependence, therefore we will need to use the double angle formulae to simplify  $\sin 2\theta$  and  $\cos 2\theta$  on the LHS.

#### Step 2: Prove the left-hand side equals the right-hand side

$$\begin{aligned}\text{LHS} &= \frac{\sin \theta + 2 \sin \theta \cos \theta}{1 + \cos \theta + (2\cos^2 \theta - 1)} \\ &= \frac{\sin \theta (1 + 2 \cos \theta)}{\cos \theta (1 + 2 \cos \theta)} \quad (\text{factorise}) \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta \\ &= \text{RHS}\end{aligned}$$

#### Step 3: Identify restricted values of $\theta$

We know that  $\tan \theta$  is undefined for  $\theta = 90^\circ + k \cdot 180^\circ, k \in \mathbb{Z}$ .

Note that division by zero on the LHS is not allowed, so the identity will also be undefined for:

$$\begin{aligned}1 + \cos \theta + \cos 2\theta &= 0 \\ \cos \theta (1 + 2 \cos \theta) &= 0 \\ \therefore \cos \theta &= 0 \text{ or } 1 + 2 \cos \theta = 0\end{aligned}$$

$$\text{For } \cos \theta = 0, \quad \theta = 90^\circ + k \cdot 180^\circ$$

$$\begin{aligned}\text{For } 1 + 2 \cos \theta &= 0, \quad \cos \theta = -\frac{1}{2} \\ \therefore \theta &= 120^\circ + k \cdot 360^\circ \\ \text{or } \theta &= 240^\circ + k \cdot 360^\circ\end{aligned}$$

for  $k \in \mathbb{Z}$ .

### Exercise 4 – 3: Double angle identities

1. Given  $5 \cos \theta = -3$  and  $\theta < 180^\circ$ . Determine the value of the following, without a calculator:
  - a)  $\cos 2\theta$
  - b)  $\sin(180^\circ - 2\theta)$
  - c)  $\tan 2\theta$
2. Given  $\cos 40^\circ = t$ , determine (without a calculator):
  - a)  $\cos 140^\circ$
  - b)  $\sin 40^\circ$
  - c)  $\sin 50^\circ$
  - d)  $\cos 80^\circ$
  - e)  $\cos 860^\circ$
  - f)  $\cos(-1160^\circ)$
3.
  - a) Prove the identity:  $\frac{1}{\sin 2A} - \frac{1}{\tan 2A} = \tan A$
  - b) Hence, solve the equation  $\frac{1}{\sin 2A} - \frac{1}{\tan 2A} = 0,75$  for  $0^\circ < A < 360^\circ$ .
4. Without using a calculator, find the value of the following:
  - a)  $\sin 22,5^\circ$
  - b)  $\cos 67,5^\circ$
5.
  - a) Prove the identity:  $\tan 2x + \frac{1}{\cos 2x} = \frac{\sin x + \cos x}{\cos x - \sin x}$
  - b) Explain why the identity is undefined for  $x = 45^\circ$
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28NJ   2. 28NK   3. 28NM   4a. 28NN   4b. 28NP   5. 28NQ



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## 4.4 Solving equations

EMCGH

### The general solution

EMCGJ

The periodicity of the trigonometric functions means that there are an infinite number of positive and negative angles that satisfy an equation. If we do not restrict the solution, then we need to determine the general solution to the equation. We know that the sine and cosine functions have a period of  $360^\circ$  and the tangent function has a period of  $180^\circ$ .

#### Method for finding the solution:

1. Simplify the equation using algebraic methods and trigonometric identities.
2. Determine the reference angle (use a positive value).
3. Use the CAST diagram to determine where the function is positive or negative (depending on the given equation/information).
4. Restricted values: find the angles that lie within a specified interval by adding/-subtracting multiples of the appropriate period.
5. General solution: find the angles in the interval  $[0^\circ; 360^\circ]$  that satisfy the equation and add multiples of the period to each answer.
6. Check answers using a calculator.

### General solutions:

1.

$$\text{If } \sin \theta = x$$

$$\theta = \sin^{-1} x + k \cdot 360^\circ$$

$$\text{or } \theta = (180^\circ - \sin^{-1} x) + k \cdot 360^\circ$$

2.

$$\text{If } \cos \theta = x$$

$$\theta = \cos^{-1} x + k \cdot 360^\circ$$

$$\text{or } \theta = (360^\circ - \cos^{-1} x) + k \cdot 360^\circ$$

3.

$$\text{If } \tan \theta = x$$

$$\theta = \tan^{-1} x + k \cdot 180^\circ$$

for  $k \in \mathbb{Z}$ .

### Worked example 9: Finding the general solution

#### QUESTION

Determine the general solution for  $\sin \theta = 0,3$  (correct to one decimal place).

#### SOLUTION

**Step 1: Use a calculator to find the reference angle**

$$\sin \theta = 0,3$$

$$\begin{aligned}\therefore \text{ref } \angle &= \sin^{-1} 0,3 \\ &= 17,5^\circ\end{aligned}$$

**Step 2: Use CAST diagram to determine in which quadrants  $\sin \theta$  is positive**

The CAST diagram indicates that  $\sin \theta$  is positive in the first and second quadrants.

Using reduction formulae, we know that  $\sin(180^\circ - \theta) = \sin \theta$ .

In the first quadrant:

$$\theta = 17,5^\circ$$

$$\therefore \theta = 17,5^\circ + k \cdot 360^\circ$$

In the second quadrant:

$$\theta = 180^\circ - 17,5^\circ$$

$$\therefore \theta = 162,5^\circ + k \cdot 360^\circ$$

where  $k \in \mathbb{Z}$ .

**Step 3: Check that the solution satisfies the original equation**

We can select random values of  $k$  to check that the answers satisfy the original equation.

Let  $k = 4$ :

$$\theta = 17,5^\circ + 4(360)^\circ$$

$$\therefore \theta = 1457,5^\circ$$

$$\text{And } \sin 1457,5^\circ = 0,3007 \dots$$

This solution is correct.

Similarly, if we let  $k = -2$ :

$$\theta = 162,5^\circ - 2(360)^\circ$$

$$\therefore \theta = -557,5^\circ$$

$$\text{And } \sin(-557,5^\circ) = 0,3007 \dots$$

This solution is also correct.

**Step 4: Write the final answer**

$$\theta = 17,5^\circ + k \cdot 360^\circ \text{ or } \theta = 162,5^\circ + k \cdot 360^\circ \text{ for } k \in \mathbb{Z}.$$

**Worked example 10: Trigonometric equations****QUESTION**

Solve the following equation for  $y$ , without using a calculator:

$$\frac{1 - \sin y - \cos 2y}{\sin 2y - \cos y} = -1$$

**SOLUTION****Step 1: Simplify the equation**

We first simplify the left-hand side of the equation using the double angle formulae. To solve this equation, we need to manipulate the given equation to be of the form:

single trigonometric ratio = constant

$$\frac{1 - \sin y - (1 - 2\sin^2 y)}{2 \sin y \cos y - \cos y} = -1$$

$$\frac{2\sin^2 y - \sin y}{\cos y (2 \sin y - 1)} = -1$$

$$\frac{\sin y (2 \sin y - 1)}{\cos y (2 \sin y - 1)} = -1$$

$$\frac{\sin y}{\cos y} = -1$$

$$\therefore \tan y = -1$$

**Step 2: Use a calculator to find the reference angle**

$$\begin{aligned}\tan y &= -1 \\ \therefore \text{ref } \angle &= \tan^{-1}(1) \\ &= 45^\circ\end{aligned}$$

**Step 3: Use CAST diagram to determine in which quadrants  $\tan y$  is negative**

The CAST diagram indicates that  $\tan y$  is negative in the second and fourth quadrants.

$$\begin{aligned}y &= (180^\circ - 45^\circ) + k \cdot 180^\circ \\ \therefore y &= 135^\circ + k \cdot 180^\circ, \quad k \in \mathbb{Z}\end{aligned}$$

Notice that for  $k = 1$ ,  $y = 135^\circ + 180^\circ = 315^\circ$ , which is the angle in the fourth quadrant.

**Step 4: Check that the solution satisfies the original equation****Step 5: Write the final answer**

$$y = 135^\circ + k \cdot 180^\circ \text{ where } k \in \mathbb{Z}.$$

**Worked example 11: Trigonometric equations****QUESTION**

Prove  $8 \cos^4 x - 8 \cos^2 x + 1 = \cos 4x$  and hence solve  $8 \cos^4 x - 8 \cos^2 x + 1 = 0,8$  (correct to one decimal place).

**SOLUTION****Step 1: Prove the identity**

Expand the right-hand side of the identity and show that it is equal to the left-hand side:

$$\begin{aligned}\text{RHS} &= \cos 4x \\ &= 2 \cos^2 2x - 1 \\ &= 2 (2 \cos^2 x - 1)^2 - 1 \\ &= 2 (4 \cos^4 x - 4 \cos^2 x + 1) - 1 \\ &= 8 \cos^4 x - 8 \cos^2 x + 2 - 1 \\ &= 8 \cos^4 x - 8 \cos^2 x + 1 \\ &= \text{LHS}\end{aligned}$$

**Step 2: Solve the equation**

$$\begin{aligned}8 \cos^4 x - 8 \cos^2 x + 1 &= 0,8 \\ \therefore \cos 4x &= 0,8\end{aligned}$$

**Step 3: Use a calculator to find the reference angle**

$$\begin{aligned}\cos 4x &= 0,8 \\ \therefore \text{ref } \angle &= \cos^{-1}(0,8) \\ &= 36,9^\circ\end{aligned}$$

**Step 4: Use CAST diagram to determine in which quadrants  $\cos 4x$  is positive**

The CAST diagram indicates that  $\cos 4x$  is positive in the first and fourth quadrants.

In the first quadrant:

$$\begin{aligned}4x &= 36,9^\circ + k \cdot 360^\circ \\ \therefore x &= 9,2^\circ + k \cdot 90^\circ\end{aligned}$$

**Important:** remember to also divide  $k \cdot 360^\circ$  by 4.

In the fourth quadrant:

$$\begin{aligned}4x &= (360^\circ - 36,9^\circ) + k \cdot 360^\circ \\ &= 323,1^\circ + k \cdot 360^\circ \\ \therefore x &= 80,8^\circ + k \cdot 90^\circ\end{aligned}$$

where  $k \in \mathbb{Z}$ .

**Step 5: Check that the solution satisfies the original equation**

**Step 6: Write the final answer**

$$\begin{aligned}x &= 9,2^\circ + k \cdot 90^\circ \\ \text{or } x &= 80,8^\circ + k \cdot 90^\circ, \quad k \in \mathbb{Z}\end{aligned}$$

**Worked example 12: Trigonometric equations**

**QUESTION**

Find the general solution for  $\sin \theta \cos^2 \theta = \sin^3 \theta$ .

**SOLUTION**

**Step 1: Simplify the given equation**

Do not divide both sides of the equation by  $\sin \theta$ :

- part of the solution would be lost;
- we would need to restrict the values of  $\theta$  to those where  $\sin \theta \neq 0$  (division by zero is not permitted).

$$\begin{aligned}\sin \theta \cos^2 \theta &= \sin^3 \theta \\ \sin \theta \cos^2 \theta - \sin^3 \theta &= 0 \\ \sin \theta (\cos^2 \theta - \sin^2 \theta) &= 0 \\ \sin \theta (\cos \theta - \sin \theta) (\cos \theta + \sin \theta) &= 0\end{aligned}$$

**Step 2: Apply the zero product law and solve for  $\theta$**

$$\sin \theta (\cos \theta - \sin \theta) (\cos \theta + \sin \theta) = 0$$

$$\begin{aligned}\sin \theta &= 0 \\ \therefore \theta &= 0^\circ + k \cdot 360^\circ \\ \text{or } \theta &= 180^\circ + k \cdot 360^\circ, \quad k \in \mathbb{Z}\end{aligned}$$

$$\begin{aligned}\cos \theta - \sin \theta &= 0 \\ \cos \theta &= \sin \theta \\ \cos \theta &= \cos (90^\circ - \theta) \\ \therefore \theta &= (90^\circ - \theta) + k \cdot 360^\circ \\ 2\theta &= 90^\circ + k \cdot 360^\circ \\ \therefore \theta &= 45^\circ + k \cdot 180^\circ, \quad k \in \mathbb{Z}\end{aligned}$$

$$\begin{aligned}\cos \theta + \sin \theta &= 0 \\ \cos \theta &= -\sin \theta \\ \sin (90^\circ + \theta) &= \sin(-\theta) \\ \therefore 90^\circ + \theta &= -\theta + k \cdot 360^\circ \\ 2\theta &= -90^\circ + k \cdot 360^\circ \\ \therefore \theta &= -45^\circ + k \cdot 180^\circ, \quad k \in \mathbb{Z}\end{aligned}$$

**Step 3: Write the final answer**

$$\begin{aligned}\theta &= 0^\circ + k \cdot 360^\circ \\ \text{or } \theta &= 180^\circ + k \cdot 360^\circ \\ \text{or } \theta &= \pm 45^\circ + k \cdot 180^\circ, \quad k \in \mathbb{Z}\end{aligned}$$

#### Exercise 4 – 4: Solving trigonometric equations

1. Find the general solution for each of the following equations (correct to two decimal places):

a)  $\sin 2x = \tan 28^\circ$

b)  $\cos y = \sin 2y$

c)  $\sin 2\alpha = \cos 2\alpha$

d)  $\sin 3p = \sin 2p$

e)  $\tan A = \frac{1}{\tan A}$

f)  $\sin x \tan x = 1$

g)  $\sin t \cdot \sin 2t + \cos 2t = 1$

h)  $\sin 60^\circ \cos x + \cos 60^\circ \sin x = 1$

2. Given:  $\sin x \cos x = \sqrt{3} \sin^2 x$

a) Solve the equation for  $x \in [0^\circ; 360^\circ]$ , without using a calculator.

b) Draw a graph and indicate the solution on the diagram.

3. Given:  $1 + \tan^2 2A = 5 \tan 2A - 5$

a) Determine the general solution.

b) How many solutions does the given equation have in the interval  $[-90^\circ; 360^\circ]$ ?

4. Without using a calculator, solve  $\cos(A - 25^\circ) + \cos(A + 25^\circ) = \cos 25^\circ$  in  $[-360^\circ; 360^\circ]$ .

5. a) Find the general solution for  $\sin x \cos 3x + \cos x \sin 3x = \tan 140^\circ$ .

b) Use a graph to illustrate the solution for the interval  $[0^\circ; 90^\circ]$ .

6. Explain why the general solution for the equation  $\cos \theta = a$  is  $\theta = \cos^{-1} a + k \cdot 360^\circ$  and the general solution for  $\tan \theta = a$  is  $\theta = \tan^{-1} a + k \cdot 180^\circ$ . Why are they different?

7. Solve for  $x$ :  $\sqrt{3} \sin x + \cos x = 2$

8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. [28NR](#) 1b. [28NS](#) 1c. [28NT](#) 1d. [28NV](#) 1e. [28NW](#) 1f. [28NX](#)

1g. [28NY](#) 1h. [28NZ](#) 2. [28P2](#) 3. [28P3](#) 4. [28P4](#) 5. [28P5](#)

6. [28P6](#) 7. [28P7](#)

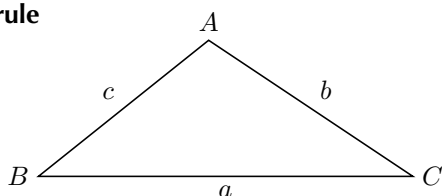


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## Area, sine and cosine rule



| Area rule                                                 | Sine rule                                                | Cosine rule                    |
|-----------------------------------------------------------|----------------------------------------------------------|--------------------------------|
| $\text{area } \triangle ABC = \frac{1}{2}bc \sin \hat{A}$ | $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$ | $a^2 = b^2 + c^2 - 2bc \cos A$ |
| $\text{area } \triangle ABC = \frac{1}{2}ac \sin \hat{B}$ | $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ | $b^2 = a^2 + c^2 - 2ac \cos B$ |
| $\text{area } \triangle ABC = \frac{1}{2}ab \sin \hat{C}$ |                                                          | $c^2 = a^2 + b^2 - 2ab \cos C$ |

## How to determine which rule to use:

## 1. Area rule:

- no perpendicular height is given

## 2. Sine rule:

- no right angle is given
- two sides and an angle are given (**not** the included angle)
- two angles and a side are given

## 3. Cosine rule:

- no right angle is given
- two sides and the included angle are given
- three sides are given

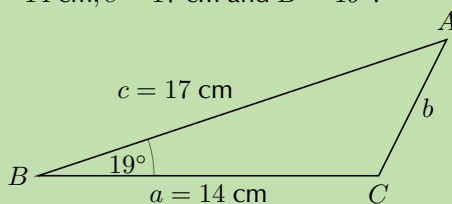
## Problems in two dimensions

## EMCGM

## Worked example 13: Area, sine and cosine rule

## QUESTION

Given  $\triangle ABC$  with  $a = 14$  cm,  $c = 17$  cm and  $\hat{B} = 19^\circ$ .



Calculate the following:

1.  $b$
2.  $\hat{C}$
3. Area  $\triangle ABC$

### SOLUTION

**Step 1: Use the cosine rule to determine the length of  $b$**

$$\begin{aligned}b^2 &= a^2 + c^2 - 2ac \cos \hat{B} \\&= (14)^2 + (17)^2 - 2(14)(17) \cos 19^\circ \\&= 34,93 \dots \\ \therefore b &= 5,9 \text{ cm}\end{aligned}$$

**Step 2: Use the sine rule to determine  $\hat{C}$**

$$\begin{aligned}\frac{\sin \hat{C}}{c} &= \frac{\sin \hat{B}}{b} \\ \frac{\sin \hat{C}}{17} &= \frac{\sin 19^\circ}{5,9} \\ \sin \hat{C} &= \frac{17 \times \sin 19^\circ}{5,9} \\ \therefore \sin \hat{C} &= 0,938 \dots\end{aligned}$$

$$\text{First quadrant: } \hat{C} = 69,7^\circ$$

$$\begin{aligned}\text{Second quadrant: } \hat{C} &= 180^\circ - 69,7^\circ \\ &= 110,3^\circ\end{aligned}$$

From the diagram, we see that  $\hat{C} > 90^\circ$ , therefore  $\hat{C} = 110,3^\circ$ .

**Step 3: Calculate the area  $\triangle ABC$**

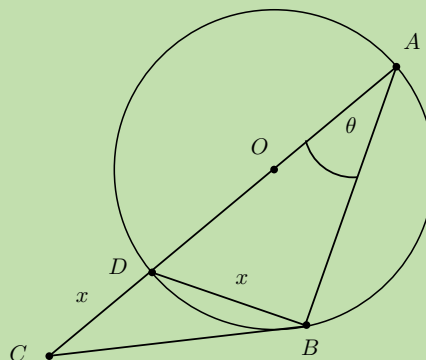
$$\begin{aligned}\text{Area } \triangle ABC &= \frac{1}{2} ac \sin \hat{B} \\ &= \frac{1}{2} (14)(17) \sin 19^\circ \\ &= 38,7 \text{ cm}^2\end{aligned}$$

### Worked example 14: Problem in two dimensions

#### QUESTION

In the figure below,  $CD = BD = x$  and  $\hat{BAD} = \theta$ .

Show that  $BC^2 = 2x^2 (1 + \sin \theta)$ .



## SOLUTION

### Step 1: Consider the given information

Use the given information to determine as many of the unknown angles as possible.

$$CD = BD = x \quad (\text{given})$$

$$\hat{B}AD = \theta \quad (\text{given})$$

$$\hat{D}BA = 90^\circ \quad (\angle \text{ in semi circle})$$

$$\hat{B}DA = 180^\circ - 90^\circ - \theta \quad (\angle \text{ s sum of } \triangle ABD)$$

$$= 90^\circ - \theta$$

$$\hat{B}DC = 90^\circ + \theta \quad (\angle \text{ s on a str. line})$$

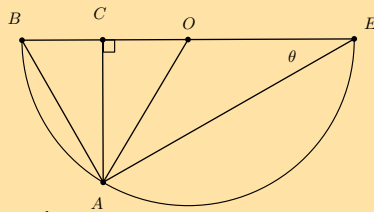
### Step 2: Determine the expression for $BC$

To derive the required expression, we need to write  $BC$  in terms of  $x$  and  $\theta$ . In  $\triangle CDB$ , we can use the cosine rule to determine  $BC$ :

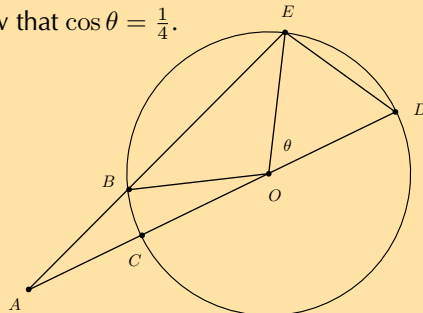
$$\begin{aligned} BC^2 &= CD^2 + BD^2 - 2 \cdot CD \cdot BD \cdot \cos(\hat{B}DC) \\ &= x^2 + x^2 - 2x^2 \cos(90^\circ + \theta) \\ &= 2x^2 - 2x^2(-\sin \theta) \\ &= 2x^2 + 2x^2 \sin \theta \\ &= 2x^2(1 + \sin \theta) \end{aligned}$$

## Exercise 4 – 5: Problems in two dimensions

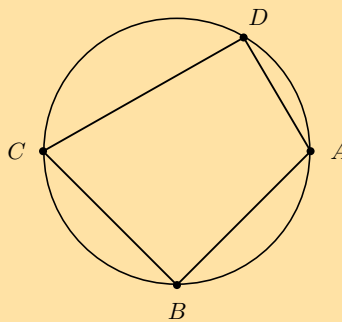
1. In the diagram below,  $O$  is the centre of the semi-circle  $BAE$ .



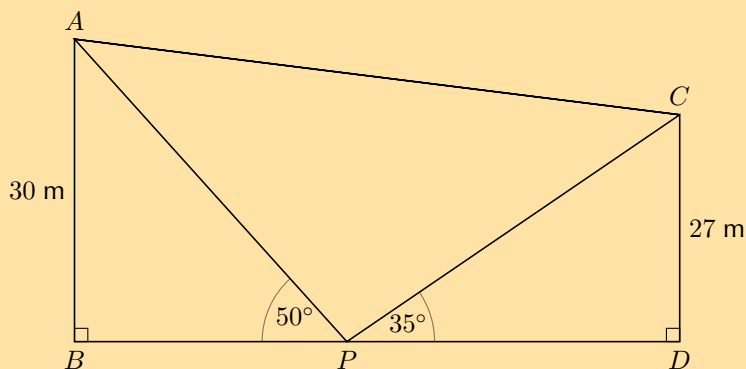
- Find  $\hat{AOC}$  in terms of  $\theta$ .
  - In  $\triangle ABE$ , determine an expression for  $\cos \theta$ .
  - In  $\triangle ACE$ , determine an expression for  $\sin \theta$ .
  - In  $\triangle ACO$ , determine an expression for  $\sin 2\theta$ .
  - Use the results from the previous questions to show that  $\sin 2\theta = 2 \sin \theta \cos \theta$ .
2.  $DC$  is a diameter of the circle with centre  $O$  and radius  $r$ .  $CA = r$ ,  $AE = 2DE$  and  $\hat{DOE} = \theta$ . Show that  $\cos \theta = \frac{1}{4}$ .



3. The figure below shows a cyclic quadrilateral with  $\frac{BC}{CD} = \frac{AD}{AB}$ .



- Show that the area of the cyclic quadrilateral is  $DC \cdot DA \cdot \sin \hat{D}$ .
  - Write down two expressions for  $CA^2$ : one in terms of  $\cos \hat{D}$  and one in terms of  $\cos \hat{B}$ .
  - Show that  $2CA^2 = CD^2 + DA^2 + AB^2 + BC^2$ .
  - Suppose that  $BC = 10$  units,  $CD = 15$  units,  $AD = 4$  units and  $AB = 6$  units. Calculate  $CA^2$  (correct to one decimal place).
  - Find the angle  $\hat{B}$ . Hence, calculate the area of  $ABCD$  (correct to one decimal place).
4. Two vertical towers  $AB$  and  $CD$  are 30 m and 27 m high, respectively. Point  $P$  lies between the two towers. The angle of elevation from  $P$  to  $A$  is  $50^\circ$  and from  $P$  to  $C$  is  $35^\circ$ . A cable is needed to connect  $A$  and  $C$ .



- Determine the minimum length of cable needed to connect  $A$  and  $C$  (to the nearest metre).
  - How far apart are the bases of the two towers (to the nearest metre)?
5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. 28P8    1b. 28P9    1c. 28PB    1d. 28PC    1e. 28PD    2. 28PF  
 3a. 28PG    3b. 28PH    3c. 28PJ    3d. 28PK    3e. 28PM    4. 28PN



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Trigonometric formulae are useful for solving problems in two dimensions. However, in the real world all objects are three dimensional, so it is important that we extend the application of the area, sine and cosine formulae to three dimensional situations.

Drawing a three dimensional diagram is a crucial step in finding the solution to a problem. Interpreting the given information and sketching a three dimensional diagram are skills that need to be practised.

### Worked example 15: Problems in three dimensions - height of a pole

#### QUESTION

$T$  is the top of a pole and its base,  $F$ , is in the same horizontal plane as the points  $A$  and  $B$ . The angle of elevation measured from  $B$  to  $T$  is  $25^\circ$ .  $AB = 120$  m,  $\hat{FAB} = 40^\circ$  and  $\hat{FBA} = 30^\circ$ .

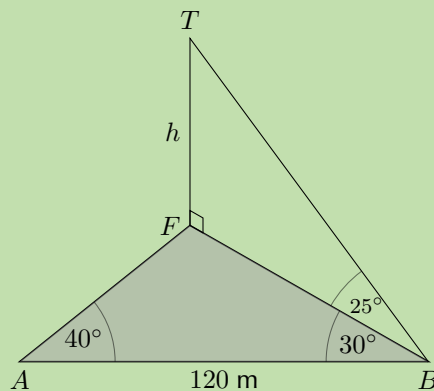
Use the given information to calculate  $h$ , the height of the pole, to the nearest metre.

#### SOLUTION

##### Step 1: Draw a sketch

It is a challenge to analyze this situation when only a description is given. A diagram is very useful for representing three dimensional problems.

Draw a sketch using the given information. Indicate all right angles on the sketch, even those that do not look as though they are a  $90^\circ$  angle (for example,  $\hat{TFB} = 90^\circ$ ). It is also helpful to shade the horizontal plane ( $\triangle FAB$ ), as it adds depth to the diagram and gives a better visualisation of the situation.



##### Step 2: Consider the given information

There are two triangles to consider:  $\triangle FAB$  and  $\triangle TFB$ . We are required to find the length of  $FT$ , but the only information we have for  $\triangle TFB$  is  $\hat{BT} = 25^\circ$ .

**Important:** notice that  $FB$  is a side of  $\triangle TFB$  and it is also a side of  $\triangle FAB$  - it is a link or bridge between the two triangles.

Therefore, to determine the height of the pole:

1. Write down all the given information on the sketch.
2. Use the information for  $\triangle FAB$  to calculate  $FB$ .
3. Use  $FB$  and the information for  $\triangle TFB$  to calculate  $FT$ .

**Step 3: Determine the length of  $FB$**

We notice that  $\triangle FAB$  does not have a right-angle, so we use the sine rule to determine  $FB$ .

$$\begin{aligned}\text{In } \triangle FAB : \quad \hat{F} &= 180^\circ - 40^\circ - 30^\circ && (\angle\text{s sum of } \triangle FAB) \\ &= 110^\circ\end{aligned}$$

$$\begin{aligned}\frac{FB}{\sin \hat{A}} &= \frac{AB}{\sin \hat{F}} \\ \frac{FB}{\sin 40^\circ} &= \frac{120}{\sin 110^\circ} \\ FB &= \frac{120 \times \sin 40^\circ}{\sin 110^\circ} \\ \therefore FB &= 82,084 \dots\end{aligned}$$

**Step 4: Determine the length of  $FT$**

In  $\triangle TFB$ :

$$\begin{aligned}\hat{F} &= 90^\circ && (\text{vertical pole}) \\ \hat{B} &= 25^\circ && (\text{given}) \\ \tan \hat{B} &= \frac{FT}{FB} \\ \tan \hat{B} \times FB &= FT \\ \tan 25^\circ \times 82,084 \dots &= FT \\ \therefore FT &= 38,246 \dots \\ &\approx 38 \text{ m} && (\text{to nearest metre})\end{aligned}$$

**Note:** do not round off answers in the intermediate steps as this will affect the accuracy of the final answer. Always try to do the calculation in one complete step and only round off the final answer.

**Step 5: Write the final answer**

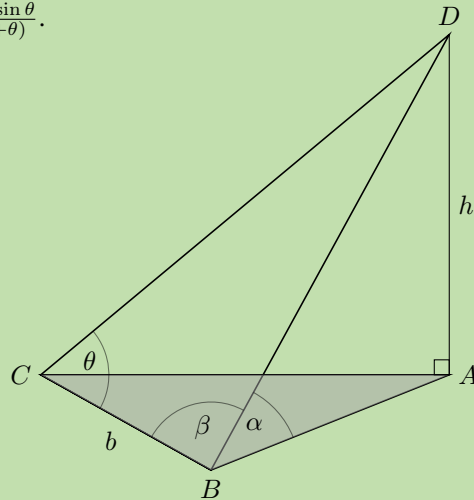
The height of the pole,  $h$ , is 38 m.

The calculation in the above worked example is only applicable for the specific numbers given. However, if we derived a general formula for this contextual situation, then it could be applied to any suitable set of numerical values.

### QUESTION

$D$  is the top of a building of height  $h$ . The base of the building is at  $A$  and  $\triangle ABC$  lies on the ground (a horizontal plane).  $BC = b$ ,  $\hat{D}BA = \alpha$ ,  $\hat{D}BC = \beta$  and  $\hat{D}CB = \theta$ .

Show that  $h = \frac{b \sin \alpha \sin \theta}{\sin(\beta + \theta)}$ .



### SOLUTION

#### Step 1: Consider the given information

We know that  $\triangle ABD$  is right-angled and we are required to find a formula to calculate the length of  $AD$ . In  $\triangle BCD$ , we are given two angles and a length of  $BC$ . We identify the side  $BD$  as the link between  $\triangle ABD$  and  $\triangle BCD$ .

#### Step 2: Determine an expression for $BD$

$\triangle BCD$  does not have a right-angle but two angles and a side are given, so we use the sine rule to determine  $BD$ .

$$\begin{aligned} \text{In } \triangle BCD : \quad \hat{BDC} &= 180^\circ - (\beta + \theta) && (\angle \text{sum of } \triangle BCD) \\ \therefore \sin(180^\circ - (\beta + \theta)) &= \sin(\beta + \theta) \\ \frac{BD}{\sin \theta} &= \frac{b}{\sin(\hat{BDC})} \\ BD &= \frac{b \sin \theta}{\sin(\hat{BDC})} \\ &= \frac{b \sin \theta}{\sin(\beta + \theta)} \end{aligned}$$

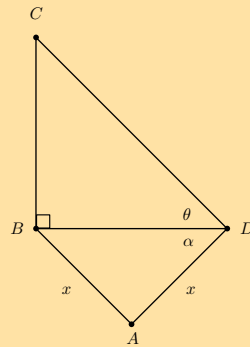
#### Step 3: Determine an expression for $AD$

In  $\triangle ABD$ :

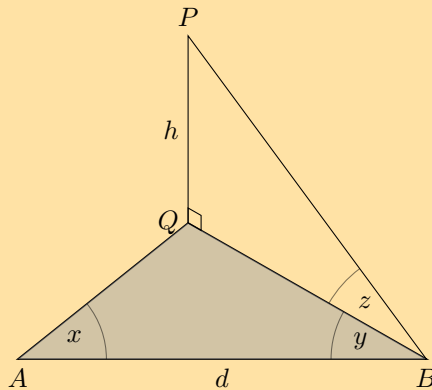
$$\begin{aligned} \hat{BAD} &= 90^\circ && (\text{building vertical}) \\ \hat{DBA} &= \alpha && (\text{given}) \\ \frac{h}{BD} &= \sin \alpha \\ h &= BD \sin \alpha \\ \therefore h &= \frac{b \sin \alpha \sin \theta}{\sin(\beta + \theta)} \end{aligned}$$

### Exercise 4 – 6: Problems in three dimensions

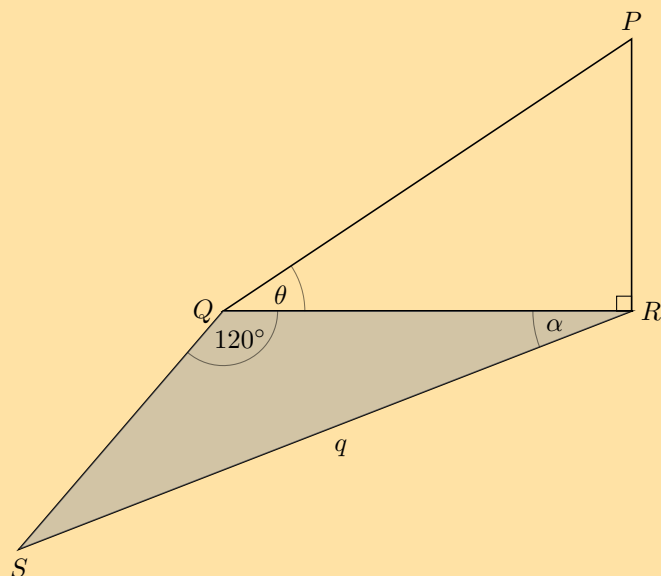
1. The line  $BC$  represents a tall tower, with  $B$  at its base. The angle of elevation from  $D$  to  $C$  is  $\theta$ . A man stands at  $A$  such that  $BA = AD = x$  and  $\hat{ADB} = \alpha$ .



- Find the height of the tower  $BC$  in terms of  $x$ ,  $\tan \theta$  and  $\cos \alpha$ .
  - Find  $BC$  if we are given that  $x = 140$  m,  $\alpha = 21^\circ$  and  $\theta = 9^\circ$ .
2.  $P$  is the top of a mast and its base,  $Q$ , is in the same horizontal plane as the points  $A$  and  $B$ . The angle of elevation measured from  $B$  to  $P$  is  $z$ .  $AB = d$ ,  $\hat{QAB} = x$  and  $\hat{QBA} = y$ .



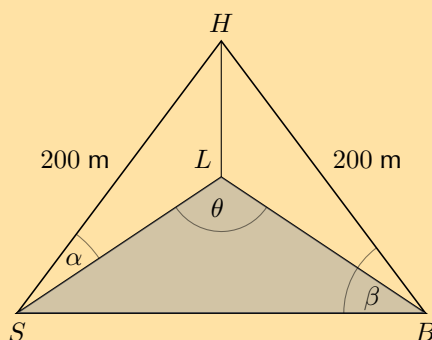
- Use the given information to derive a general formula for  $h$ , the height of the mast.
  - If  $d = 50$  m,  $x = 46^\circ$ ,  $y = 15^\circ$  and  $z = 20^\circ$ , calculate  $h$  (to the nearest metre).
3.  $PR$  is the height of a block of flats with  $R$  at the base and  $P$  at the top of the building.  $S$  is a point in the same horizontal plane as points  $Q$  and  $R$ .  $SR = q$  units,  $\hat{SQR} = 120^\circ$ ,  $\hat{SRQ} = \alpha$  and  $\hat{RQP} = \theta$ .



- a) Show that the height of the block of flats,  $PR$ , can be expressed as:

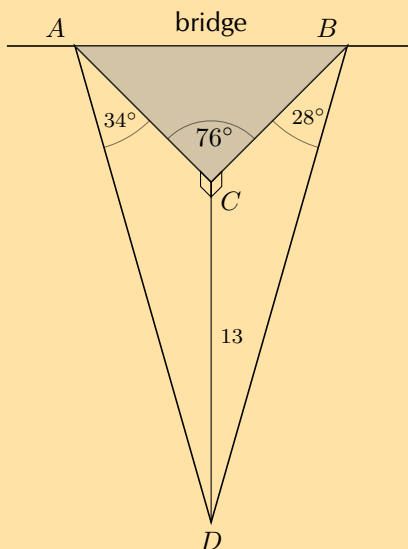
$$PR = q \tan \theta \left( \cos \alpha - \frac{\sqrt{3} \sin \alpha}{3} \right)$$

- b) If  $SR = 35$  m,  $\hat{SRQ} = 16^\circ$  and  $\hat{RQP} = 30^\circ$ , calculate  $PR$  (correct to one decimal place).  
 c) Assuming each level is 2,5 m high, estimate the number of levels in the block of flats.
4. Two ships at sea can see a lighthouse on the shore. The distance from the top of the lighthouse ( $H$ ) to ship  $S$  and to ship  $B$  is 200 m. The angle of elevation from  $S$  to  $H$  is  $\alpha$ ,  $\hat{HBS} = \beta$  and  $\hat{SLB} = \theta$

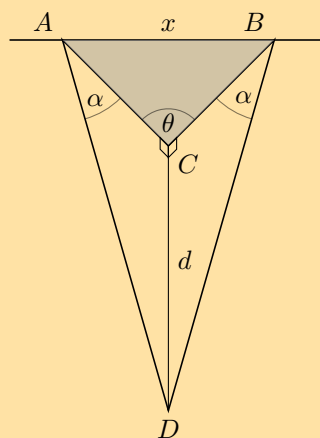


- a) Show that the distance between the two ships is given by  $SB = 400 \cos \beta$ .  
 b) Show that the area of the sea included in  $\triangle LSB$  is given by area  $\triangle LSB = 2000 \cos^2 \alpha \sin \theta$ .  
 c) Calculate the triangular area of the sea if the angle of inclination from the ship to the top of the lighthouse is  $10^\circ$  and the angle between the direct lines from the base of the lighthouse to each ship is  $85^\circ$ .

5. A triangular look-out platform ( $\triangle ABC$ ) is attached to a bridge that extends over a deep gorge. The vertical depth of the gorge, the distance from the edge of the look-out  $C$  to the bottom of the gorge  $D$ , is 13 m. The angle of depression from  $A$  to  $D$  is  $34^\circ$  and from  $B$  to  $D$  is  $28^\circ$ . The angle at the edge of the platform,  $\hat{C}$  is  $76^\circ$ .



- a) Calculate the area of the look-out platform (to the nearest  $\text{m}^2$ ).
- b) If the platform is constructed so that the two angles of depression,  $\hat{C}AD$  and  $\hat{C}BD$ , are both equal to  $45^\circ$  and the vertical depth of the gorge  $CD = d$ ,  $AB = x$  and  $\hat{ACB} = \theta$ , show that  $\cos \theta = 1 - \frac{x^2}{2d^2}$ .



- c) If  $AB = 25$  m and  $CD = 13$  m, calculate  $\hat{ACB}$  (to the nearest integer).
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28PP   2. 28PQ   3. 28PR   4. 28PS   5. 28PT

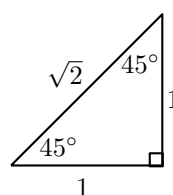
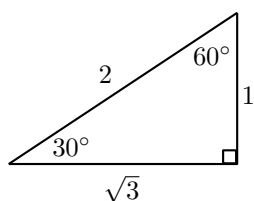


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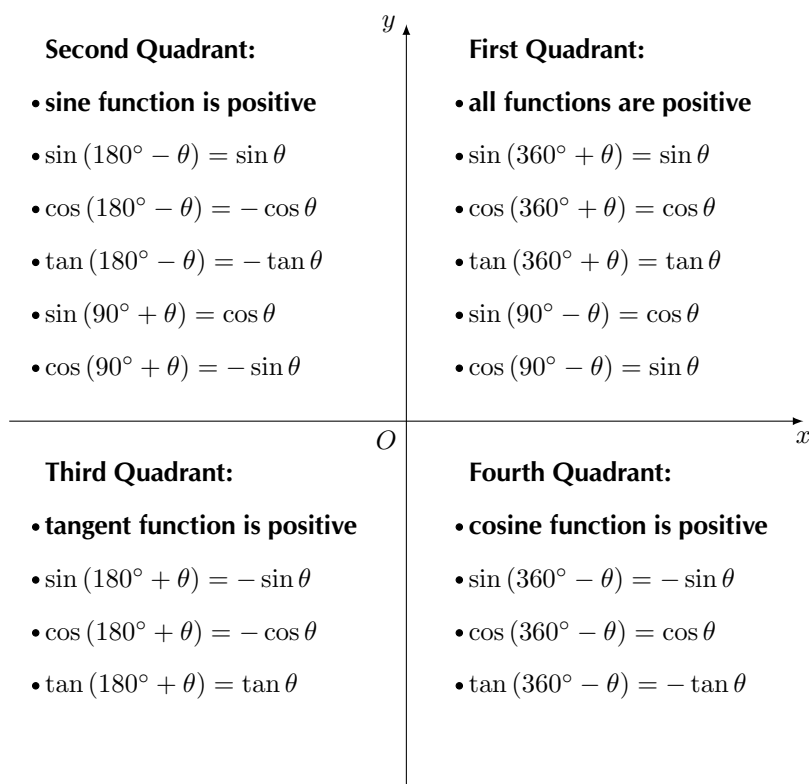


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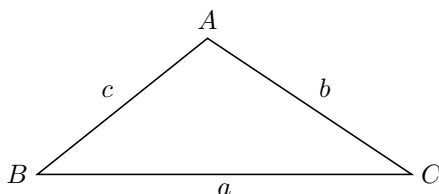
| Pythagorean Identities              | Ratio Identities                                          |
|-------------------------------------|-----------------------------------------------------------|
| $\cos^2 \theta + \sin^2 \theta = 1$ | $\tan \theta = \frac{\sin \theta}{\cos \theta}$           |
| $\cos^2 \theta = 1 - \sin^2 \theta$ | $\frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$ |
| $\sin^2 \theta = 1 - \cos^2 \theta$ |                                                           |

**Special angle triangles**

| $\theta$      | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|---------------|-----------|----------------------|----------------------|----------------------|------------|
| $\cos \theta$ | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0          |
| $\sin \theta$ | 0         | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| $\tan \theta$ | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | undef      |

**CAST diagram and reduction formulae**

| Negative angles                | Periodicity Identities                     | Cofunction Identities                    |
|--------------------------------|--------------------------------------------|------------------------------------------|
| $\sin(-\theta) = -\sin \theta$ | $\sin(\theta \pm 360^\circ) = \sin \theta$ | $\sin(90^\circ - \theta) = \cos \theta$  |
| $\cos(-\theta) = \cos \theta$  | $\cos(\theta \pm 360^\circ) = \cos \theta$ | $\cos(90^\circ - \theta) = \sin \theta$  |
| $\tan(-\theta) = -\tan \theta$ | $\tan(\theta \pm 180^\circ) = \tan \theta$ | $\sin(90^\circ + \theta) = \cos \theta$  |
|                                |                                            | $\cos(90^\circ + \theta) = -\sin \theta$ |



| Area Rule                           | Sine Rule                                                                  | Cosine Rule                          |
|-------------------------------------|----------------------------------------------------------------------------|--------------------------------------|
| Area = $\frac{1}{2}bc \sin \hat{A}$ | $\frac{\sin \hat{A}}{a} = \frac{\sin \hat{B}}{b} = \frac{\sin \hat{C}}{c}$ | $a^2 = b^2 + c^2 - 2bc \cos \hat{A}$ |
| Area = $\frac{1}{2}ab \sin \hat{C}$ | $a \sin \hat{B} = b \sin \hat{A}$                                          | $b^2 = a^2 + c^2 - 2ac \cos \hat{B}$ |
| Area = $\frac{1}{2}ac \sin \hat{B}$ | $b \sin \hat{C} = c \sin \hat{B}$                                          | $c^2 = a^2 + b^2 - 2ab \cos \hat{C}$ |
|                                     | $a \sin \hat{C} = c \sin \hat{A}$                                          |                                      |

| Compound Angle Identities                                                | Double Angle Identities                             |
|--------------------------------------------------------------------------|-----------------------------------------------------|
| $\sin(\theta + \beta) = \sin \theta \cos \beta + \cos \theta \sin \beta$ | $\sin(2\theta) = 2 \sin \theta \cos \theta$         |
| $\sin(\theta - \beta) = \sin \theta \cos \beta - \cos \theta \sin \beta$ | $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$     |
| $\cos(\theta + \beta) = \cos \theta \cos \beta - \sin \theta \sin \beta$ | $\cos(2\theta) = 1 - 2\sin^2 \theta$                |
| $\cos(\theta - \beta) = \cos \theta \cos \beta + \sin \theta \sin \beta$ | $\cos(2\theta) = 2\cos^2 \theta - 1$                |
|                                                                          | $\tan(2\theta) = \frac{\sin 2\theta}{\cos 2\theta}$ |

► See video: 28N2 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

#### Exercise 4 – 7: End of chapter exercises

1. Determine the following without using a calculator:

- |                    |                                                              |
|--------------------|--------------------------------------------------------------|
| a) $\cos 15^\circ$ | d) $\cos 3^\circ \cos 42^\circ - \sin 3^\circ \sin 42^\circ$ |
| b) $\cos 75^\circ$ | e) $1 - 2\sin^2(22,5^\circ)$                                 |
| c) $\tan 75^\circ$ |                                                              |

2. Given  $\cos \theta = 0,7$ . Using a diagram, find  $\cos 2\theta$  and  $\cos 4\theta$ .

3. Given  $7 \sin \alpha = 3$  for  $\alpha > 90^\circ$ .

Determine the following (leave answers in surd form):

- a)  $\cos 2\alpha$   
b)  $\tan 2\alpha$

4. If  $4 \tan A + 3 = 0$  for  $A < 270^\circ$ , determine, without the use of a calculator:

$$\left( \sin \frac{A}{2} - \cos \frac{A}{2} \right) \left( \sin \frac{A}{2} + \cos \frac{A}{2} \right)$$

5. Simplify:  $\cos 67^\circ \cos 7^\circ + \cos 23^\circ \cos 83^\circ$

6. Solve the equation:

$$\cos 3\theta \cos \theta - \sin 3\theta \sin \theta = -\frac{1}{2} \text{ for } \theta \in [-90^\circ; 90^\circ].$$

7. Find the general solution, without a calculator, for the following equations:

a)  $3 \sin \theta = 2 \cos^2 \theta$

b)  $2 \sin 2x - 2 \cos x = \sqrt{2} - 2\sqrt{2} \sin x$

c)  $\cos x \cos 10^\circ + \sin x \cos 100^\circ = 1 - 2 \sin^2 x$

d)  $6 \sin^2 \alpha + 2 \sin 2\alpha - 1 = 0$

8. a) Prove:  $\sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$

b) Hence, solve the equation  $3 \sin \theta - \sin 3\theta = 2$  for  $\theta \in [0^\circ; 360^\circ]$ .

9. Prove the following identities:

a)  $\cos^2 \alpha (1 - \tan^2 \alpha) = \cos 2\alpha$

b)  $4 \sin \theta \cos \theta \cos 2\theta = \sin 4\theta$

c)  $4 \cos^3 x - 3 \cos x = \cos 3x$

d)  $\cos 2A + 2 \sin 2A + 2 = (3 \cos A + \sin A)(\cos A + \sin A)$

e)  $\frac{\cos 2x}{(\cos x + \sin x)^3} = \frac{\cos x - \sin x}{1 + \sin 2x}$

10. a) Prove:  $\tan y = \frac{\sin 2y}{\cos 2y + 1}$

b) For which values of  $y$  is the identity undefined?

11. Given:  $1 + \tan^2 3\theta - 3 \tan 3\theta = 5$

a) Find the general solution.

b) Find the solution for  $\theta \in [0^\circ; 90^\circ]$ .

c) Draw a graph of  $y = \tan 3\theta$  for  $\theta \in [0^\circ; 90^\circ]$  and indicate the solutions to the equation on the graph.

d) Use the graph to determine where  $\tan 3\theta < -1$ .

12. a) Show that:

$$\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$$

b) Use this result to solve  $\sin 3x - \sin x = 0$  for  $x \in [-180^\circ; 360^\circ]$ .

c) On the same system of axes, draw two graphs to solve graphically:  $\sin 3x - \sin x = 0$  for  $x \in [0^\circ; 360^\circ]$ . Indicate the solutions on the graph using the letters  $A, B, \dots$  etc.

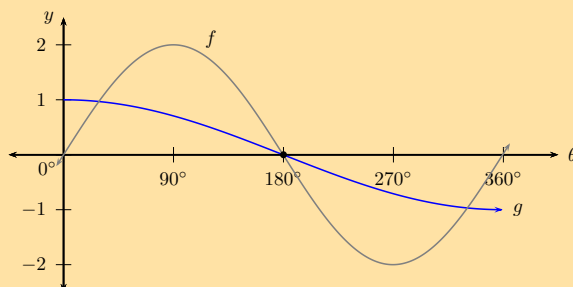
13. Given:  $\cos 2x = \sin x$  for  $x \in [0^\circ; 360^\circ]$

a) Solve for  $x$  algebraically.

b) Verify the solution graphically by sketching two graphs on the same system of axes.

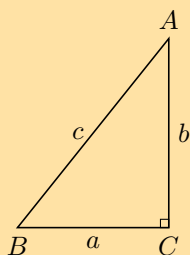
14. The following graphs are given below:

$$\begin{aligned} f &:= a \sin x \\ g &:= \cos bx \quad (x \in [0^\circ; 360^\circ]) \end{aligned}$$



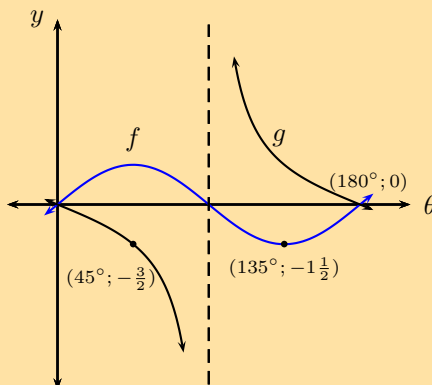
- Explain why  $a = 2$  and  $b = \frac{1}{2}$ .
- For how many  $x$ -values in  $[0^\circ; 360^\circ]$  will  $f(x) - g(x) = 0$ ?
- Use the graph to solve  $f(x) - g(x) = 1$ .
- Solve  $a \sin x = \cos bx$  for  $x \in [0^\circ; 360^\circ]$  using trigonometric identities.
- For which values of  $x$  will  $\frac{1}{2} \cos\left(\frac{x}{2}\right) \leq \sin x$  for  $x \in [0^\circ; 360^\circ]$ ?

15. In  $\triangle ABC$ ,  $AB = c$ ,  $BC = a$ ,  $CA = b$  and  $\hat{C} = 90^\circ$ .



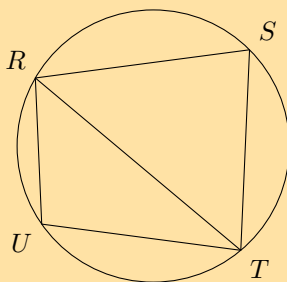
- Prove that  $\sin 2A = \frac{2ab}{c^2}$ .
- Show that  $\cos 2A = \frac{b^2 - a^2}{c^2}$ .

16. Given the graphs of  $f(\theta) = p \sin k\theta$  and  $g(\theta) = q \tan \theta$ , determine the values of  $p$ ,  $k$  and  $q$ .

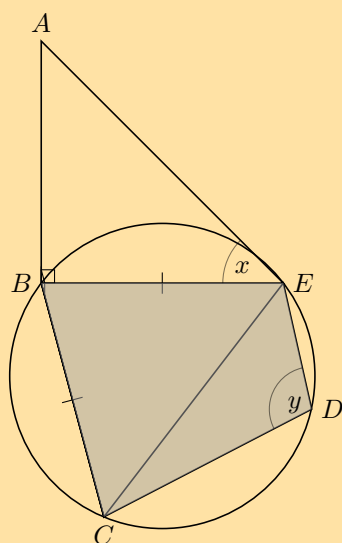


17.  $\triangle RST$  is an acute angled triangle with  $RS = ST = t$ . Show that area  $\triangle RST = t^2 \sin \hat{T} \cos \hat{T}$ .

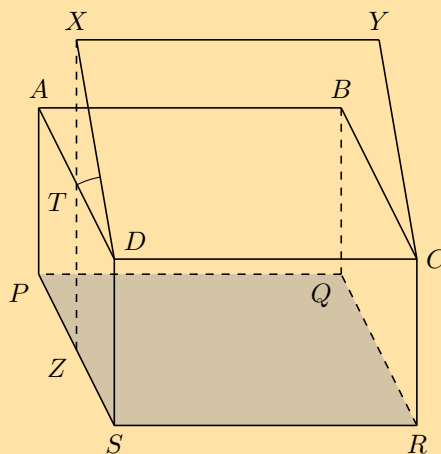
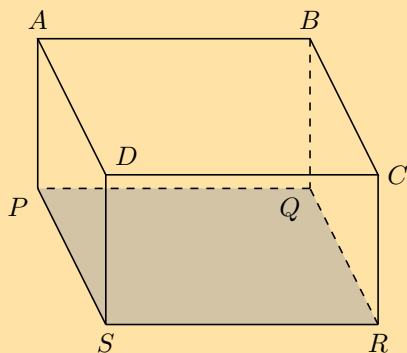
18.  $RSTU$  is a cyclic quadrilateral with  $RU = 6$  cm,  $UT = 7,5$  cm,  $RT = 11$  cm and  $RS = 9,5$  cm.



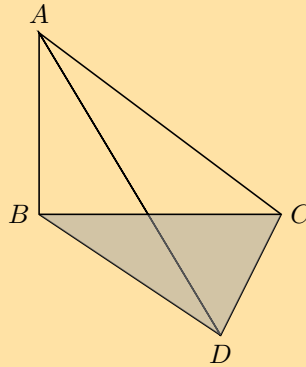
- Calculate  $\hat{U}$ .
  - Determine  $\hat{S}$ .
  - Find  $\hat{RTS}$ .
19.  $BCDE$  is a cyclic quadrilateral that lies in a horizontal plane.  $AB$  is a vertical pole with base  $B$ . The angle of elevation from  $E$  to  $A$  is  $x^\circ$  and  $\hat{CDE} = y^\circ$ .  $\triangle BEC$  is an isosceles triangle with  $BE = BC$ .



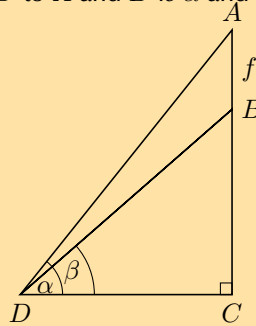
- Show that  $\hat{BCE} = \frac{1}{2}y$ .
  - Show that  $CE = 2BE \cos\left(\frac{y}{2}\right)$ .
  - If  $AB = 2,6$  m,  $x = 37^\circ$  and  $y = 109^\circ$ , calculate the length of  $CE$ .
20. The first diagram shows a rectangular box with  $SR = 8$  cm,  $PS = 6$  cm and  $PA = 4$  cm. The lid of the box,  $ABCD$ , is opened  $30^\circ$  to the position  $XYCD$ , as shown in the second diagram.



- a) Write down the dimensions (length, breadth and diagonal) of the lid  $XYCD$ .
- b) Calculate  $XZ$ , the perpendicular height of  $X$  above the base of the box.
- c) Calculate the ratio  $\frac{\sin \hat{XZC}}{\sin \hat{XZ}}.$
21.  $AB$  is a vertical pole on a horizontal plane  $BCD$ .  $DC$  is  $a$  metres and the angle of elevation from  $D$  to  $A$  is  $\theta$ .  $\hat{ACD} = \alpha$  and  $\hat{ADC} = \beta$ .



- a) Name the two right angles in the diagram.
- b) Show that  $AB = \frac{a \sin \alpha \sin \theta}{\sin(\alpha + \beta)}.$
- c) If it is given that  $AD = AC$ , show that the height of the pole is given by  $AB = \frac{a \sin \theta}{2 \cos \alpha}.$
- d) Calculate the height of the pole if  $a = 13$  m,  $\theta = 33^\circ$ ,  $\alpha = \beta = 65^\circ$ .
22.  $AB$  is a flagpole on top of a government building  $BC$ .  $AB = f$  units and  $D$  is a point on the ground in the same horizontal plane as the base of the building,  $C$ . The angle of elevation from  $D$  to  $A$  and  $B$  is  $\alpha$  and  $\beta$ , respectively.



- a) Show that  $f = \frac{BC \sin(\alpha - \beta)}{\sin \beta \cos \alpha}$
- b) Calculate the height of the flagpole (to the nearest metre) if the building is 7 m,  $\alpha = 63^\circ$  and  $\beta = 57^\circ$ .
- Check answers online with the exercise code below or click on 'show me the answer'.

- |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1a. 28PV | 1b. 28PW | 1c. 28PX | 1d. 28PY | 1e. 28PZ | 2. 28Q2  |
| 3a. 28Q3 | 3b. 28Q4 | 4. 28Q5  | 5. 28Q6  | 6. 28Q7  | 7a. 28Q8 |
| 7b. 28Q9 | 7c. 28QB | 7d. 28QC | 8. 28QD  | 9a. 28QF | 9b. 28QG |
| 9c. 28QH | 9d. 28QJ | 9e. 28QK | 10. 28QM | 11. 28QN | 12. 28QP |
| 13. 28QQ | 14. 28QR | 15. 28QS | 16. 28QT | 17. 28QV | 18. 28QW |
| 19. 28QX | 20. 28QY | 21. 28QZ | 22. 28R2 |          |          |



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## *Polynomials*

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## 5.1 Revision

EMCGQ

## Identifying polynomials

EMCGR

| Terminology: |                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polynomial   | An expression that involves one or more variables having different powers and coefficients.<br>$a_n x^n + \dots + a_2 x^2 + a_1 x + a_0$ , where $n \in \mathbb{N}_0$            |
| Monomial     | A polynomial with one term.<br>For example, $7a^2b$ or $15xyz^2$ .                                                                                                               |
| Binomial     | A polynomial that has two terms.<br>For example, $2x + 5z$ or $26 - g^2k$ .                                                                                                      |
| Trinomial    | A polynomial that has three terms.<br>For example, $a - b + c$ or $4x^2 + 17xy - y^3$ .                                                                                          |
| Degree/Order | The degree, also called the order, of a univariate polynomial is the value of the highest exponent in the polynomial.<br>For example, $7p - 12p^2 + 3p^5 + 8$ has a degree of 5. |

It is important to notice that the definition of a polynomial states that all exponents of the variables must be elements of the set of natural numbers. If an expression contains terms with exponents that are not natural numbers, then it is not a polynomial.

The following examples are not polynomials:

$$\frac{3}{y} - 4y^2 + 1$$

$$5\sqrt{k} + k - 2k^2$$

$$x^{-2} + 3 + 7x^2$$

$$t^2 - 4t + 6t^{\frac{1}{3}}$$

**Investigation: More on polynomials**

Discuss whether the following statements are true or false:

1. The expression  $3y^2 + 2y - 4$  is a trinomial of degree 2.
2.  $25z^5 - 36\sqrt{z}$  is a binomial of order 5.
3. 25 is a constant polynomial of degree 0.
4.  $3x^2 - 2x - 5$  is a quadratic polynomial.
5. The expression  $23b^{-2}$  is a monomial because it only has one term.
6. 0 is a constant polynomial of undefined degree.
7. A cubic polynomial has three terms and all the exponents are natural numbers.
- 8.

Given the expression :  $\frac{1}{t} - 3t^2 + 1$

If we multiply by  $t$  :  $1 - 3t^3 + t$ ,  
we get a trinomial of degree 3

## Exercise 5 – 1: Identifying polynomials

- Given  $f(x) = 2x^3 + 3x^2 - 1$ , determine whether the following statements are true or false. If false, provide the correct statement.
  - $f(x)$  is a trinomial.
  - The coefficient of the  $x$  is zero.
  - $f\left(\frac{1}{2}\right) = \frac{1}{12}$ .
  - $f(x)$  is of degree 3.
  - The constant term is 1.
  - $f(x)$  will have 3 real roots.
- Given  $g(x) = 2x^3 - 9x^2 + 7x + 6$ , determine the following:
  - the number of terms in  $g(x)$ .
  - the degree of  $g(x)$ .
  - the coefficient of the  $x^2$  term.
  - the constant term.
- Determine which of the following expressions are polynomials and which are not. For those that are not polynomials, give reasons.

|                          |                                               |
|--------------------------|-----------------------------------------------|
| a) $y^3 + \sqrt{5}$      | f) $(\sqrt{m} - 1)(\sqrt{m} + 1)$             |
| b) $-x^2 - x - 1$        | g) $t^0 - 1$                                  |
| c) $4\sqrt{k} - 9$       | h) $16y^7$                                    |
| d) $\frac{2}{p} + p + 3$ | i) $-\frac{x^3}{2} + 5x^2 + \frac{x}{3} - 11$ |
| e) $x(x - 1)(x - 2) - 2$ | j) $4b^0 + 3b^{-1} + 5b^2 - b^3$              |

4. Peter's mathematics homework is shown below. Find and correct his mistakes.

### Homework:

Given  $p(x) = x + \frac{4}{x} - 5$ , answer the following questions:

- Simplify the expression.
- Is  $p(x)$  a polynomial?
- What is the coefficient of the  $x$  term?

### Peter's answers:

- $$\begin{aligned} p(x) &= x + \frac{4}{x} - 5 && (\text{restrictions: } x \neq 0) \\ &= x^2 + 4 - 5x && (\text{multiply through by } x) \\ &= x^2 - 5x + 4 && (\text{write in descending order}) \\ &= (x - 1)(x + 4) && (\text{factorise, quadratic has two roots}) \end{aligned}$$
  - Yes, because it can be simplified to have exponents that are all natural numbers. It is a quadratic binomial because the highest exponent is two and there are only two terms;  $(x - 1)$  and  $(x + 4)$ .
  - Before I simplified, the coefficient of the  $x$  term was nothing and after I simplified it became 5.

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28R3    2. 28R4    3a. 28R5    3b. 28R6    3c. 28R7    3d. 28R8  
3e. 28R9    3f. 28RB    3g. 28RC    3h. 28RD    3i. 28RF    3j. 28RG  
4. 28RH



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In earlier grades we learnt the following useful techniques and methods for factorising an expression:

- taking out a common factor
- factorising the difference of two squares
- grouping in pairs
- factorising the sum and difference of two cubes

We also looked at the different methods for factorising quadratic expressions:

- factorising by inspection
- completing the square
- using the quadratic formula
- making a suitable substitution

It is important to revise these methods; we use the quadratic formula for factorising cubic polynomials and we also use completing the square to find the equation of a circle in Chapter 7.

| Terminology:          |                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| Variable              | A symbol used to represent an unknown numerical value. For example, $a, b, x, y, \alpha, \theta$ .            |
| Coefficient           | The number or parameter that is multiplied by the variable of an expression.                                  |
| Expression            | A term or group of terms consisting of numbers, variables and the basic operators (+, −, ×, ÷).               |
| Univariate expression | An expression containing only one variable.                                                                   |
| Equation              | A mathematical statement that asserts that two expressions are equal.                                         |
| Identity              | A mathematical relationship that equates one expression with another.                                         |
| Solution              | A value or set of values that satisfies the original problem statement.                                       |
| Root/Zero             | A root, also referred to as the “zero”, of an equation is the value of $x$ such that $f(x) = 0$ is satisfied. |

### Worked example 1: Solving quadratic equations using factorisation

#### QUESTION

Solve for  $x$ :

$$\frac{3x}{x+2} + 1 = \frac{4}{x+1}$$

#### SOLUTION

##### Step 1: Determine the restrictions

The restrictions are the values for  $x$  that would result in the denominator being equal to 0, which would make the fraction undefined. Therefore  $x \neq -2$  and  $x \neq -1$ .

**Step 2: Determine the lowest common denominator**

The lowest common denominator is  $(x + 2)(x + 1)$ .

**Step 3: Multiply each term in the equation by the lowest common denominator and simplify**

$$\begin{aligned}\frac{3x(x+2)(x+1)}{x+2} + (x+2)(x+1) &= \frac{4(x+2)(x+1)}{x+1} \\ 3x(x+1) + (x+2)(x+1) &= 4(x+2) \\ 3x^2 + 3x + x^2 + 3x + 2 &= 4x + 8 \\ 4x^2 + 2x - 6 &= 0 \\ 2x^2 + x - 3 &= 0\end{aligned}$$

**Step 4: Factorise and solve the equation**

$$\begin{aligned}(2x+3)(x-1) &= 0 \\ 2x+3 &= 0 \text{ or } x-1 = 0 \\ x &= -\frac{3}{2} \text{ or } x = 1\end{aligned}$$

**Step 5: Check the solution by substituting both answers back into the original equation****Step 6: Write the final answer**

Therefore  $x = -1\frac{1}{2}$  or  $x = 1$ .

**Worked example 2: Using the quadratic formula****QUESTION**

Find the roots of the function  $f(x) = 3x^2 + 4x - 8$ .

**SOLUTION****Step 1: Finding the roots**

To determine the roots of  $f(x)$ , we let  $3x^2 + 4x - 8 = 0$ .

**Step 2: Check whether the expression can be factorised**

The expression cannot be factorised, so the general quadratic formula must be used.

**Step 3: Identify the coefficients to substitute into the formula**

$$a = 3; \quad b = 4; \quad c = -8$$

**Step 4: Apply the quadratic formula**

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\&= \frac{-(4) \pm \sqrt{(4)^2 - 4(3)(-8)}}{2(3)} \\&= \frac{-4 \pm \sqrt{16 + 96}}{6} \\&= \frac{-4 \pm \sqrt{112}}{6} \\&= \frac{-4 \pm \sqrt{16 \times 7}}{6} \\&= \frac{-4 \pm 4\sqrt{7}}{6} \\&= \frac{-2 \pm 2\sqrt{7}}{3}\end{aligned}$$

**Step 5: Write the final answer**

Therefore  $x = \frac{-2 + 2\sqrt{7}}{3}$  or  $x = \frac{-2 - 2\sqrt{7}}{3}$

**Worked example 3: Solving quadratic equations by completing the square****QUESTION**

Solve by completing the square:  $y^2 - 10y - 11 = 0$

**SOLUTION**

**Step 1:** The equation is already in the form  $ax^2 + bx + c = 0$

**Step 2:** Make sure the coefficient of the  $y^2$  term is equal to 1

$$y^2 - 10y - 11 = 0$$

**Step 3:** Take half the coefficient of the  $y$  term and square it, then add and subtract it from the equation

The coefficient of the  $y$  term is  $-10$ . Half of the coefficient of the  $y$  term is  $-5$  and the square of it is  $25$ . Therefore  $y^2 - 10y + 25 - 25 - 11 = 0$ .

**Step 4:** Write the trinomial as a perfect square

$$\begin{aligned}(y^2 - 10y + 25) - 25 - 11 &= 0 \\(y - 5)^2 - 36 &= 0\end{aligned}$$

**Step 5: Method 1: Take square roots on both sides of the equation**

$$(y - 5)^2 = 36$$
$$y - 5 = \pm\sqrt{36}$$

**Important:** When taking a square root always remember that there is a positive and negative answer, since  $(6)^2 = 36$  and  $(-6)^2 = 36$ .

$$y - 5 = \pm 6$$

**Step 6: Solve for  $y$**

$$\text{If } y - 5 = 6$$
$$y = 11$$
$$\text{Or if } y - 5 = -6$$
$$y = -1$$

Therefore  $y = 11$  or  $y = -1$ .

**Step 7: Method 2: Factorise the expression as a difference of two squares**

$$(y - 5)^2 - (6)^2 = 0$$
$$[(y - 5) + 6][(y - 5) - 6] = 0$$

**Step 8: Simplify and solve for  $y$**

$$(y + 1)(y - 11) = 0$$
$$\therefore y = -1 \text{ or } y = 11$$

**Step 9: Write the final answer**

$$y = -1 \text{ or } y = 11$$

Notice that both methods produce the same answer. These roots are rational because 36 is a perfect square.

### Exercise 5 – 2: Quadratic polynomials

1. Solve the following quadratic equations by factorisation. Answers may be left in surd form, where applicable.

a)  $7p^2 + 14p = 0$

d)  $(x - 1)(x + 10) + 24 = 0$

b)  $k^2 + 5k - 36 = 0$

e)  $y^2 - 5ky + 4k^2 = 0$

c)  $400 = 16h^2$

2. Solve the following equations by completing the square:

a)  $p^2 + 10p - 2 = 0$

d)  $f^2 + 30 = 2(10 - 8f)$

b)  $2(6y + y^2) = -4$

e)  $3x^2 + 6x - 2 = 0$

c)  $x^2 + 5x + 9 = 0$

3. Solve the following using the quadratic formula.

a)  $3m^2 + m - 4 = 0$

c)  $y^2 - 4y + 2 = 0$

b)  $2t^2 + 6t + 5 = 0$

d)  $3f - 2 = -2f^2$

4. Factorise the following:

a)  $27p^3 - 1$

b)  $16 + \frac{2}{x^3}$

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. [28RJ](#)   1b. [28RK](#)   1c. [28RM](#)   1d. [28RN](#)   1e. [28RP](#)   2a. [28RQ](#)  
2b. [28RR](#)   2c. [28RS](#)   2d. [28RT](#)   2e. [28RV](#)   3a. [28RW](#)   3b. [28RX](#)  
3c. [28RY](#)   3d. [28RZ](#)   4a. [28S2](#)   4b. [28S3](#)



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## 5.2 Cubic polynomials

EMCGT

### Investigation: Simple division

Consider the following and answer the questions below:

1. 6 students are at a product promotion and there are 15 free gifts to be given away. Each student must receive the same number of gifts.
  - a) Determine how many gifts each student will get?
  - b) How many gifts will be left over?
  - c) Use the following variables to express the above situation as a mathematical equation:
    - $a$  = total number of gifts
    - $b$  = total number of students
    - $q$  = number of gifts for each student
    - $r$  = number of gifts left over

2. A group of students go to dinner together at a restaurant and the total bill is R 510. Each student contributes R 120 towards the bill. They count the money and find that they are still R 30 short.
- Assign variables to the known and unknown values.
  - Write a mathematical equation to describe the situation.
  - Use this equation to determine how many students went to dinner.

We know that 11 divided by 2 gives an answer of 5 with a remainder 1.

$$\frac{11}{2} = 5 \text{ remainder } 1$$

$$\frac{11}{2} = 5 + \frac{1}{2}$$

This means that:

$$\begin{array}{ccc}
 & \text{divisor} & \text{remainder} \\
 & \downarrow & \downarrow \\
 11 = 2 \times 5 + 1 \\
 \uparrow & & \uparrow \\
 \text{dividend} & & \text{quotient}
 \end{array}$$

We can write a general expression for the rule of division; if an integer  $a$  is divided by an integer  $b$ , then the answer is  $q$  with a remainder  $r$ .

$$\frac{a}{b} = q + \frac{r}{b}$$

$$a = b \times q + r$$

where  $b \neq 0$  and  $0 \leq r < b$ .

This rule can be extended to include the division of polynomials; if a polynomial  $a(x)$  is divided by a polynomial  $b(x)$ , then the answer is  $Q(x)$  with a remainder  $R(x)$ .

$$a(x) = b(x) \times Q(x) + R(x)$$

where  $b(x) \neq 0$ .

In words: the dividend is equal to the divisor multiplied by the quotient, plus the remainder.

A cubic polynomial is an expression with the highest power equal to 3; we say that the degree of the polynomial is 3.

The general form of a cubic polynomial is

$$y = ax^3 + bx^2 + cx + d$$

where  $a \neq 0$ .

In Grade 10 we learnt how to factorise the sum and difference of two cubes by first finding a factor (the first bracket) and then using inspection (the second bracket). For example,

$$p^3 + 8 = (p + 2)(p^2 - 2p + 4)$$

$$k^3 - 1 = (k - 1)(k^2 + k + 1)$$

In this section we focus on factorising cubic polynomials with one variable (univariate) where  $b$  and  $c$  are not zero.

We use the following methods for factorising cubic polynomials:

- long division
- synthetic division
- inspection

#### Worked example 4: Long division

##### QUESTION

Use the long division method to determine the quotient  $Q(x)$  and the remainder  $R(x)$  if  $a(x) = 2x^3 - x^2 - 6x + 16$  is divided by  $b(x) = x - 1$ . Write your answer in the form  $a(x) = b(x) \cdot Q(x) + R(x)$ .

##### SOLUTION

**Step 1: Write down the known and unknown expressions**

$$a(x) = b(x) \cdot Q(x) + R(x)$$

$$2x^3 - x^2 - 6x + 16 = (x - 1) \cdot Q(x) + R(x)$$

**Step 2: Use long division method to determine  $Q(x)$  and  $R(x)$**

Make sure that  $a(x)$  and  $b(x)$  are written in descending order of the exponents. If a term of a certain degree is missing from  $a(x)$ , then write the term with a coefficient of 0.

$$\begin{array}{r}
 2x^2 + x - 5 \\
 x - 1 \overline{) 2x^3 - x^2 - 6x + 16} \\
 \underline{-(2x^3 - 2x^2)} \phantom{+ 16} \\
 0 + x^2 - 6x \phantom{+ 16} \\
 \underline{-(x^2 - x)} \phantom{+ 16} \\
 0 - 5x + 16 \\
 \underline{-(-5x + 5)} \\
 0 + 11
 \end{array}$$

**Step 3: Write the final answer**

$$Q(x) = 2x^2 + x - 5$$

$$R(x) = 11$$

and  $a(x) = b(x) \cdot Q(x) + R(x)$

$$\therefore a(x) = (x - 1)(2x^2 + x - 5) + 11$$

Synthetic division is a simpler and more efficient method for dividing polynomials. It allows us to determine the quotient and the remainder by considering the coefficients of the terms in each of the polynomials without needing to rewrite the variable and exponent for each term. If a term of a certain degree is missing from  $a(x)$ , then write the term with a coefficient of 0. For example,  $a(x) = 5x^3 + 6x - 1$  should be written as  $a(x) = 5x^3 + 0x^2 + 6x - 1$ .

Notice that for synthetic division:

- the coefficients of the dividend ( $a(x)$ ) are written below the horizontal line.
- the coefficients of the quotient ( $Q(x)$ ) are written above the horizontal line.
- we add coefficients instead of subtracting as is the case with long division
- we use the opposite sign of the divisor ( $b(x)$ ); the divisor is  $(x - 1)$  and we use  $+1$ .
- the coefficient of the  $x$  term in the divisor is 1, so  $q_2 = a_3$ .

► See video: 2854 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 5: Synthetic division

#### QUESTION

Use the synthetic division method to determine the quotient  $Q(x)$  and the remainder  $R(x)$  if  $a(x) = 2x^3 - x^2 - 6x + 16$  is divided by  $b(x) = x - 1$ . Write your answer in the form  $a(x) = b(x) \cdot Q(x) + R(x)$ .

#### SOLUTION

**Step 1: Write down the known and unknown expressions**

$$\begin{aligned} a(x) &= b(x) \cdot Q(x) + R(x) \\ 2x^3 - x^2 - 6x + 16 &= (x - 1) \cdot Q(x) + R(x) \end{aligned}$$

**Step 2: Use synthetic division to determine  $Q(x)$  and  $R(x)$**

$$\begin{array}{r|rrrr} & 2 & +1 & -5 & 11 \\ 1 & 2 & -1 & -6 & 16 \end{array}$$

$$\begin{aligned} q_2 &= 2 \\ q_1 &= -1 + (2)(1) = 1 \\ q_0 &= -6 + (1)(1) = -5 \\ R &= 16 + (-5)(1) = 11 \end{aligned}$$

**Step 3: Write the final answer**

The quotient will be one degree lower than the dividend if we divide by a linear expression, therefore we have:

$$\begin{aligned} Q(x) &= 2x^2 + x - 5 \\ R(x) &= 11 \\ \text{and } a(x) &= b(x) \cdot Q(x) + R(x) \\ \therefore a(x) &= (x - 1)(2x^2 + x - 5) + 11 \end{aligned}$$

**General method:** given the dividend  $a(x) = a_3x^3 + a_2x^2 + a_1x^1 + a_0x^0$  and the divisor  $(cx - d)$ , we determine the quotient  $Q(x) = q_2x^2 + q_1x^1 + q_0x^0$  and the remainder  $R(x)$  using:

$$\frac{d}{c} \overline{) \begin{array}{cccc} & q_2 & q_1 & q_0 & R \\ a_3 & a_2 & a_1 & a_0 \end{array}}$$

We determine the coefficients of the quotient by calculating:

$$\begin{aligned} q_2 &= a_3 + \left( q_3 \times \frac{d}{c} \right) \\ &= a_3 \quad (\text{since } q_3 = 0) \\ q_1 &= a_2 + \left( q_2 \times \frac{d}{c} \right) \\ q_0 &= a_1 + \left( q_1 \times \frac{d}{c} \right) \\ R &= a_0 + \left( q_0 \times \frac{d}{c} \right) \end{aligned}$$

Important note:  $a(x)$  is a function and  $a_3, a_2, a_1$ , and  $a_0$  are coefficients.

► See video: 28S5 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 6: Synthetic division

#### QUESTION

Use the synthetic division method to determine the quotient  $Q(x)$  and the remainder  $R(x)$  if  $a(x) = 6x^3 + x^2 - 4x + 5$  is divided by  $b(x) = 2x - 1$ .

#### SOLUTION

**Step 1: Write down the known and unknown expressions**

$$\begin{aligned} a(x) &= b(x) \cdot Q(x) + R(x) \\ 6x^3 + x^2 - 4x + 5 &= (2x - 1) \cdot Q(x) + R(x) \end{aligned}$$

**Step 2: Use synthetic division to determine  $Q(x)$  and  $R(x)$**

Make the leading coefficient of the divisor equal to 1:

$$b(x) = (2x - 1) = 2 \left( x - \frac{1}{2} \right)$$

$$\begin{array}{r} 6 \quad 1 \quad -4 \quad 5 \\ \frac{1}{2} \overline{) 6 \quad 1 \quad -4 \quad 5} \end{array}$$

$$q_2 = 6$$

$$q_1 = 1 + (6) \left( \frac{1}{2} \right) = 4$$

$$q_0 = -4 + (4) \left( \frac{1}{2} \right) = -2$$

$$R = 5 + (-2) \left( \frac{1}{2} \right) = 4$$

**Step 3: Write the final answer**

$$Q(x) = 6x^2 + 4x - 2$$

$$= 2(3x^2 + 2x - 1)$$

$$R = 4$$

and  $a(x) = \frac{1}{2}b(x) \cdot Q(x) + R(x)$

$$\therefore a(x) = \frac{1}{2} \cdot 2 \left( x - \frac{1}{2} \right) (6x^2 + 4x - 2) + 4$$

$$= \frac{1}{2} \cdot 2 \left( x - \frac{1}{2} \right) (2)(3x^2 + 2x - 1) + 4$$

$$= (2x - 1)(3x^2 + 2x - 1) + 4$$

► See video: 28S6 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Exercise 5 – 3: Cubic polynomials

1. Factorise the following:

- |               |                    |
|---------------|--------------------|
| a) $p^3 - 1$  | d) $k - 125k^4$    |
| b) $t^3 + 27$ | e) $8a^6 - b^9$    |
| c) $64 - m^3$ | f) $8 - (p + q)^3$ |

2. For each of the following:

- Use long division to determine the quotient  $Q(x)$  and the remainder  $R(x)$ .
- Write  $a(x)$  in the form  $a(x) = b(x) \cdot Q(x) + R(x)$ .
- Check your answer by expanding the brackets to get back to the original cubic polynomial.

- a)  $a(x) = x^3 + 2x^2 + 3x + 7$  is divided by  $(x + 1)$
- b)  $a(x) = 1 + 4x^2 - 5x - x^3$  and  $b(x) = x + 2$
- c)  $a(x) = 2x^3 + 3x^2 + x - 6$  and  $b(x) = x - 1$

d)  $a(x) = x^3 + 2x^2 + 5$  and  $b(x) = x - 1$

e)  $(x - 1)$  is divided into  $a(x) = x^4 + 2x^3 - 3x^2 + 5x + 4$

f)  $\frac{a(x)}{b(x)} = \frac{5x^4 + 3x^3 + 6x^2 + x + 2}{x^2 - 2}$

g)  $a(x) = 3x^3 - x^2 + 2x + 1$  is divided by  $(3x - 1)$

h)  $a(x) = 2x^5 + x^3 + 3x^2 - 4$  and  $b(x) = x + 2$

3. Use synthetic division to determine the quotient  $Q(x)$  and the remainder  $R(x)$  when  $f(x)$  is divided by  $g(x)$ :

a)  $f(x) = x^2 + 5x + 1$   
 $g(x) = x + 2$

e)  $f(x) = x^3 + 2x^2 + x - 10$   
 $g(x) = x - 1$

b)  $f(x) = x^2 - 5x - 7$   
 $g(x) = x - 1$

f)  $f(x) = 2x^3 + 7x^2 + 2x - 3$   
 $g(x) = x + 3$

c)  $f(x) = 2x^3 + 5x - 4$   
 $g(x) = x - 1$

g)  $f(x) = 4x^3 + 4x^2 - x - 2$   
 $g(x) = 2x - 1$

d)  $f(x) = 19 + x^2 + 8x$   
 $g(x) = x + 3$

h)  $f(x) = 5x + 22 + 2x^3 + x^2$   
 $g(x) = 2x + 3$

4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

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2g. [28SN](#) 2h. [28SP](#) 3a. [28SQ](#) 3b. [28SR](#) 3c. [28SS](#) 3d. [28ST](#)  
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**Investigation: Remainder theorem**

Given the following functions:

- $f(x) = x^3 + 3x^2 + 4x + 12$
- $k(x) = x - 1$
- $g(x) = 4x^3 - 2x^2 + x - 7$
- $h(x) = x + 2$

1. Determine  $\frac{f(x)}{k(x)}$  and  $\frac{g(x)}{h(x)}$ .
2. Write your answers in the general form:  $a(x) = b(x).Q(x) + R(x)$ .
3. Determine  $f(1)$  and  $g(-2)$ .
4. What do you notice?
5. Consider the degree of the quotient and the remainder - is there a rule?
6. What conclusions can you draw?
7. Write a mathematical equation to describe your conclusions.
8. Complete the following sentence: a cubic function divided by a linear polynomial gives a quotient with a degree of ..... and a remainder with a degree of ....., which is called a constant.

**The Remainder theorem**

A polynomial  $p(x)$  divided by  $cx - d$  gives a remainder of  $p\left(\frac{d}{c}\right)$ .

In words: the value of the remainder  $R$  is obtained by substituting  $x = \frac{d}{c}$  into the polynomial  $p(x)$ .

$$R = p\left(\frac{d}{c}\right)$$

**NOTE: PROOF NOT FOR EXAMS**

Let the quotient be  $Q(x)$  and let the remainder be  $R$ . Therefore we can write:

$$\begin{aligned}
 p(x) &= (cx - d) \cdot Q(x) + R \\
 \therefore p\left(\frac{d}{c}\right) &= \left[c\left(\frac{d}{c}\right) - d\right] \cdot Q\left(\frac{d}{c}\right) + R \\
 &= (d - d) \cdot Q\left(\frac{d}{c}\right) + R \\
 &= 0 \cdot Q\left(\frac{d}{c}\right) + R \\
 &= R \\
 \therefore p\left(\frac{d}{c}\right) &= R
 \end{aligned}$$

### Worked example 7: Finding the remainder

#### QUESTION

Use the remainder theorem to determine the remainder when  $p(x) = 3x^3 + 5x^2 - x + 1$  is divided by the following linear polynomials:

1.  $x + 2$
2.  $2x - 1$
3.  $x + m$

#### SOLUTION

##### Step 1: Determine the remainder for each linear divisor

The remainder theorem states that any polynomial  $p(x)$  that is divided by  $cx - d$  gives a remainder of  $p\left(\frac{d}{c}\right)$ :

1.

$$\begin{aligned}p(x) &= 3x^3 + 5x^2 - x + 1 \\p(-2) &= 3(-2)^3 + 5(-2)^2 - (-2) + 1 \\&= 3(-8) + 5(4) + 2 + 1 \\&= -24 + 20 + 3 \\&\therefore R = -1\end{aligned}$$

2.

$$\begin{aligned}p(x) &= 3x^3 + 5x^2 - x + 1 \\p\left(\frac{1}{2}\right) &= 3\left(\frac{1}{2}\right)^3 + 5\left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right) + 1 \\&= 3\left(\frac{1}{8}\right) + 5\left(\frac{1}{4}\right) - \left(\frac{1}{2}\right) + 1 \\&= \frac{3}{8} + \frac{5}{4} + \frac{1}{2} \\&= \frac{3}{8} + \frac{10}{8} + \frac{4}{8} \\&\therefore R = \frac{17}{8}\end{aligned}$$

3.

$$\begin{aligned}p(x) &= 3x^3 + 5x^2 - x + 1 \\p(m) &= 3(-m)^3 + 5(-m)^2 - (-m) + 1 \\&\therefore R = -3m^3 + 5m^2 + m + 1\end{aligned}$$

### Worked example 8: Using the remainder to solve for an unknown variable

#### QUESTION

Given that  $f(x) = 2x^3 + x^2 + kx + 5$  divided by  $2x - 3$  gives a remainder of  $9\frac{1}{2}$ , use the remainder theorem to determine the value of  $k$ .

#### SOLUTION

**Step 1: Use the remainder theorem to determine the unknown variable  $k$**

From the remainder theorem we know that  $f\left(\frac{3}{2}\right) = 9\frac{1}{2}$  and we can therefore solve for  $k$ :

$$\begin{aligned}f(x) &= 2x^3 + x^2 + kx + 5 \\f\left(\frac{3}{2}\right) &= 2\left(\frac{3}{2}\right)^3 + \left(\frac{3}{2}\right)^2 + k\left(\frac{3}{2}\right) + 5 \\9\frac{1}{2} &= 2\left(\frac{27}{8}\right) + \left(\frac{9}{4}\right) + k\left(\frac{3}{2}\right) + 5 \\9\frac{1}{2} &= \frac{27}{4} + \frac{9}{4} + \frac{3k}{2} + 5 \\9\frac{1}{2} &= \frac{36}{4} + \frac{3k}{2} + 5 \\\therefore 9\frac{1}{2} - 9 - 5 &= \frac{3k}{2} \\-4\frac{1}{2} &= \frac{3k}{2} \\-\frac{9}{2} \times \frac{2}{3} &= k \\\therefore -3 &= k\end{aligned}$$

**Step 2: Write the final answer**

Therefore  $k = -3$  and  $f(x) = 2x^3 + x^2 - 3x + 5$ .

### Exercise 5 – 4: Remainder theorem

1. Use the remainder theorem to determine the remainder  $R$  when  $g(x)$  is divided by  $h(x)$ :

a) 
$$\begin{aligned} g(x) &= x^3 + 4x^2 + 11x - 5 \\ h(x) &= x - 1 \end{aligned}$$

e) 
$$\begin{aligned} g(x) &= x^4 + 5x^2 + 2x - 8 \\ h(x) &= x + 1 \end{aligned}$$

b) 
$$\begin{aligned} g(x) &= 2x^3 - 5x^2 + 8 \\ h(x) &= 2x - 1 \end{aligned}$$

f) 
$$\begin{aligned} g(x) &= 3x^5 - 8x^4 + x^2 + 2 \\ h(x) &= 2 - x \end{aligned}$$

c) 
$$\begin{aligned} g(x) &= 4x^3 + 5x^2 + 6x - 1 \\ h(x) &= x + 2 \end{aligned}$$

g) 
$$\begin{aligned} g(x) &= 2x^{100} - x - 1 \\ h(x) &= x + 1 \end{aligned}$$

d) 
$$\begin{aligned} g(x) &= -5x^3 - x^2 - 10x + 9 \\ h(x) &= 5x + 1 \end{aligned}$$

2. Determine the value of  $t$  if  $x^3 + tx^2 + 8x + 21$  divided by  $x + 1$  gives a remainder of 16.
3. Calculate the value of  $m$  if  $2x^3 - 7x^2 + mx - 26$  is divided by  $x - 2$  and gives a remainder of  $-24$ .
4. If  $x^5 - 2x^3 - kx - 1$  is divided by  $x - 1$  and the remainder is  $-\frac{1}{2}$ , find the value of  $k$ .
5. Determine the value of  $p$  if  $18x^3 + px^2 - 8x + 9$  is divided by  $2x - 1$  and gives a remainder of 6.
6. If  $x^3 + x^2 - x + b$  is divided by  $x - 2$  and the remainder is  $2\frac{1}{2}$ , calculate the value of  $b$ .
7. Calculate the value of  $h$  if  $3x^5 + hx^4 + 10x^2 - 21x + 12$  is divided by  $x - 2$  and gives a remainder of 10.
8. If  $x^3 + 8x^2 + mx - 5$  is divided by  $x + 1$  and the remainder is  $n$ , express  $m$  in terms of  $n$ .
9. When the polynomial  $2x^3 + px^2 + qx + 1$  is divided by  $x + 1$  or  $x - 4$ , the remainder is 5. Determine the values of  $p$  and  $q$ .
10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

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1g. [28T7](#)   2. [28T8](#)   3. [28T9](#)   4. [28TB](#)   5. [28TC](#)   6. [28TD](#)  
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If an integer  $a$  is divided by an integer  $b$ , and the answer is  $q$  with the remainder  $r = 0$ , then we know that  $b$  is a factor of  $a$ .

$$a = b \times q + r$$

$$\text{and } r = 0$$

$$\text{then we know that } a = b \times q$$

$$\text{and also that } \frac{a}{b} = q$$

This is also true of polynomials; if a polynomial  $a(x)$  is divided by a polynomial  $b(x)$ , and the answer is  $Q(x)$  with the remainder  $R(x) = 0$ , then we know that  $b(x)$  is a factor of  $a(x)$ .

$$a(x) = b(x) \cdot Q(x) + R(x)$$

$$\text{and } R(x) = 0$$

$$\text{then we know that } a(x) = b(x) \cdot Q(x)$$

$$\text{and also that } \frac{a(x)}{b(x)} = Q(x)$$

The factor theorem describes the relationship between the root of a polynomial and a factor of the polynomial.

#### The Factor theorem

If the polynomial  $p(x)$  is divided by  $cx - d$  and the remainder, given by  $p\left(\frac{d}{c}\right)$ , is equal to zero, then  $cx - d$  is a factor of  $p(x)$ .

**Converse:** if  $(cx - d)$  is a factor of  $p(x)$ , then  $p\left(\frac{d}{c}\right) = 0$ .

#### Worked example 9: Factor theorem

##### QUESTION

Using the factor theorem, show that  $y + 4$  is a factor of  $g(y) = 5y^4 + 16y^3 - 15y^2 + 8y + 16$ .

##### SOLUTION

##### Step 1: Determine how to approach the problem

For  $y + 4$  to be a factor,  $g(-4)$  must be equal to 0.

##### Step 2: Calculate $g(-4)$

$$\begin{aligned} g(y) &= 5y^4 + 16y^3 - 15y^2 + 8y + 16 \\ \therefore g(-4) &= 5(-4)^4 + 16(-4)^3 - 15(-4)^2 + 8(-4) + 16 \\ &= 5(256) + 16(-64) - 15(16) + 8(-4) + 16 \\ &= 1280 - 1024 - 240 - 32 + 16 \\ &= 0 \end{aligned}$$

##### Step 3: Conclusion

Since  $g(-4) = 0$ ,  $y + 4$  is a factor of  $g(y)$ .

In general, to factorise a cubic polynomial we need to do the following:

- Find one factor by trial and error: consider the coefficients of the given cubic polynomial  $p(x)$  and guess a possible root ( $\frac{c}{d}$ ).
- Use the factor theorem to confirm that  $\frac{c}{d}$  is a root; show that  $p(\frac{c}{d}) = 0$ .
- Divide  $p(x)$  by the factor  $(cx - d)$  to obtain a quadratic polynomial (remember to be careful with the signs).
- Apply the standard methods of factorisation to determine the two factors of the quadratic polynomial.

### Worked example 10: Factor theorem

#### QUESTION

Use the factor theorem to determine if  $y - 1$  is a factor of  $f(y) = 2y^4 + 3y^2 - 5y + 7$ .

#### SOLUTION

##### Step 1: Determine how to approach the problem

For  $y - 1$  to be a factor,  $f(1)$  must be equal to 0.

##### Step 2: Calculate $f(1)$

$$\begin{aligned}f(y) &= 2y^4 + 3y^2 - 5y + 7 \\ \therefore f(1) &= 2(1)^4 + 3(1)^2 - 5(1) + 7 \\ &= 2 + 3 - 5 + 7 \\ &= 7\end{aligned}$$

##### Step 3: Conclusion

Since  $f(1) \neq 0$ ,  $y - 1$  is not a factor of  $f(y) = 2y^4 + 3y^2 - 5y + 7$ .

### Worked example 11: Factorising cubic polynomials

#### QUESTION

Factorise completely:  $f(x) = x^3 + x^2 - 9x - 9$

#### SOLUTION

##### Step 1: Find a factor by trial and error

$$\text{Try } f(1) = (1)^3 + (1)^2 - 9(1) - 9 = 1 + 1 - 9 - 9 = -16$$

Therefore  $(x - 1)$  is not a factor.

We consider the coefficients of the given polynomial and try:

$$f(-1) = (-1)^3 + (-1)^2 - 9(-1) - 9 = -1 + 1 + 9 - 9 = 0$$

Therefore  $(x + 1)$  is a factor, because  $f(-1) = 0$ .

### Step 2: Factorise by inspection

Now divide  $f(x)$  by  $(x + 1)$  using inspection:

$$\text{Write } x^3 + x^2 - 9x - 9 = (x + 1)(\dots)$$

The first term in the second bracket must be  $x^2$  to give  $x^3$  and make the polynomial a cubic.

The last term in the second bracket must be  $-9$  because  $(+1)(-9) = -9$ .

$$\text{So we have } x^3 + x^2 - 9x - 9 = (x + 1)(x^2 + ?x - 9)$$

Now, we must find the coefficient of the middle term:

$(+1)(x^2)$  gives the  $x^2$  in the original polynomial. So, the coefficient of the  $x$ -term must be 0.

$$\therefore f(x) = (x + 1)(x^2 - 9).$$

### Step 3: Write the final answer

We can factorise the last bracket as a difference of two squares:

$$\begin{aligned} f(x) &= (x + 1)(x^2 - 9) \\ &= (x + 1)(x - 3)(x + 3) \end{aligned}$$

## Worked example 12: Factorising cubic polynomials

### QUESTION

Use the factor theorem to factorise  $f(x) = x^3 - 2x^2 - 5x + 6$ .

### SOLUTION

#### Step 1: Find a factor by trial and error

$$\text{Try } f(1) = (1)^3 - 2(1)^2 - 5(1) + 6 = 1 - 2 - 5 + 6 = 0$$

Therefore  $(x - 1)$  is a factor.

#### Step 2: Factorise by inspection

$$x^3 - 2x^2 - 5x + 6 = (x - 1)(\dots)$$

The first term in the second bracket must be  $x^2$  to give  $x^3$  if we work backwards.

The last term in the second bracket must be  $-6$  because  $(-1)(-6) = +6$ .

$$\text{So we have } x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 + ?x - 6)$$

Now, we must find the coefficient of the middle term:

$(-1)(x^2)$  gives  $-x^2$ . So, the coefficient of the  $x$ -term in the second bracket must be  $-1$  to give another  $-x^2$  so that overall we have  $-x^2 - x^2 = -2x^2$ .

So  $f(x) = (x - 1)(x^2 - x - 6)$ .

Make sure that the expression has been factorised correctly by checking that the coefficient of the  $x$ -term also works out:  $(x)(-6) + (-1)(-x) = -5x$ , which is correct.

### Step 3: Write the final answer

We can factorise the last bracket as:

$$\begin{aligned} f(x) &= (x - 1)(x^2 - x - 6) \\ &= (x - 1)(x - 3)(x + 2) \end{aligned}$$

## Exercise 5 – 5: Factorising cubic polynomials

1. Find the remainder when  $4x^3 - 4x^2 + x - 5$  is divided by  $x + 1$ .
2. Use the factor theorem to factorise  $x^3 - 3x^2 + 4$  completely.
3.  $f(x) = 2x^3 + x^2 - 5x + 2$ 
  - a) Find  $f(1)$ .
  - b) Factorise  $f(x)$  completely.
4. Use the factor theorem to determine all the factors of the following expression:  
 $x^3 + x^2 - 17x + 15$
5. Complete: If  $f(x)$  is a polynomial and  $p$  is a number such that  $f(p) = 0$ , then  $(x - p)$  is...
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

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Now that we know how to factorise cubic polynomials, it is also easy to solve cubic equations of the form  $ax^3 + bx^2 + cx + d = 0$ .

### Worked example 13: Solving cubic equations

#### QUESTION

Solve:  $6x^3 - 5x^2 - 17x + 6 = 0$

#### SOLUTION

##### Step 1: Find one factor using the factor theorem

Let  $f(x) = 6x^3 - 5x^2 - 17x + 6$

Try  $f(1) = 6(1)^3 - 5(1)^2 - 17(1) + 6 = 6 - 5 - 17 + 6 = -10$

Therefore  $(x - 1)$  is not a factor.

Try  $f(2) = 6(2)^3 - 5(2)^2 - 17(2) + 6 = 48 - 20 - 34 + 6 = 0$

Therefore  $(x - 2)$  is a factor.

##### Step 2: Factorise by inspection

$$6x^3 - 5x^2 - 17x + 6 = (x - 2)(6x^2 + 7x - 3)$$

##### Step 3: Factorise fully

$$6x^3 - 5x^2 - 17x + 6 = (x - 2)(2x + 3)(3x - 1)$$

##### Step 4: Solve the equation

$$\begin{aligned} 6x^3 - 5x^2 - 17x + 6 &= 0 \\ (x - 2)(2x + 3)(3x - 1) &= 0 \\ x = 2 \text{ or } x = -\frac{3}{2} \text{ or } x = \frac{1}{3} \end{aligned}$$

Sometimes it is not possible to factorise a quadratic expression using inspection, in which case we use the quadratic formula to fully factorise and solve the cubic equation.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

### Worked example 14: Solving cubic equations

#### QUESTION

Solve for  $x$ :  $0 = x^3 - 2x^2 - 6x + 4$

#### SOLUTION

##### Step 1: Use the factor theorem to determine a factor

Let  $f(x) = x^3 - 2x^2 - 6x + 4$

$$\text{Try } f(1) = (1)^3 - 2(1)^2 - 6(1) + 4 = 1 - 2 - 6 + 4 = -3$$

Therefore  $(x - 1)$  is not a factor.

$$\text{Try } f(2) = (2)^3 - 2(2)^2 - 6(2) + 4 = 8 - 8 - 12 + 4 = -8$$

Therefore  $(x - 2)$  is not a factor.

$$f(-2) = (-2)^3 - 2(-2)^2 - 6(-2) + 4 = -8 - 8 + 12 + 4 = 0$$

Therefore  $(x + 2)$  is a factor.

##### Step 2: Factorise by inspection

$$x^3 - 2x^2 - 6x + 4 = (x + 2)(x^2 - 4x + 2)$$

$x^2 - 4x + 2$  cannot be factorised any further and we are left with

$$(x + 2)(x^2 - 4x + 2) = 0$$

##### Step 3: Solve the equation

$$\begin{aligned}(x + 2)(x^2 - 4x + 2) &= 0 \\ (x + 2) &= 0 \text{ or } (x^2 - 4x + 2) = 0\end{aligned}$$

##### Step 4: Apply the quadratic formula for the second bracket

Always write down the formula first and then substitute the values of  $a$ ,  $b$  and  $c$ .

$$\begin{aligned}a &= 1; & b &= -4; & c &= 2 \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(2)}}{2(1)} \\ &= \frac{4 \pm \sqrt{8}}{2} \\ &= 2 \pm \sqrt{2}\end{aligned}$$

##### Step 5: Final solutions

$$x = -2 \text{ or } x = 2 \pm \sqrt{2}$$

## Exercise 5 – 6: Solving cubic equations

Solve the following cubic equations:

1.  $x^3 + x^2 - 16x = 16$

2.  $-n^3 - n^2 + 22n + 40 = 0$

3.  $y(y^2 + 2y) = 19y + 20$

4.  $k^3 + 9k^2 + 26k + 24 = 0$

5.  $x^3 + 2x^2 - 50 = 25x$

6.  $-p^3 + 19p = 30$

7.  $6x^2 - x^3 = 5x + 12$

Check answers online with the exercise code below or click on 'show me the answer'.

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7. [28TY](#)



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## 5.6 Summary

EMCGY

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| Terminology:          |                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expression            | A term or group of terms consisting of numbers, variables and the basic operators (+, −, ×, ÷).                                                                                  |
| Univariate expression | An expression containing only one variable.                                                                                                                                      |
| Root/Zero             | A root, also referred to as the "zero", of an equation is the value of $x$ such that $f(x) = 0$ is satisfied.                                                                    |
| Polynomial            | An expression that involves one or more variables having different powers and coefficients.<br>$a_n x^n + \dots + a_2 x^2 + a_1 x + a_0$ , where $n \in \mathbb{N}_0$            |
| Monomial              | A polynomial with one term.<br>For example, $7a^2b$ or $15xyz^2$ .                                                                                                               |
| Binomial              | A polynomial that has two terms.<br>For example, $2x + 5z$ or $26 - g^2k$ .                                                                                                      |
| Trinomial             | A polynomial that has three terms.<br>For example, $a - b + c$ or $4x^2 + 17xy - y^3$ .                                                                                          |
| Degree/Order          | The degree, also called the order, of a univariate polynomial is the value of the highest exponent in the polynomial.<br>For example, $7p - 12p^2 + 3p^5 + 8$ has a degree of 5. |

- Quadratic formula:  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Remainder theorem: a polynomial  $p(x)$  divided by  $cx - d$  gives a remainder of  $p\left(\frac{d}{c}\right)$ .
- Factor theorem: if the polynomial  $p(x)$  is divided by  $cx - d$  and the remainder,  $p\left(\frac{d}{c}\right)$ , is equal to zero, then  $cx - d$  is a factor of  $p(x)$ .
- Converse of the factor theorem: if  $cx - d$  is a factor of  $p(x)$ , then  $p\left(\frac{d}{c}\right) = 0$ .

- Synthetic division:

$$\begin{array}{r|rrrr} & q_2 & q_1 & q_0 & R \\ \frac{d}{c} & a_3 & a_2 & a_1 & a_0 \end{array}$$

We determine the coefficients of the quotient by calculating:

$$\begin{aligned} q_2 &= a_3 + \left( q_3 \times \frac{d}{c} \right) \\ &= a_3 \quad (\text{since } q_3 = 0) \\ q_1 &= a_2 + \left( q_2 \times \frac{d}{c} \right) \\ q_0 &= a_1 + \left( q_1 \times \frac{d}{c} \right) \\ R &= a_0 + \left( q_0 \times \frac{d}{c} \right) \end{aligned}$$

### Exercise 5 – 7: End of chapter exercises

- Solve for  $x$ :  $x^3 + x^2 - 5x + 3 = 0$
- Solve for  $y$ :  $y^3 = 3y^2 + 16y + 12$
- Solve for  $m$ :  $m(m^2 - m - 4) = -4$
- Solve for  $x$ :  $x^3 - x^2 = 3(3x + 2)$
- Solve for  $x$  if  $2x^3 - 3x^2 - 8x = 3$ .
- Solve for  $x$ :  $16(x + 1) = x^2(x + 1)$
- Show that  $x - 2$  is a factor of  $3x^3 - 11x^2 + 12x - 4$ .
  - Hence, by factorising completely, solve the equation:  
 $3x^3 - 11x^2 + 12x - 4 = 0$
- $2x^3 - x^2 - 2x + 2 = Q(x) \cdot (2x - 1) + R$  for all values of  $x$ .  
 What is the value of  $R$ ?
- Use the factor theorem to solve the following equation for  $m$ :  
 $8m^3 + 7m^2 - 17m + 2 = 0$
  - Hence, or otherwise, solve for  $x$ :  
 $2^{3x+3} + 7 \cdot 2^{2x} + 2 = 17 \cdot 2^x$
- Find the value of  $R$  if  $x - 1$  is a factor of  $h(x) = (x - 6) \cdot Q(x) + R$  and  $Q(x)$  divided by  $x - 1$  gives a remainder of 8.
- Determine the values of  $p$  for which the function

$$f(x) = 3p^3 - (3p - 7)x^2 + 5x - 3$$

leaves a remainder of 9 when it is divided by  $(x - p)$ .

- Calculate  $t$  and  $Q(x)$  if  $x^2 + tx + 3 = (x + 4) \cdot Q(x) - 17$ .
- More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 28V2
- 28V3
- 28V4
- 28V5
- 28V6
- 28V7
- 28V8
- 28V9
- 28VB
- 28VC
- 28VD
- 28VF



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## *Differential calculus*

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## 6.1 Limits

EMCGZ

Calculus is one of the central branches of mathematics and was developed from algebra and geometry. It is built on the concept of limits, which will be discussed in this chapter. Calculus consists of two related ideas: differential calculus and integral calculus. We will only be dealing with differential calculus in this chapter and will explore how it can be used to solve optimisation problems and finding rates of change.

## The tale of Achilles and the tortoise

EMCH2

Zeno (born about 490 BC) was a philosopher of southern Italy who is famous for his paradoxes (a “paradox” is a statement that seems contradictory and yet may be true).

One of Zeno’s paradoxes can be summarised as:

Achilles and a tortoise agree to a race, but the tortoise is unhappy because Achilles is very fast. So, the tortoise asks Achilles for a head start. Achilles agrees to give the tortoise a 1000 m head start. Does Achilles overtake the tortoise?

To solve this problem, we start by writing:

$$\text{Achilles: } x_A = v_A t$$

$$\text{Tortoise: } x_T = 1000 \text{ m} + v_T t$$

where

- $x_A$  is the distance covered by Achilles
- $v_A$  is Achilles’ speed
- $t$  is the time taken by Achilles to overtake the tortoise
- $x_T$  is the distance covered by the tortoise
- $v_T$  is the tortoise’s speed

Achilles will overtake the tortoise when both of them have covered the same distance. If we assume that Achilles runs at  $2 \text{ m.s}^{-1}$  and the tortoise runs at  $0,25 \text{ m.s}^{-1}$ , then this means that Achilles will overtake the tortoise at a time calculated as:

$$\begin{aligned} x_A &= x_T \\ v_A t &= 1000 + v_T t \\ 2t &= 1000 + 0,25t \\ 2 - 0,25t &= 1000 \\ \frac{7}{4}t &= 1000 \\ t &= \frac{4000}{7} \\ &= 571,43 \text{ s} \end{aligned}$$

However, Zeno looked at it as follows: Achilles takes  $t = \frac{1000 \text{ m}}{2 \text{ m.s}^{-1}} = 500 \text{ s}$  to travel the 1000 m head start that he gave the tortoise. However, in these 500 s, the tortoise has travelled a further  $x = 500 \text{ s} \times 0,25 \text{ m.s}^{-1} = 125 \text{ m}$ .

Achilles then takes another  $t = \frac{125 \text{ m}}{2 \text{ m.s}^{-1}} = 62,5 \text{ s}$  to travel the 125 m. In these 62,5 s, the tortoise travels a further  $x = 62,5 \text{ s} \times 0,25 \text{ m.s}^{-1} = 15,625 \text{ m}$ .

Zeno saw that Achilles would always get closer and closer but wouldn't actually overtake the tortoise.

So what does Zeno, Achilles and the tortoise have to do with calculus? Consider our earlier studies of sequences and series:

We know that the sequence  $0; \frac{1}{2}; \frac{2}{3}; \frac{3}{4}; \frac{4}{5}; \dots$  can be defined by the expression  $T_n = 1 - \frac{1}{n}$  and that the terms get closer to 1 as  $n$  gets larger.

Similarly, the sequence  $1; \frac{1}{2}; \frac{1}{3}; \frac{1}{4}; \frac{1}{5}; \dots$  can be defined by the expression  $T_n = \frac{1}{n}$  and the terms get closer to 0 as  $n$  gets larger.

We have also seen that an infinite geometric series can have a finite sum.

$$S_{\infty} = \sum_{i=1}^{\infty} a.r^{i-1} = \frac{a}{1-r} \quad \text{for } -1 < r < 1$$

where  $a$  is the first term of the series and  $r$  is the common ratio.

We see that there are some functions where the value of the function gets close to or **approaches** a certain value as the number of terms increases.

## Limits

EMCH3

Consider the function:  $y = \frac{x^2+4x-12}{x+6}$

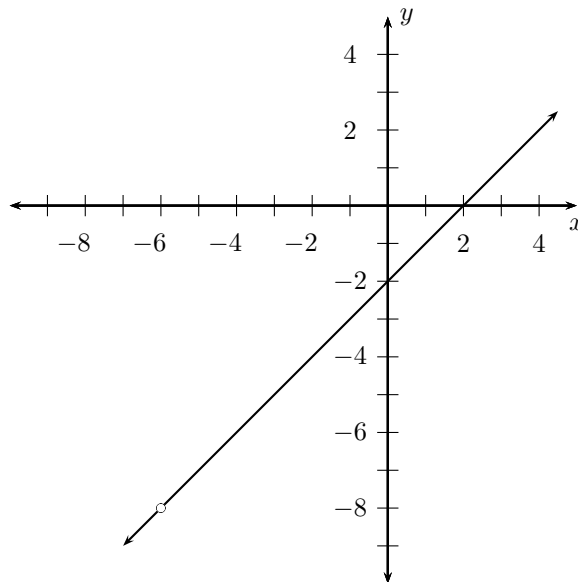
The numerator of the function can be factorised as:  $y = \frac{(x+6)(x-2)}{x+6}$ .

Then we can cancel the  $x + 6$  from numerator and denominator and we are left with:  $y = x - 2$ .

However, we are only able to cancel the  $x + 6$  term if  $x \neq -6$ . If  $x = -6$ , then the denominator becomes 0 and the function is not defined. This means that the domain of the function does not include  $x = -6$ . But we can examine what happens to the values for  $y$  as  $x$  gets closer to  $-6$ . The list of values shows that as  $x$  gets closer to  $-6$ ,  $y$  gets closer and closer to  $-8$ .

| $x$   | $y = \frac{(x+6)(x-2)}{x+6}$ |
|-------|------------------------------|
| -9    | -11                          |
| -8    | -10                          |
| -7    | -9                           |
| -6,5  | -8,5                         |
| -6,4  | -8,4                         |
| -6,3  | -8,3                         |
| -6,2  | -8,2                         |
| -6,1  | -8,1                         |
| -6,09 | -8,09                        |
| -6,08 | -8,08                        |
| -6,01 | -8,01                        |
| -5,9  | -7,9                         |
| -5,8  | -7,8                         |
| -5,7  | -7,7                         |
| -5,6  | -7,6                         |
| -5,5  | -7,5                         |
| -5    | -7                           |
| -4    | -6                           |
| -3    | -5                           |

The graph of this function is shown below. The graph is a straight line with slope 1 and  $y$ -intercept  $-2$ , but with a hole at  $x = -6$ . As  $x$  approaches  $-6$  from the left, the  $y$ -value approaches  $-8$  and as  $x$  approaches  $-6$  from the right, the  $y$ -value approaches  $-8$ . Since the function approaches the same  $y$ -value from the left and from the right, the limit exists.



## Notation

EMCH4

We can now introduce new notation. For the function  $y = \frac{(x+6)(x-2)}{x+6}$ , we can write:

$$\lim_{x \rightarrow -6} \frac{(x+6)(x-2)}{x+6} = -8.$$

This is read: the limit of  $\frac{(x+6)(x-2)}{x+6}$  as  $x$  tends to  $-6$  (from both the left and the right) is equal to  $-8$ .

## Investigation: Limits

If  $f(x) = x + 1$ , determine:

|            |  |
|------------|--|
| $f(-0,1)$  |  |
| $f(-0,05)$ |  |
| $f(-0,04)$ |  |
| $f(-0,03)$ |  |
| $f(-0,02)$ |  |
| $f(-0,01)$ |  |
| $f(0,00)$  |  |
| $f(0,01)$  |  |
| $f(0,02)$  |  |
| $f(0,03)$  |  |
| $f(0,04)$  |  |
| $f(0,05)$  |  |
| $f(0,1)$   |  |

What do you notice about the value of  $f(x)$  as  $x$  gets closer and closer to 0?

### Worked example 1: Limit notation

#### QUESTION

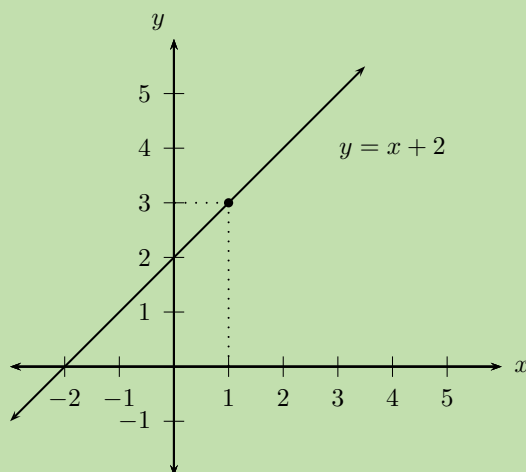
Write the following using limit notation: as  $x$  gets close to 1, the value of the function  $y = x + 2$  approaches 3.

#### SOLUTION

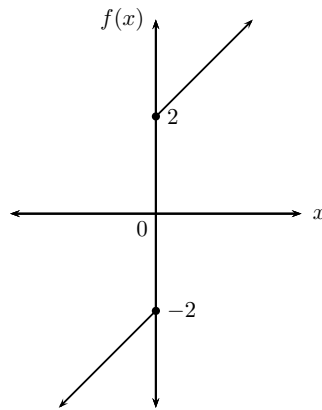
This is written as:

$$\lim_{x \rightarrow 1} (x + 2) = 3$$

This is illustrated in the diagram below:



We can also have the situation where a function tends to a different limit depending on whether  $x$  approaches from the left or the right.



As  $x \rightarrow 0$  from the left,  $f(x)$  approaches  $-2$ . As  $x \rightarrow 0$  from the right,  $f(x)$  approaches  $2$ .

The limit for  $x$  approaching 0 from the left is:

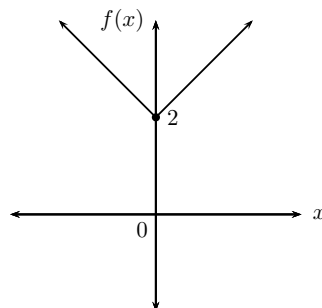
$$\lim_{x \rightarrow 0^-} f(x) = -2$$

and for  $x$  approaching 0 from the right:

$$\lim_{x \rightarrow 0^+} f(x) = 2$$

where  $0^-$  means  $x$  approaches zero from the left and  $0^+$  means  $x$  approaches zero from the right.

Therefore, since  $f(x)$  does not approach the same value from both sides, we can conclude that the limit as  $x$  tends to zero does not exist.



In the diagram above, as  $x$  tends to 0 from the left, the function approaches 2 and as  $x$  tends to 0 from the right, the function approaches 2. Since the function approaches the same value from both sides, the limit as  $x$  tends to 0 exists and is equal to 2.

📺 See video: [28VG](https://www.everythingmaths.co.za) at [www.everythingmaths.co.za](https://www.everythingmaths.co.za)

## Worked example 2: Limits

### QUESTION

Determine:

1.  $\lim_{x \rightarrow 1} 10$
2.  $\lim_{x \rightarrow 2} (x + 4)$

Illustrate answers graphically.

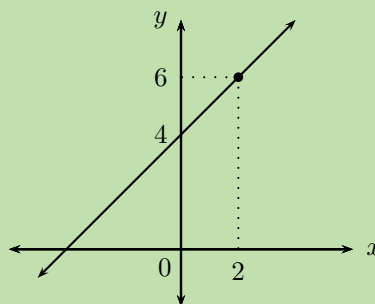
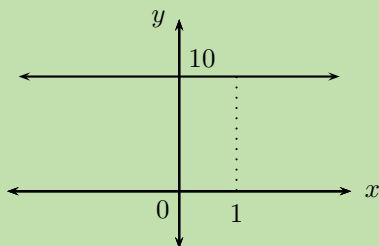
### SOLUTION

#### Step 1: Simplify the expression and cancel all common terms

We cannot simplify further and there are no terms to cancel.

#### Step 2: Calculate the limit

1.  $\lim_{x \rightarrow 1} 10 = 10$
2.  $\lim_{x \rightarrow 2} (x + 4) = 2 + 4 = 6$



## Worked example 3: Limits

### QUESTION

Determine the following and illustrate the answer graphically:

$$\lim_{x \rightarrow 10} \frac{x^2 - 100}{x - 10}$$

### SOLUTION

#### Step 1: Simplify the expression

Factorise the numerator:  $\frac{x^2 - 100}{x - 10} = \frac{(x + 10)(x - 10)}{x - 10}$

As  $x \rightarrow 10$ , the denominator  $(x - 10) \rightarrow 0$ , therefore the expression is not defined for  $x = 10$  since division by zero is not permitted.

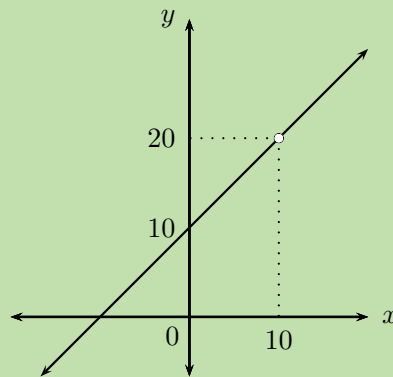
**Step 2: Cancel all common terms**

$$\frac{(x+10)(x-10)}{x-10} = x+10$$

**Step 3: Calculate the limit**

$$\begin{aligned}\lim_{x \rightarrow 10} \frac{x^2 - 100}{x - 10} &= \lim_{x \rightarrow 10} (x + 10) \\ &= 10 + 10 \\ &= 20\end{aligned}$$

**Step 4: Draw the graph**



▶ See video: 28YM at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

**Exercise 6 – 1: Limits**

1. Determine the following limits and draw a rough sketch to illustrate:

a)  $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x + 3}$

b)  $\lim_{x \rightarrow 3} \frac{x + 3}{x^2 + 3x}$

2. Determine the following limits (if they exist):

a)  $\lim_{x \rightarrow 2} \frac{3x^2 - 4x}{3 - x}$

b)  $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$

c)  $\lim_{x \rightarrow 2} \left( 3x + \frac{1}{3x} \right)$

d)  $\lim_{x \rightarrow 0} \frac{1}{x}$

e)  $\lim_{y \rightarrow 1} \frac{y - 1}{y + 1}$

f)  $\lim_{y \rightarrow 1} \frac{y + 1}{y - 1}$

g)  $\lim_{h \rightarrow 0} \frac{3h + h^2}{h}$

h)  $\lim_{h \rightarrow 1} \frac{h^3 - 1}{h - 1}$

i)  $\lim_{x \rightarrow 3} \frac{\sqrt{x} - \sqrt{3}}{x - 3}$

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 28VH   1b. 28VJ   2a. 28VK   2b. 28VM   2c. 28VN   2d. 28VP

2e. 28VQ   2f. 28VR   2g. 28VS   2h. 28VT   2i. 28VV



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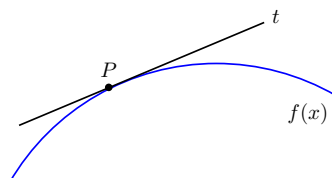
## Gradient at a point

EMCH5

### Average gradient

In Grade 11 we learnt that the average gradient between any two points on a curve is given by the gradient of the straight line that passes through both points. We also looked at the gradient at a single point on a curve and saw that it was the gradient of the tangent to the curve at the given point. In this section we learn how to determine the gradient of the tangent.

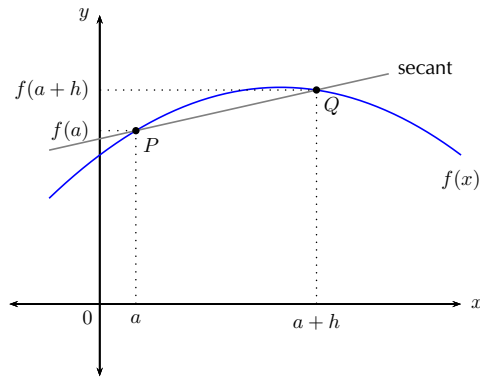
Let us consider finding the gradient of a tangent  $t$  to a curve with equation  $y = f(x)$  at a given point  $P$ .



We know how to calculate the average gradient between two points on a curve, but we need two points. The problem now is that we only have one point, namely  $P$ . To get around the problem we first consider a secant (a straight line that intersects a curve at two or more points) to the curve that passes through point  $P(x_P; y_P)$  and another point on the curve  $Q(x_Q; y_Q)$ , where  $Q$  is an arbitrary distance from  $P$ .

We can determine the average gradient of the curve between the two points:

$$m = \frac{y_Q - y_P}{x_Q - x_P}$$



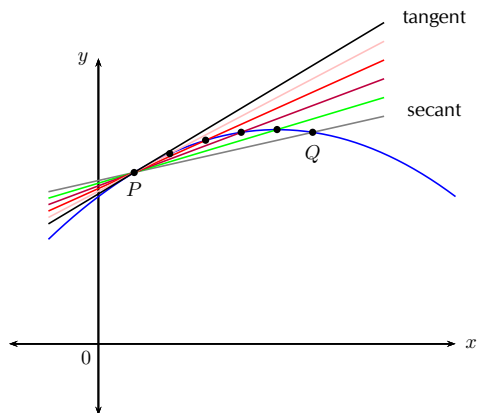
If we let the  $x$ -coordinate of  $P$  be  $a$ , then the  $y$ -coordinate is  $f(a)$ . Similarly, if the  $x$ -coordinate of  $Q$  is  $(a + h)$ , then the  $y$ -coordinate is  $f(a + h)$ .

We can now calculate the average gradient as:

$$\begin{aligned} \frac{y_Q - y_P}{x_Q - x_P} &= \frac{f(a + h) - f(a)}{(a + h) - a} \\ &= \frac{f(a + h) - f(a)}{h} \end{aligned}$$

### Gradient at a point

Imagine that  $Q$  moves along the curve, getting closer and closer to  $P$ . The secant line approaches the tangent line as its limiting position. This means that the average gradient of the secant approaches the gradient of the tangent to the curve at  $P$ .



We see that as point  $Q$  approaches point  $P$ ,  $h$  gets closer to 0. If point  $Q$  lies on point  $P$ , then  $h = 0$  and the formula for average gradient is undefined. We use our knowledge of limits to let  $h$  tend towards 0 to determine the gradient of the tangent to the curve at point  $P$ :

$$\text{Gradient at point } P = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

## Worked example 4: Gradient at a point

### QUESTION

Given  $g(x) = 3x^2$ , determine the gradient of the curve at the point  $x = -1$ .

### SOLUTION

**Step 1: Write down the formula for the gradient at a point**

$$\text{Gradient at a point} = \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h}$$

**Step 2: Determine  $g(a+h)$  and  $g(a)$**

We need to find the gradient of the curve at  $x = -1$ , therefore we let  $a = -1$ :

$$g(x) = 3x^2$$

$$\begin{aligned} g(a) &= g(-1) \\ &= 3(-1)^2 \\ &= 3 \end{aligned}$$

$$\begin{aligned} g(a+h) &= g(-1+h) \\ &= 3(-1+h)^2 \\ &= 3(1-2h+h^2) \\ &= 3-6h+3h^2 \end{aligned}$$

**Step 3: Substitute into the formula and simplify**

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} &= \lim_{h \rightarrow 0} \frac{g(-1+h) - g(-1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(3-6h+3h^2) - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{-6h+3h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(-6+3h)}{h} \\ &= \lim_{h \rightarrow 0} (-6+3h) \\ &= -6 \end{aligned}$$

Notice that we only take the limit once we have removed  $h$  from the denominator.

**Step 4: Write the final answer**

The gradient of the curve  $g(x) = 3x^2$  at  $x = -1$  is  $-6$ .

### Worked example 5: Gradient at a point

#### QUESTION

Given the function  $f(x) = 2x^2 - 5x$ , determine the gradient of the tangent to the curve at the point  $x = 2$ .

#### SOLUTION

**Step 1: Write down the formula for the gradient at a point**

$$\text{Gradient at a point} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

**Step 2: Determine  $f(a+h)$  and  $f(a)$**

We need to find the gradient of the tangent to the curve at  $x = 2$ , therefore we let  $a = 2$ :

$$f(x) = 2x^2 - 5x$$

$$\begin{aligned} f(a) &= f(2) \\ &= 2(2)^2 - 5(2) \\ &= 8 - 10 \\ &= -2 \end{aligned}$$

$$\begin{aligned} f(a+h) &= f(2+h) \\ &= 2(2+h)^2 - 5(2+h) \\ &= 2(2^2 + 4h + h^2) - 10 - 5h \\ &= 8 + 8h + 2h^2 - 10 - 5h \\ &= -2 + 3h + 2h^2 \end{aligned}$$

**Step 3: Substitute into the formula and simplify**

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(-2 + 3h + 2h^2) - (-2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2 + 3h + 2h^2 + 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{3h + 2h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(3 + 2h)}{h} \\ &= \lim_{h \rightarrow 0} (3 + 2h) \\ &= 3 \end{aligned}$$

**Step 4: Write the final answer**

The gradient of the tangent to the curve  $f(x) = 2x^2 - 5x$  at  $x = 2$  is 3.

## Worked example 6: Gradient at a point

### QUESTION

Determine the gradient of  $k(x) = -x^3 + 2x + 1$  at the point  $x = 1$ .

### SOLUTION

**Step 1: Write down the formula for the gradient at a point**

$$\text{Gradient at a point} = \lim_{h \rightarrow 0} \frac{k(a+h) - k(a)}{h}$$

**Step 2: Determine  $k(a+h)$  and  $k(a)$**

Let  $a = 1$ :

$$k(x) = -x^3 + 2x + 1$$

$$\begin{aligned} k(a) &= k(1) \\ &= -(1)^3 + 2(1) + 1 \\ &= -1 + 2 + 1 \\ &= 2 \end{aligned}$$

$$\begin{aligned} k(a+h) &= k(1+h) \\ &= -(1+h)^3 + 2(1+h) + 1 \\ &= -(1+3h+3h^2+h^3) + 2+2h+1 \\ &= -1-3h-3h^2-h^3+2+2h+1 \\ &= 2-h-3h^2-h^3 \end{aligned}$$

**Step 3: Substitute into the formula and simplify**

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{k(a+h) - k(a)}{h} &= \lim_{h \rightarrow 0} \frac{k(1+h) - k(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2-h-3h^2-h^3) - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h-3h^2-h^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(-1-3h-h^2)}{h} \\ &= \lim_{h \rightarrow 0} (-1-3h-h^2) \\ &= -1 \end{aligned}$$

**Step 4: Write the final answer**

The gradient of  $k(x) = -x^3 + 2x + 1$  at  $x = 1$  is  $-1$ .

## Exercise 6 – 2: Gradient at a point

- Given:  $f(x) = -x^2 + 7$ 
  - Find the average gradient of function  $f$ , between  $x = -1$  and  $x = 3$ .
  - Illustrate this with a graph.
  - Find the gradient of  $f$  at the point  $x = 3$  and illustrate this on your graph.
- Determine the gradient of the tangent to  $g$  if  $g(x) = \frac{3}{x}$  ( $x \neq 0$ ) at  $x = a$ .
- Determine the equation of the tangent to  $H(x) = x^2 + 3x$  at  $x = -1$ .
- More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28VX    2. 28VY    3. 28VZ



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## 6.2 Differentiation from first principles

EMCH6

We know that the gradient of the tangent to a curve with equation  $y = f(x)$  at  $x = a$  can be determined using the formula:

$$\text{Gradient at a point} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

We can use this formula to determine an expression that describes the gradient of the graph (or the gradient of the tangent to the graph) at any point on the graph. This expression (or gradient function) is called the derivative.

### DEFINITION: Derivative

The derivative of a function  $f(x)$  is written as  $f'(x)$  and is defined by:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

### DEFINITION: Differentiation

The process of determining the derivative of a given function.

This method is called differentiation from first principles or using the definition.

## Worked example 7: Differentiation from first principles

### QUESTION

Calculate the derivative of  $g(x) = 2x - 3$  from first principles.

### SOLUTION

**Step 1: Write down the formula for finding the derivative using first principles**

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

**Step 2: Determine  $g(x+h)$**

$$g(x) = 2x - 3$$

$$\begin{aligned} g(x+h) &= 2(x+h) - 3 \\ &= 2x + 2h - 3 \end{aligned}$$

**Step 3: Substitute into the formula and simplify**

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{2x + 2h - 3 - (2x - 3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2h}{h} \\ &= \lim_{h \rightarrow 0} 2 \\ &= 2 \end{aligned}$$

**Step 4: Write the final answer**

The derivative  $g'(x) = 2$ .

### Notation

There are a few different notations used to refer to derivatives. It is very important that you learn to identify these different ways of denoting the derivative and that you are consistent in your usage of them when answering questions.

If we use the common notation  $y = f(x)$ , where the dependent variable is  $y$  and the independent variable is  $x$ , then some alternative notations for the derivative are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}[f(x)] = Df(x) = D_x y$$

The symbols  $D$  and  $\frac{d}{dx}$  are called differential operators because they indicate the operation of differentiation.

$\frac{dy}{dx}$  means  $y$  differentiated with respect to  $x$ . Similarly,  $\frac{dp}{dx}$  means  $p$  differentiated with respect to  $x$ .

**Important:**  $\frac{dy}{dx}$  is not a fraction and does not mean  $dy \div dx$ .

▶ See video: 28W2 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 8: Differentiation from first principles

#### QUESTION

1. Find the derivative of  $f(x) = 4x^3$  from first principles.
2. Determine  $f'(0,5)$  and interpret the answer.

#### SOLUTION

**Step 1: Write down the formula for finding the derivative from first principles**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

**Step 2: Substitute into the formula and simplify**

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{4(x+h)^3 - 4x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{4(x^3 + 3x^2h + 3xh^2 + h^3) - 4x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{4x^3 + 12x^2h + 12xh^2 + 4h^3 - 4x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{12x^2h + 12xh^2 + 4h^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(12x^2 + 12xh + 4h^2)}{h} \\ &= \lim_{h \rightarrow 0} (12x^2 + 12xh + 4h^2) \\ &= 12x^2 \end{aligned}$$

**Step 3: Calculate  $f'(0,5)$  and interpret the answer**

$$\begin{aligned} f'(x) &= 12x^2 \\ \therefore f'(0,5) &= 12(0,5)^2 \\ &= 12\left(\frac{1}{4}\right) \\ &= 3 \end{aligned}$$

- The derivative of  $f(x)$  at  $x = 0,5$  is 3.
- The gradient of the function  $f$  at  $x = 0,5$  is equal to 3.
- The gradient of the tangent to  $f(x)$  at  $x = 0,5$  is equal to 3.

## Worked example 9: Differentiation from first principles

### QUESTION

Calculate  $\frac{dp}{dx}$  from first principles if  $p(x) = -\frac{2}{x}$ .

### SOLUTION

**Step 1: Write down the formula for finding the derivative using first principles**

$$\frac{dp}{dx} = \lim_{h \rightarrow 0} \frac{p(x+h) - p(x)}{h}$$

**Step 2: Substitute into the formula and simplify**

$$\frac{dp}{dx} = \lim_{h \rightarrow 0} \frac{-\frac{2}{x+h} - \left(-\frac{2}{x}\right)}{h}$$

It is sometimes easier to write the right-hand side of the equation as:

$$\begin{aligned}\frac{dp}{dx} &= \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{-2}{x+h} + \frac{2}{x} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{-2x + 2(x+h)}{x(x+h)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{-2x + 2x + 2h}{x(x+h)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{2h}{x^2 + xh} \right) \\ &= \lim_{h \rightarrow 0} \frac{2}{x^2 + xh} \\ &= \frac{2}{x^2}\end{aligned}$$

Notice: even though  $h$  remains in the denominator, we can take the limit since it does not result in division by 0.

**Step 3: Write the final answer**

$$\frac{dp}{dx} = \frac{2}{x^2}$$

### Worked example 10: Differentiation from first principles

#### QUESTION

Differentiate  $g(x) = \frac{1}{4}$  from first principles and interpret the answer.

#### SOLUTION

**Step 1: Write down the formula for finding the derivative from first principles**

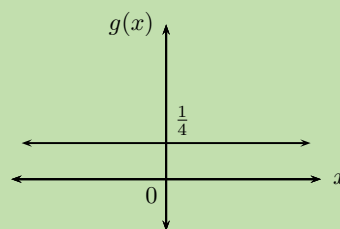
$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

**Step 2: Substitute into the formula and simplify**

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{\frac{1}{4} - \frac{1}{4}}{h} \\ &= \lim_{h \rightarrow 0} \frac{0}{h} \\ &= \lim_{h \rightarrow 0} 0 \\ &= 0 \end{aligned}$$

**Step 3: Interpret the answer**

The gradient of  $g(x)$  is equal to 0 at any point on the graph. The derivative of this constant function is equal to 0.



### Exercise 6 – 3: Differentiation from first principles

1. Given:  $g(x) = -x^2$

a) Determine  $\frac{g(x+h) - g(x)}{h}$ .

b) Hence, determine  $\lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$ .

c) Explain the meaning of your answer in (b).

2. Find the derivative of  $f(x) = -2x^2 + 3x + 1$  using first principles.

3. Determine the derivative of  $f(x) = \frac{1}{x-2}$  using first principles.

4. Determine  $g'(3)$  from first principles if  $g(x) = -5x^2$ .

5. If  $p(x) = 4x(x-1)$ , determine  $p'(x)$  using first principles.

6. Find the derivative of  $k(x) = 10x^3$  using first principles.

7. Differentiate  $f(x) = x^n$  using first principles.

(Hint: Use Pascal's triangle)

8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 28W3    1b. 28W4    1c. 28W5    2. 28W6    3. 28W7    4. 28W8

5. 28W9    6. 28WB    7. 28WC



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Determining the derivative of a function from first principles requires a long calculation and it is easy to make mistakes. However, we can use this method of finding the derivative from first principles to obtain rules which make finding the derivative of a function much simpler.

### Investigation: Rules for differentiation

1. Differentiate the following from first principles:

a)  $f(x) = x$

b)  $f(x) = -4x$

c)  $f(x) = x^2$

d)  $f(x) = 3x^2$

e)  $f(x) = -x^3$

f)  $f(x) = 2x^3$

g)  $f(x) = \frac{1}{x}$

h)  $f(x) = -\frac{2}{x}$

2. Complete the table:

| $f(x)$         | $f'(x)$ |
|----------------|---------|
| $x$            |         |
| $-4x$          |         |
| $x^2$          |         |
| $3x^2$         |         |
| $-x^3$         |         |
| $2x^3$         |         |
| $\frac{1}{x}$  |         |
| $-\frac{2}{x}$ |         |

3. Can you identify a pattern for determining the derivative?

### Rules for differentiation

- General rule for differentiation:

$$\frac{d}{dx} [x^n] = nx^{n-1}, \text{ where } n \in \mathbb{R} \text{ and } n \neq 0.$$

- The derivative of a constant is equal to zero.

$$\frac{d}{dx} [k] = 0$$

- The derivative of a constant multiplied by a function is equal to the constant multiplied by the derivative of the function.

$$\frac{d}{dx} [k \cdot f(x)] = k \frac{d}{dx} [f(x)]$$

- The derivative of a sum is equal to the sum of the derivatives.

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} [f(x)] + \frac{d}{dx} [g(x)]$$

- The derivative of a difference is equal to the difference of the derivatives.

$$\frac{d}{dx} [f(x) - g(x)] = \frac{d}{dx} [f(x)] - \frac{d}{dx} [g(x)]$$

### Worked example 11: Rules of differentiation

#### QUESTION

Use the rules of differentiation to find the derivative of each of the following:

1.  $y = 3x^5$
2.  $p = \frac{1}{4}q^2$
3.  $f(x) = 60$
4.  $y = 12x^3 + 7x$
5.  $m = \frac{3}{2}n^4 - 1$

#### SOLUTION

**Step 1: Apply the appropriate rules to determine the derivative**

1.  $\frac{dy}{dx} = 3(5x^4) = 15x^4$
2.  $\frac{dp}{dq} = \frac{1}{4}(2q) = \frac{1}{2}q$
3.  $f'(x) = 0$
4.  $\frac{dy}{dx} = 12(3x^2) + 7 = 36x^2 + 7$
5.  $\frac{dm}{dn} = \frac{3}{2}(4n^3) - 0 = 6n^3$

### Worked example 12: Rules of differentiation

#### QUESTION

Differentiate the following with respect to  $t$ :

1.  $g(t) = 4(t+1)^2(t-3)$
2.  $k(t) = \frac{(t+2)^3}{\sqrt{t}}$

#### SOLUTION

**Step 1: Expand the expression and apply the rules of differentiation**

We have not learnt a rule for differentiating a product, therefore we must expand the brackets and simplify before we can determine the derivative:

$$\begin{aligned}g(t) &= 4(t+1)^2(t-3) \\&= 4(t^2 + 2t + 1)(t-3) \\&= 4(t^3 + 2t^2 + t - 3t^2 - 6t - 3) \\&= 4(t^3 - t^2 - 5t - 3) \\&= 4t^3 - 4t^2 - 20t - 12\end{aligned}$$

$$\begin{aligned}\therefore g'(t) &= 4(3t^2) - 4(2t) - 20 - 0 \\&= 12t^2 - 8t - 20\end{aligned}$$

## Step 2: Expand the expression and apply the rules of differentiation

We have not learnt a rule for differentiating a quotient, therefore we must first simplify the expression and then we can differentiate:

$$\begin{aligned}k(t) &= \frac{(t+2)^3}{\sqrt{t}} \\&= \frac{(t+2)(t^2+4t+4)}{\sqrt{t}} \\&= \frac{t^3+6t^2+12t+8}{t^{\frac{1}{2}}} \\&= t^{-\frac{1}{2}}(t^3+6t^2+12t+8) \\&= t^{\frac{5}{2}}+6t^{\frac{3}{2}}+12t^{\frac{1}{2}}+8t^{-\frac{1}{2}} \\\\ \therefore g'(t) &= \frac{5}{2}t^{\frac{3}{2}}+6\left(\frac{3}{2}t^{\frac{1}{2}}\right)+12\left(\frac{1}{2}t^{-\frac{1}{2}}\right)+8\left(-\frac{1}{2}t^{-\frac{3}{2}}\right) \\&= \frac{5}{2}t^{\frac{3}{2}}+9t^{\frac{1}{2}}+6t^{-\frac{1}{2}}-4t^{-\frac{3}{2}}\end{aligned}$$

**Important:** always write the final answer with positive exponents.

$$g'(t) = \frac{5}{2}t^{\frac{3}{2}} + 9t^{\frac{1}{2}} + \frac{6}{t^{\frac{1}{2}}} - \frac{4}{t^{\frac{3}{2}}}$$

### When to use the rules for differentiation:

- If the question does not specify how we must determine the derivative, then we use the rules for differentiation.

### When to differentiate using first principles:

- If the question specifically states to use first principles.
- If we are required to differentiate using the definition of a derivative, then we use first principles.

## Exercise 6 – 4: Rules for differentiation

1. Differentiate the following:

a)  $y = 3x^2$

b)  $f(x) = 25x$

c)  $k(x) = -30$

d)  $y = -4x^5 + 2$

e)  $g(x) = 16x^{-2}$

f)  $y = 10(7 - 3)$

g)  $q(x) = x^4 - 6x^2 - 1$

h)  $y = x^2 + x + 4$

i)  $f(x) = \frac{1}{3}x^3 - x^2 + \frac{2}{5}$

j)  $y = 3x^{\frac{3}{2}} - 4x + 20$

k)  $g(x) = x(x + 2) + 5x$

l)  $p(x) = 200[x^3 - \frac{1}{2}x^2 + \frac{1}{5}x - 40]$

m)  $y = 14(x - 1) \left[\frac{1}{2} + x^2\right]$

2. Find  $f'(x)$  if  $f(x) = \frac{x^2 - 5x + 6}{x - 2}$ .
3. Find  $f'(y)$  if  $f(y) = \sqrt{y}$ .
4. Find  $f'(z)$  if  $f(z) = (z - 1)(z + 1)$ .
5. Determine  $\frac{dy}{dx}$  if  $y = \frac{x^3 + 2\sqrt{x} - 3}{x}$ .
6. Determine the derivative of  $y = \sqrt{x^3} + \frac{1}{3x^3}$ .
7. Find  $D_x \left[ x^{\frac{3}{2}} - \frac{3}{x^{\frac{1}{2}}} \right]^2$ .
8. Find  $\frac{dy}{dx}$  if  $x = 2y + 3$ .
9. Determine  $f'(\theta)$  if  $f(\theta) = 2(\theta^{\frac{3}{2}} - 3\theta^{-\frac{1}{2}})^2$ .
10. Find  $\frac{dp}{dt}$  if  $p(t) = \frac{(t+1)^3}{\sqrt{t}}$ .
11. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1a. 28WF | 1b. 28WG | 1c. 28WH | 1d. 28WJ | 1e. 28WK | 1f. 28WM |
| 1g. 28WN | 1h. 28WP | 1i. 28WQ | 1j. 28WR | 1k. 28WS | 1l. 28WT |
| 1m. 28WV | 2. 28WW  | 3. 28WX  | 4. 28WY  | 5. 28WZ  | 6. 28X2  |
| 7. 28X3  | 8. 28X4  | 9. 28X5  | 10. 28X6 |          |          |



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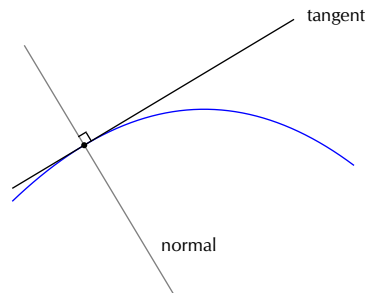


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## 6.4 Equation of a tangent to a curve

EMCH8

At a given point on a curve, the gradient of the curve is equal to the gradient of the tangent to the curve.



The derivative (or gradient function) describes the gradient of a curve at any point on the curve. Similarly, it also describes the gradient of a tangent to a curve at any point on the curve.

**To determine the equation of a tangent to a curve:**

1. Find the derivative using the rules of differentiation.
2. Substitute the  $x$ -coordinate of the given point into the derivative to calculate the gradient of the tangent.
3. Substitute the gradient of the tangent and the coordinates of the given point into an appropriate form of the straight line equation.
4. Make  $y$  the subject of the formula.

The normal to a curve is the line perpendicular to the tangent to the curve at a given point.

$$m_{\text{tangent}} \times m_{\text{normal}} = -1$$

### Worked example 13: Finding the equation of a tangent to a curve

#### QUESTION

Find the equation of the tangent to the curve  $y = 3x^2$  at the point  $(1; 3)$ . Sketch the curve and the tangent.

#### SOLUTION

##### Step 1: Find the derivative

Use the rules of differentiation:

$$y = 3x^2$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= 3(2x) \\ &= 6x\end{aligned}$$

##### Step 2: Calculate the gradient of the tangent

To determine the gradient of the tangent at the point  $(1; 3)$ , we substitute the  $x$ -value into the equation for the derivative.

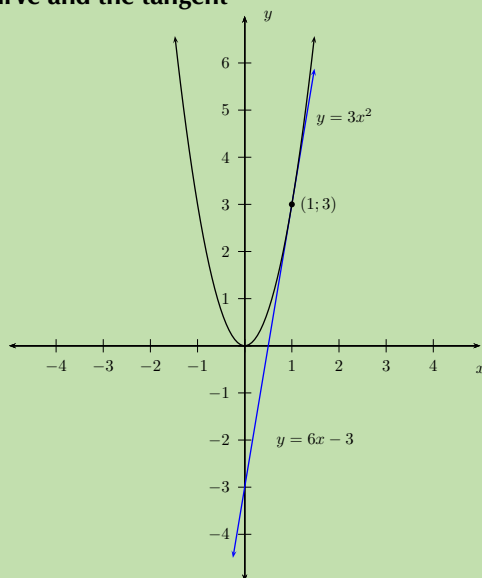
$$\begin{aligned}\frac{dy}{dx} &= 6x \\ \therefore m &= 6(1) \\ &= 6\end{aligned}$$

##### Step 3: Determine the equation of the tangent

Substitute the gradient of the tangent and the coordinates of the given point into the gradient-point form of the straight line equation.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ y - 3 &= 6(x - 1) \\ y &= 6x - 6 + 3 \\ y &= 6x - 3\end{aligned}$$

##### Step 4: Sketch the curve and the tangent



### Worked example 14: Finding the equation of a tangent to a curve

#### QUESTION

Given  $g(x) = (x + 2)(2x + 1)^2$ , determine the equation of the tangent to the curve at  $x = -1$ .

#### SOLUTION

**Step 1: Determine the  $y$ -coordinate of the point**

$$\begin{aligned}g(x) &= (x + 2)(2x + 1)^2 \\g(-1) &= (-1 + 2)[2(-1) + 1]^2 \\&= (1)(-1)^2 \\&= 1\end{aligned}$$

Therefore the tangent to the curve passes through the point  $(-1; 1)$ .

**Step 2: Expand and simplify the given function**

$$\begin{aligned}g(x) &= (x + 2)(2x + 1)^2 \\&= (x + 2)(4x^2 + 4x + 1) \\&= 4x^3 + 4x^2 + x + 8x^2 + 8x + 2 \\&= 4x^3 + 12x^2 + 9x + 2\end{aligned}$$

**Step 3: Find the derivative**

$$\begin{aligned}g'(x) &= 4(3x^2) + 12(2x) + 9 + 0 \\&= 12x^2 + 24x + 9\end{aligned}$$

**Step 4: Calculate the gradient of the tangent**

Substitute  $x = -1$  into the equation for  $g'(x)$ :

$$\begin{aligned}g'(-1) &= 12(-1)^2 + 24(-1) + 9 \\ \therefore m &= 12 - 24 + 9 \\&= -3\end{aligned}$$

**Step 5: Determine the equation of the tangent**

Substitute the gradient of the tangent and the coordinates of the point into the gradient-point form of the straight line equation.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 1 &= -3(x - (-1)) \\y &= -3x - 3 + 1 \\y &= -3x - 2\end{aligned}$$

### Worked example 15: Finding the equation of a normal to a curve

#### QUESTION

1. Determine the equation of the normal to the curve  $xy = -4$  at  $(-1; 4)$ .
2. Draw a rough sketch.

#### SOLUTION

##### Step 1: Find the derivative

Make  $y$  the subject of the formula and differentiate with respect to  $x$ :

$$\begin{aligned}y &= -\frac{4}{x} \\ &= -4x^{-1}\end{aligned}$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= -4(-1x^{-2}) \\ &= 4x^{-2} \\ &= \frac{4}{x^2}\end{aligned}$$

##### Step 2: Calculate the gradient of the normal at $(-1; 4)$

First determine the gradient of the tangent at the given point:

$$\begin{aligned}\frac{dy}{dx} &= \frac{4}{(-1)^2} \\ \therefore m &= 4\end{aligned}$$

Use the gradient of the tangent to calculate the gradient of the normal:

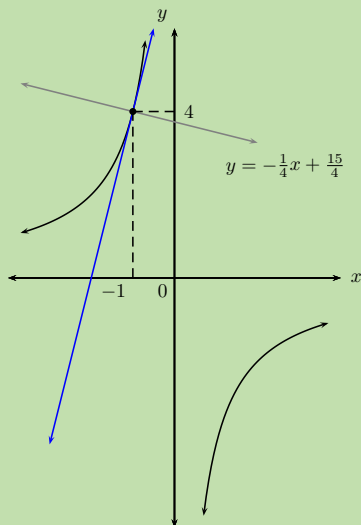
$$\begin{aligned}m_{\text{tangent}} \times m_{\text{normal}} &= -1 \\ 4 \times m_{\text{normal}} &= -1 \\ \therefore m_{\text{normal}} &= -\frac{1}{4}\end{aligned}$$

##### Step 3: Find the equation of the normal

Substitute the gradient of the normal and the coordinates of the given point into the gradient-point form of the straight line equation.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ y - 4 &= -\frac{1}{4}(x - (-1)) \\ y &= -\frac{1}{4}x - \frac{1}{4} + 4 \\ y &= -\frac{1}{4}x + \frac{15}{4}\end{aligned}$$

#### Step 4: Draw a rough sketch



#### Exercise 6 – 5: Equation of a tangent to a curve

1. Determine the equation of the tangent to the curve defined by  $F(x) = x^3 + 2x^2 - 7x + 1$  at  $x = 2$ .
2. Determine the point where the gradient of the tangent to the curve:
  - a)  $f(x) = 1 - 3x^2$  is equal to 5.
  - b)  $g(x) = \frac{1}{3}x^2 + 2x + 1$  is equal to 0.
3. Determine the point(s) on the curve  $f(x) = (2x - 1)^2$  where the tangent is:
  - a) parallel to the line  $y = 4x - 2$ .
  - b) perpendicular to the line  $2y + x - 4 = 0$ .
4. Given the function  $f: y = -x^2 + 4x - 3$ .
  - a) Draw a graph of  $f$ , indicating all intercepts and turning points.
  - b) Find the equations of the tangents to  $f$  at:
    - i. the  $y$ -intercept of  $f$ .
    - ii. the turning point of  $f$ .
    - iii. the point where  $x = 4,25$ .
  - c) Draw the three tangents above on your graph of  $f$ .
  - d) Write down all observations about the three tangents to  $f$ .
5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28X7    2a. 28X8    2b. 28X9    3. 28XB    4. 28XC



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The second derivative of a function is the derivative of the first derivative and it indicates the change in gradient of the original function. The sign of the second derivative tells us if the gradient of the original function is increasing, decreasing or remaining constant.

To determine the second derivative of the function  $f(x)$ , we differentiate  $f'(x)$  using the rules for differentiation.

$$f''(x) = \frac{d}{dx}[f'(x)]$$

We also use the following notation for determining the second derivative of  $y$ :

$$y'' = \frac{d}{dx} \left[ \frac{dy}{dx} \right] = \frac{d^2y}{dx^2}$$

### Worked example 16: Finding the second derivative

#### QUESTION

Calculate the second derivative for each of the following:

1.  $k(x) = 2x^3 - 4x^2 + 9$

2.  $y = \frac{3}{x}$

#### SOLUTION

1.

$$\begin{aligned} k'(x) &= 2(3x^2) - 4(2x) + 0 \\ &= 6x^2 - 8x \end{aligned}$$

$$\begin{aligned} k''(x) &= 6(2x) - 8 \\ &= 12x - 8 \end{aligned}$$

2.

$$y = 3x^{-1}$$

$$\begin{aligned} \frac{dy}{dx} &= 3(-1x^{-2}) \\ &= -3x^{-2} \\ &= -\frac{3}{x^2} \end{aligned}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= -3(-2x^{-3}) \\ &= \frac{6}{x^3} \end{aligned}$$

## Exercise 6 – 6: Second derivative

1. Calculate the second derivative for each of the following:

a)  $g(x) = 5x^2$

b)  $y = 8x^3 - 7x$

c)  $f(x) = x(x - 6) + 10$

d)  $y = x^5 - x^3 + x - 1$

e)  $k(x) = (x^2 + 1)(x - 1)$

f)  $p(x) = -\frac{10}{x^2}$

g)  $q(x) = \sqrt{x} + 5x^2$

2. Find the first and second derivatives of  $f(x) = 5x(2x + 3)$ .

3. Find  $\frac{d^2}{dx^2} [6\sqrt[3]{x^2}]$ .

4. Given the function  $g: y = (1 - 2x)^3$ .

a) Determine  $g'$  and  $g''$ .

b) What type of function is:

i.  $g'$

ii.  $g''$

c) Find the value of  $g''(\frac{1}{2})$ .

d) What do you observe about the degree (highest power) of each of the derived functions?

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 28XD    1b. 28XF    1c. 28XG    1d. 28XH    1e. 28XJ    1f. 28XK

1g. 28XM    2. 28XN    3. 28XP    4. 28XQ



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## 6.6 Sketching graphs

EMCHB

Functions of the form  $y = ax^3 + bx^2 + cx + d$

EMCHC

### Investigation: The effects of $a$ on a cubic function

Complete the table below and plot the graphs of  $f(x)$  and  $g(x)$  on the same system of axes.

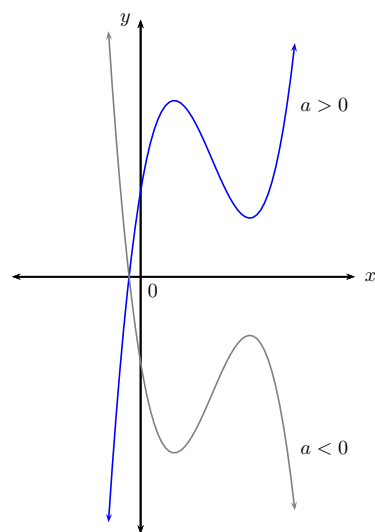
Be careful to choose a suitable scale for the  $y$ -axis.

$$f(x) = 2x^3 - 5x^2 - 14x + 8$$

$$g(x) = -2x^3 + 5x^2 + 14x - 8$$

| $x$    | -3 | -2 | -1 | 0 | 1 | 2 | 3   | 4 | 5 |
|--------|----|----|----|---|---|---|-----|---|---|
| $f(x)$ |    |    |    |   |   |   | -25 |   |   |
| $g(x)$ |    |    |    |   |   |   | 25  |   |   |

The effects of  $a$  on a cubic graph



Intercepts

EMCHD

Investigation: Number of intercepts

Complete the table:

|                        | $y = x + 1$ | $y = x^2 - x - 6$ | $y = x^3 + x^2 - 26x + 24$ |
|------------------------|-------------|-------------------|----------------------------|
| Degree of function     |             |                   |                            |
| Type of function       |             |                   |                            |
| Factorised form        |             |                   |                            |
| No. of $x$ -intercepts |             |                   |                            |
| No. of $y$ -intercepts |             |                   |                            |

Worked example 17: Determining the intercepts

QUESTION

Given  $f(x) = -x^3 + 4x^2 + x - 4$ , find the  $x$ - and  $y$ -intercepts.

SOLUTION

Step 1: Determine the  $y$ -intercept

The  $y$ -intercept is obtained by letting  $x = 0$ :

$$\begin{aligned} y &= -(0)^3 + 4(0)^2 + (0) - 4 \\ &= -4 \end{aligned}$$

This gives the point  $(0; -4)$ .

**Step 2: Use the factor theorem to factorise the expression**

We use the factor theorem to find a factor of  $f(x)$  by trial and error:

$$\begin{aligned}f(x) &= -x^3 + 4x^2 + x - 4 \\f(1) &= -(1)^3 + 4(1)^2 + (1) - 4 \\&= 0 \\ \therefore (x - 1) &\text{ is a factor of } f(x)\end{aligned}$$

Factorise further by inspection:

$$\begin{aligned}f(x) &= (x - 1)(-x^2 + 3x + 4) \\&= -(x - 1)(x^2 - 3x - 4) \\&= -(x - 1)(x + 1)(x - 4)\end{aligned}$$

The  $x$ -intercepts are obtained by letting  $f(x) = 0$ :

$$\begin{aligned}0 &= -(x - 1)(x + 1)(x - 4) \\ \therefore x &= -1, x = 1 \text{ or } x = 4\end{aligned}$$

This gives the points  $(-1; 0)$ ,  $(1; 0)$  and  $(4; 0)$ .

**Exercise 6 – 7: Intercepts**

- Given the function  $f(x) = x^3 + x^2 - 10x + 8$ .
  - Determine the  $x$ - and  $y$ -intercepts of  $f(x)$ .
  - Draw a rough sketch of the graph.
  - Is the function increasing or decreasing at  $x = -5$ ?
- Determine the  $x$ - and  $y$ -intercepts for each of the following:
  - $y = -x^3 - 5x^2 + 9x + 45$
  - $y = x^3 - \frac{5}{4}x^2 - \frac{7}{4}x + \frac{1}{2}$
  - $y = x^3 - x^2 - 12x + 12$
  - $y = x^3 - 16x$
  - $y = x^3 - 5x^2 + 6$
- Determine all intercepts for  $g(x) = x^3 + 3x^2 - 10x$  and draw a rough sketch of the graph.
- More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28XR   2a. 28XS   2b. 28XT   2c. 28XV   2d. 28XW   2e. 28XX  
3. 28XY



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**Investigation:**

1. Complete the table below for the quadratic function  $f(x)$ :

$$f(x) = x^2 + 2x + 1$$

$$f'(x) = \dots\dots\dots$$

| $x$ -value                         | -3 | -1 | 0 | 1 | 3 |
|------------------------------------|----|----|---|---|---|
| Gradient of $f$                    |    |    |   |   |   |
| Sign of gradient                   |    |    |   |   |   |
| Increasing function ( $\nearrow$ ) |    |    |   |   |   |
| Decreasing function ( $\searrow$ ) |    |    |   |   |   |
| Maximum TP ( $\cap$ )              |    |    |   |   |   |
| Minimum TP ( $\cup$ )              |    |    |   |   |   |

2. Use the table to draw a rough sketch of the graph of  $f(x)$ .

3. Solve for  $x$  if  $f'(x) = 0$ .

4. Indicate solutions to  $f'(x) = 0$  on the graph.

5. Complete the table below for the cubic function  $g(x)$ :

$$g(x) = 2x^3 + 3x^2 - 12x$$

$$g'(x) = \dots\dots\dots$$

| $x$ -value                         | -3 | -2 | 0 | 1 | 3 |
|------------------------------------|----|----|---|---|---|
| Gradient of $g$                    |    |    |   |   |   |
| Sign of gradient                   |    |    |   |   |   |
| Increasing function ( $\nearrow$ ) |    |    |   |   |   |
| Decreasing function ( $\searrow$ ) |    |    |   |   |   |
| Maximum TP ( $\cap$ )              |    |    |   |   |   |
| Minimum TP ( $\cup$ )              |    |    |   |   |   |

6. Use the table to draw a rough sketch of the graph of  $g(x)$ .

7. Solve for  $x$  if  $g'(x) = 0$ .

8. Indicate solutions to  $g'(x) = 0$  on the graph.

9. Complete the following sentence:

The derivative describes the ..... of a tangent to a curve at a given point and we have seen that the ..... of a curve at its stationary point(s) is equal to ..... Therefore, we can use ..... as a tool for finding the stationary points of the graphs of quadratic and cubic functions.

To determine the coordinates of the stationary point(s) of  $f(x)$ :

- Determine the derivative  $f'(x)$ .
- Let  $f'(x) = 0$  and solve for the  $x$ -coordinate(s) of the stationary point(s).
- Substitute value(s) of  $x$  into  $f(x)$  to calculate the  $y$ -coordinate(s) of the stationary point(s).

### Worked example 18: Finding stationary points

#### QUESTION

Calculate the stationary points of the graph of  $p(x) = x^3 - 6x^2 + 9x - 4$ .

#### SOLUTION

**Step 1: Determine the derivative of  $p(x)$**

Using the rules of differentiation we get:

$$p'(x) = 3x^2 - 12x + 9$$

**Step 2: Let  $p'(x) = 0$  and solve for  $x$**

$$3x^2 - 12x + 9 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

$$\therefore x = 1 \text{ or } x = 3$$

Therefore, the  $x$ -coordinates of the turning points are  $x = 1$  and  $x = 3$ .

**Step 3: Substitute the  $x$ -values into  $p(x)$**

We use the  $x$ -coordinates to calculate the corresponding  $y$ -coordinates of the stationary points.

$$\begin{aligned} p(1) &= (1)^3 - 6(1)^2 + 9(1) - 4 \\ &= 1 - 6 + 9 - 4 \\ &= 0 \end{aligned}$$

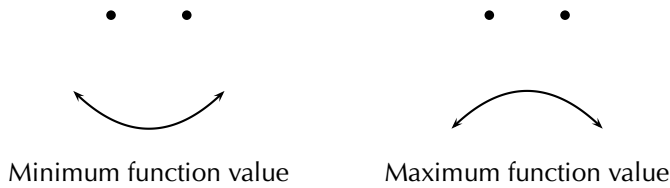
$$\begin{aligned} p(3) &= (3)^3 - 6(3)^2 + 9(3) - 4 \\ &= 27 - 54 + 27 - 4 \\ &= -4 \end{aligned}$$

**Step 4: Write final answer**

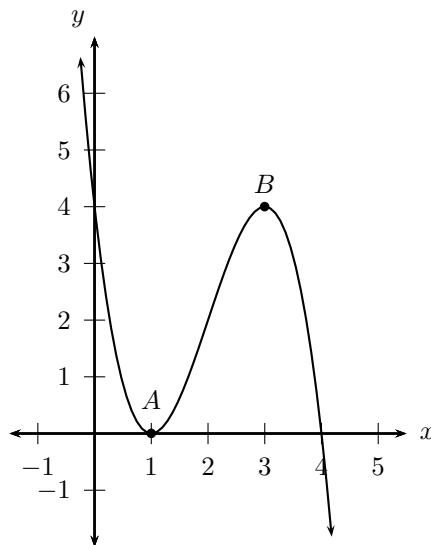
The turning points of the graph of  $p(x) = x^3 - 6x^2 + 9x - 4$  are  $(1; 0)$  and  $(3; -4)$ .

## Local maximum and local minimum

We have seen that the graph of a quadratic function can have either a minimum turning point ("smile") or a maximum turning point ("frown").



For cubic functions, we refer to the turning (or stationary) points of the graph as local minimum or local maximum turning points. The diagram below shows local minimum turning point  $A(1; 0)$  and local maximum turning point  $B(3; 4)$ . These points are described as a local (or relative) minimum and a local maximum because there are other points on the graph with lower and higher function values.



### Exercise 6 – 8: Stationary points

1. Use differentiation to determine the stationary point(s) for  $g(x) = -x^2 + 5x - 6$ .
2. Determine the  $x$ -values of the stationary points for  $f(x) = -\frac{1}{3}x^3 + \frac{1}{2}x^2 + 6x + 5$ .
3. Find the coordinates of the stationary points of the following functions using the rules of differentiation:
  - a)  $y = (x - 1)^3$
  - b)  $y = x^3 - 5x^2 + 1$
  - c)  $y + 7x = 1$

4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28XZ   2. 28Y2   3a. 28Y3   3b. 28Y4   3c. 28Y5



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**General method for sketching cubic graphs:**

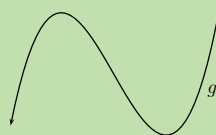
1. Consider the sign of  $a$  and determine the general shape of the graph.
2. Determine the  $y$ -intercept by letting  $x = 0$ .
3. Determine the  $x$ -intercepts by factorising  $ax^3 + bx^2 + cx + d = 0$  and solving for  $x$ .
4. Find the  $x$ -coordinates of the turning points of the function by letting  $f'(x) = 0$  and solving for  $x$ .
5. Determine the  $y$ -coordinates of the turning points by substituting the  $x$ -values into  $f(x)$ .
6. Plot the points and draw a smooth curve.

**Worked example 19: Sketching cubic graphs****QUESTION**

Sketch the graph of  $g(x) = x^3 - 3x^2 - 4x$ .

**SOLUTION****Step 1: Determine the shape of the graph**

The coefficient of the  $x^3$  term is greater than zero, therefore the graph will have the following shape:

**Step 2: Determine the intercepts**

The  $y$ -intercept is obtained by letting  $x = 0$ :

$$\begin{aligned} g(0) &= (0)^3 - 3(0)^2 - 4(0) \\ &= 0 \end{aligned}$$

This gives the point  $(0; 0)$ .

The  $x$ -intercept is obtained by letting  $g(x) = 0$  and solving for  $x$ :

$$\begin{aligned} 0 &= x^3 - 3x^2 - 4x \\ &= x(x^2 - 3x - 4) \\ &= x(x - 4)(x + 1) \\ \therefore x &= -1, x = 0 \text{ or } x = 4 \end{aligned}$$

This gives the points  $(-1; 0)$ ,  $(0; 0)$  and  $(4; 0)$ .

### Step 3: Calculate the stationary points

Find the  $x$ -coordinates of the stationary points by setting  $g'(x) = 0$ :

$$g'(x) = 3x^2 - 6x - 4$$

$$0 = 3x^2 - 6x - 4$$

$$\text{Using the quadratic formula } x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(3)(-4)}}{2(3)}$$

$$= \frac{6 \pm \sqrt{36 + 48}}{6}$$

$$\therefore x = 2,53 \text{ or } x = -0,53$$

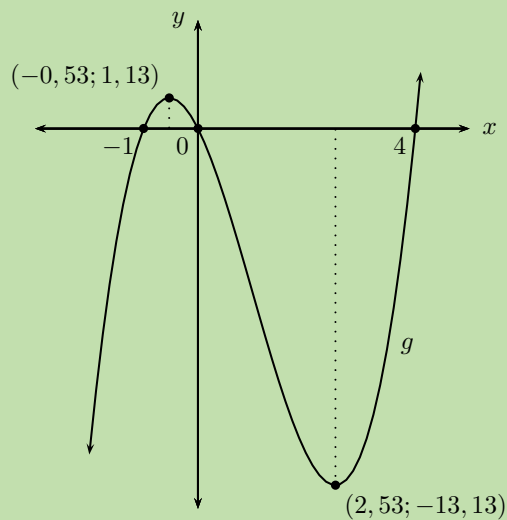
Substitute these  $x$ -coordinates into  $g(x)$  to determine the corresponding  $y$ -coordinates:

$$\begin{aligned} g(x) &= (2,53)^3 - 3(2,53)^2 - 4(2,53) \\ &= -13,13 \end{aligned}$$

$$\begin{aligned} g(x) &= (-0,53)^3 - 3(-0,53)^2 - 4(-0,53) \\ &= 1,13 \end{aligned}$$

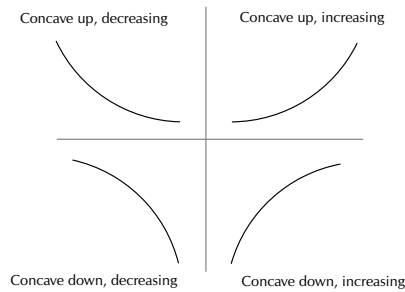
Therefore, the stationary points are  $(2,53; -13,13)$  and  $(-0,53; 1,13)$ .

### Step 4: Draw a neat sketch



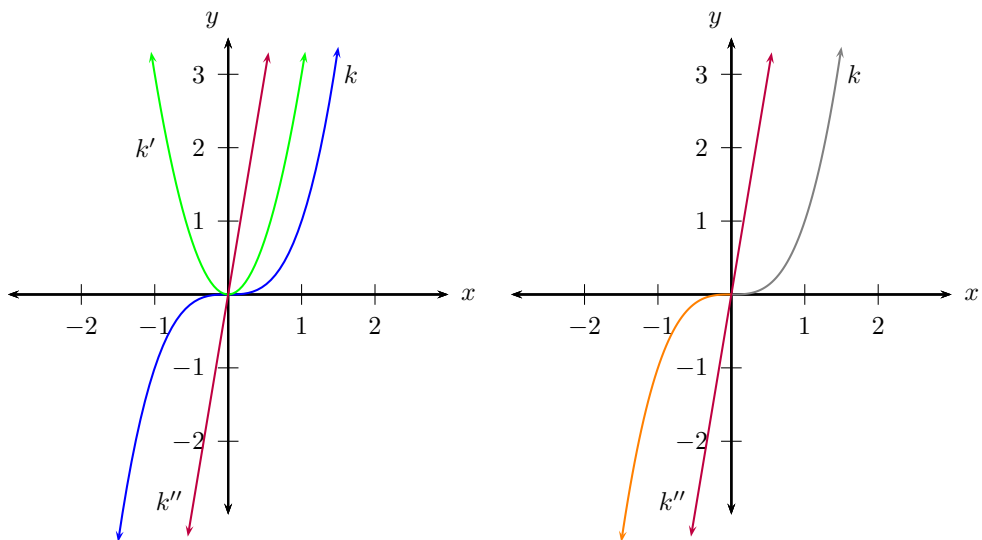
## Concavity

Concavity indicates whether the gradient of a curve is increasing, decreasing or stationary.



- Concave up: the gradient of the curve increases as  $x$  increases.
- Concave down: the gradient of the curve decreases as  $x$  increases.
- Zero concavity: the gradient of the curve is constant.

The diagram below shows the graph of the cubic function  $k(x) = x^3$ . The first derivative of  $k(x)$  is a quadratic function,  $k'(x) = 3x^2$  and the second derivative is a linear function,  $k''(x) = 6x$ .



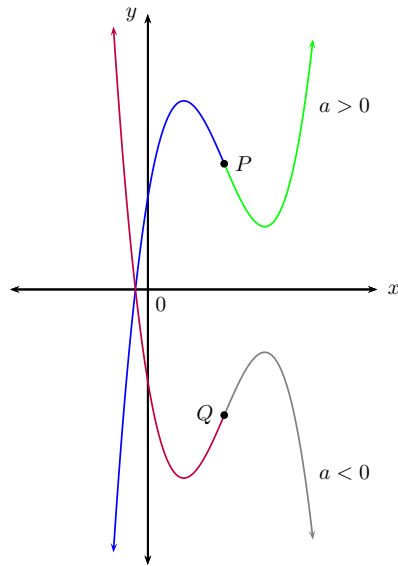
Notice the following:

- $k''(x) > 0$ , the graph is concave up.
- $k''(x) < 0$ , the graph is concave down.
- $k''(x) = 0$ , change in concavity (point of inflection).

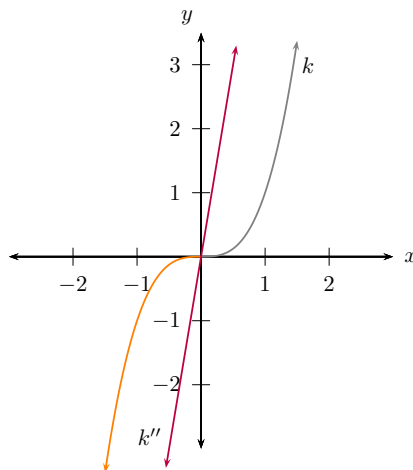
## Points of inflection

This is the point where the concavity of a curve changes, as shown in the diagram below. If  $a < 0$ , then the concavity changes from concave up (purple) to concave down (grey) and if  $a > 0$ , concavity changes from concave down (blue) to concave up (green). Unlike a turning point, the gradient of the curve on the left-hand side of

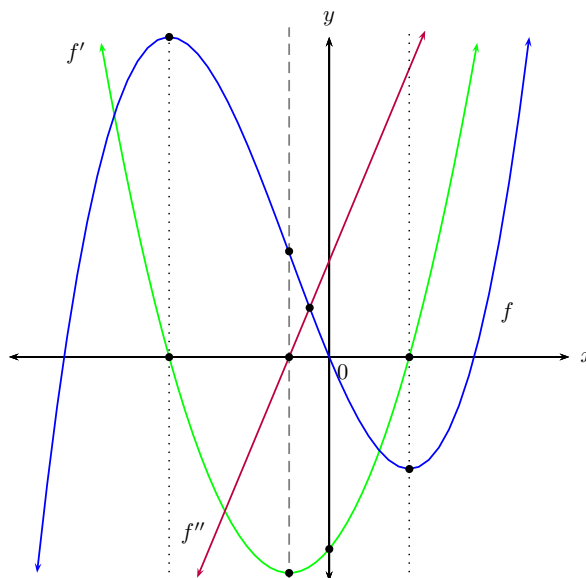
an inflection point ( $P$  and  $Q$ ) has the same sign as the gradient of the curve on the right-hand side.



A graph has a horizontal point of inflection where the derivative is zero but the sign of the gradient of the curve does not change. That means the graph (shown below) will continue to increase or decrease after the stationary point.



In the example above, the equation  $k'(x) = 3x^2$  indicates that the gradient of this curve will always be positive (except where  $x = 0$ ). Therefore, the stationary point is a point of inflection.



|                                      |                                            |                                        |
|--------------------------------------|--------------------------------------------|----------------------------------------|
| $f$ : cubic function<br>(blue graph) | $f'$ : quadratic function<br>(green graph) | $f''$ : linear function<br>(red graph) |
| turning points $\rightarrow$         | $x$ -intercepts                            |                                        |
| point of inflection                  | $\leftarrow$ turning point $\rightarrow$   | $x$ -intercept                         |

### Exercise 6 – 9: Concavity and points of inflection

Complete the following for each function:

- Determine and discuss the change in gradient of the function.
- Determine the concavity of the graph.
- Find the inflection point.
- Draw a sketch of the graph.

1.  $f : y = -2x^3$

2.  $g(x) = \frac{1}{8}x^3 + 1$

3.  $h : x \rightarrow (x - 2)^3$

4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28Y6    2. 28Y7    3. 28Y8



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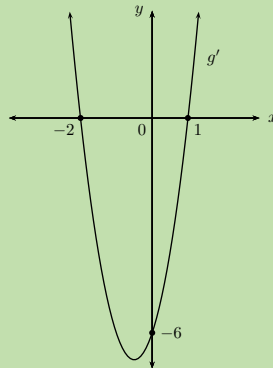


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## Worked example 20: Interpreting graphs

## QUESTION

Consider the graph of the derivative of  $g(x)$ .



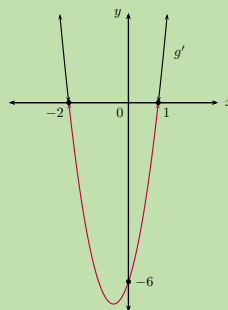
1. For which values of  $x$  is  $g(x)$  decreasing?
2. Determine the  $x$ -coordinate(s) of the turning point(s) of  $g(x)$ .
3. Given that  $g(x) = ax^3 + bx^2 + cx$ , calculate  $a$ ,  $b$  and  $c$ .

## SOLUTION

**Step 1: Examine the parabolic graph and interpret the given information**

We know that  $g'(x)$  describes the gradient of  $g(x)$ . To determine where the cubic function is decreasing, we must find the values of  $x$  for which  $g'(x) < 0$ :

$$\{x : -2 < x < 1; x \in \mathbb{R}\} \text{ or we can write } x \in (-2; 1)$$

**Step 2: Determine the  $x$ -coordinate(s) of the turning point(s)**

To determine the turning points of a cubic function, we let  $g'(x) = 0$  and solve for the  $x$ -values. These  $x$ -values are the  $x$ -intercepts of the parabola and are indicated on the given graph:

$$x = -2 \text{ or } x = 1$$

**Step 3: Determine the equation of  $g(x)$**

$$g(x) = ax^3 + bx^2 + cx$$

$$g'(x) = 3ax^2 + 2bx + c$$

From the graph, we see that the  $y$ -intercept of  $g'(x)$  is  $-6$ .

$$\therefore c = -6$$

$$g'(x) = 3ax^2 + 2bx - 6$$

$$\text{Substitute } x = -2 : g'(-2) = 3a(-2)^2 + 2b(-2) - 6$$

$$0 = 12a - 4b - 6 \dots\dots (1)$$

$$\text{Substitute } x = 1 : g'(1) = 3a(1)^2 + 2b(1) - 6$$

$$0 = 3a + 2b - 6 \dots\dots (2)$$

$$\text{Eqn. (1)} - 4 \text{ Eqn. (2)} : 0 = 0 - 12b + 18$$

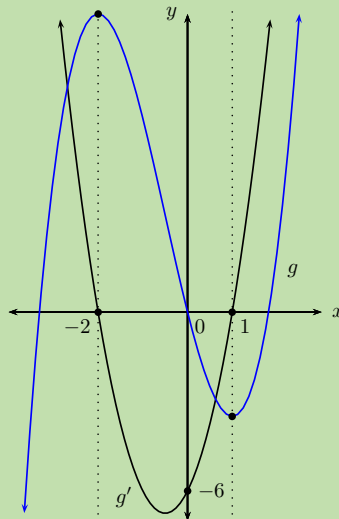
$$\therefore b = \frac{3}{2}$$

$$\text{And } 0 = 3a + 2\left(\frac{3}{2}\right) - 6$$

$$0 = 3a - 3$$

$$\therefore a = 1$$

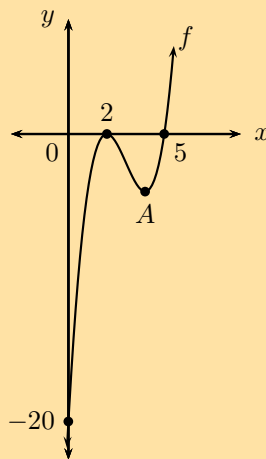
$$g(x) = x^3 + \frac{3}{2}x^2 - 6x$$



▶ See video: 28VV at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

## Exercise 6 – 10: Mixed exercises on cubic graphs

1. Given  $f(x) = x^3 + x^2 - 5x + 3$ .
  - a) Show that  $(x - 1)$  is a factor of  $f(x)$  and hence factorise  $f(x)$ .
  - b) Determine the coordinates of the intercepts and the turning points.
  - c) Sketch the graph.
2.
  - a) Sketch the graph of  $f(x) = -x^3 + 4x^2 + 11x - 30$ . Show all the turning points and intercepts with the axes.
  - b) Given  $g(x) = x^3 - 4x^2 - 11x + 30$ , sketch the graph of  $g$  without any further calculations. Describe the method for drawing the graph.
3. The sketch shows the graph of a cubic function,  $f$ , with a turning point at  $(2; 0)$ , going through  $(5; 0)$  and  $(0; -20)$ .



- a) Find the equation of  $f$ .
  - b) Find the coordinates of turning point  $A$ .
4.
  - a) Find the intercepts and stationary point(s) of  $f(x) = -\frac{1}{3}x^3 + 2$  and draw a sketch of the graph.
  - b) For which values of  $x$  will:
    - i.  $f(x) < 0$
    - ii.  $f'(x) < 0$
    - iii.  $f''(x) < 0$Motivate each answer.
5. Use the information below to sketch a graph of each cubic function (do not find the equations of the functions).

a)

$$g(-6) = g(-1,5) = g(2) = 0$$

$$g'(-4) = g'(1) = 0$$

$$g'(x) > 0 \text{ for } x < -4 \text{ or } x > 1$$

$$g'(x) < 0 \text{ for } -4 < x < 1$$

b)

$$h(-3) = 0$$

$$h(0) = 4$$

$$h(-1) = 3$$

$$h'(-1) = 0$$

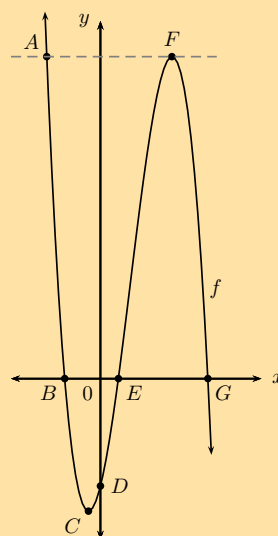
$$h''(-1) = 0$$

$$h'(x) > 0 \text{ for all } x \text{ values except } x = -1$$

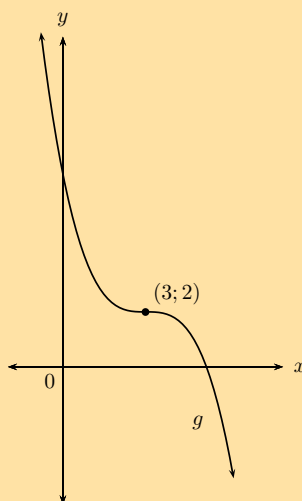
6. The sketch below shows the curve of  $f(x) = -(x+2)(x-1)(x-6)$  with turning points at  $C$  and  $F$ .  $AF$  is parallel to the  $x$ -axis.

Determine the following:

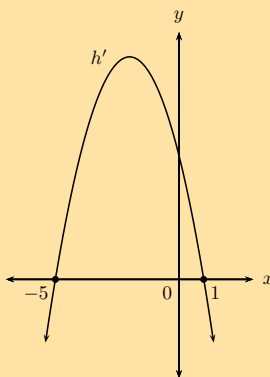
- length  $OB$
- length  $OE$
- length  $EG$
- length  $OD$
- coordinates of  $C$  and  $F$
- length  $AF$
- average gradient between  $E$  and  $F$
- the equation of the tangent to the graph at  $E$



7. Given the graph of a cubic function with the stationary point  $(3; 2)$ , sketch the graph of the derivative function if it is also given that the gradient of the graph is  $-5$  at  $x = 0$ .



8. The sketch below shows the graph of  $h'(x)$  with  $x$ -intercepts at  $-5$  and  $1$ . Draw a sketch graph of  $h(x)$  if  $h(-5) = 2$  and  $h(1) = 6$ .



9. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28Y9   2. 28YB   3. 28YC   4. 28YD   5a. 28YF   5b. 28YG  
6. 28YH   7. 28YJ   8. 28YK



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## 6.7 Applications of differential calculus

EMCHH

### Optimisation problems

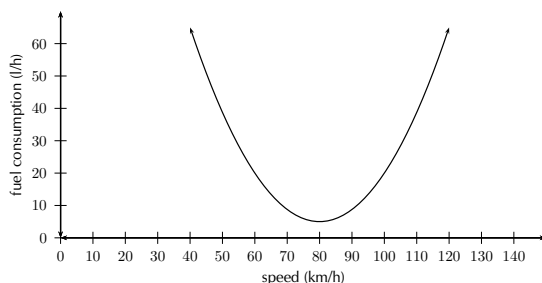
EMCHJ

We have seen that differential calculus can be used to determine the stationary points of functions, in order to sketch their graphs. Calculating stationary points also lends itself to the solving of problems that require some variable to be maximised or minimised. These are referred to as optimisation problems.

The fuel used by a car is defined by  $f(v) = \frac{3}{80}v^2 - 6v + 245$ , where  $v$  is the travelling speed in km/h.

What is the most economical speed of the car? In other words, determine the speed of the car which uses the least amount of fuel.

If we draw the graph of this function we find that the graph has a minimum. The speed at the minimum would then give the most economical speed.



We have seen that the coordinates of the turning point can be calculated by differentiating the function and finding the  $x$ -coordinate (speed in the case of the example) for which the derivative is 0.

$$f'(v) = \frac{3}{40}v - 6$$

If we set  $f'(v) = 0$  we can calculate the speed that corresponds to the turning point:

$$\begin{aligned} f'(v) &= \frac{3}{40}v - 6 \\ 0 &= \frac{3}{40}v - 6 \\ v &= \frac{6 \times 40}{3} \\ &= 80 \end{aligned}$$

This means that the most economical speed is 80 km/h.

#### Finding the optimum point:

Let  $f'(x) = 0$  and solve for  $x$  to find the optimum point.

To check whether the optimum point at  $x = a$  is a local minimum or a local maximum, we find  $f''(x)$ :

- If  $f''(a) < 0$ , then the point is a local maximum.
- If  $f''(a) > 0$ , then the point is a local minimum.

### Worked example 21: Optimisation problems

#### QUESTION

The sum of two positive numbers is 10. One of the numbers is multiplied by the square of the other. If each number is greater than 0, find the numbers that make this product a maximum.

Draw a graph to illustrate the answer.

#### SOLUTION

##### Step 1: Analyse the problem and formulate the equations that are required

Let the two numbers be  $a$  and  $b$  and the product be  $P$ .

$$a + b = 10 \dots\dots (1)$$

$$P = a \times b^2 \dots\dots (2)$$

Make  $b$  the subject of equation (1) and substitute into equation (2):

$$\begin{aligned} P &= a(10 - a)^2 \\ &= a(100 - 20a + a^2) \\ \therefore P(a) &= 100a - 20a^2 + a^3 \end{aligned}$$

##### Step 2: Differentiate with respect to $a$

$$P'(a) = 100 - 40a + 3a^2$$

**Step 3: Determine the stationary points by letting  $P'(a) = 0$** 

We find the value of  $a$  which makes  $P$  a maximum:

$$\begin{aligned}P'(a) &= 3a^2 - 40a + 100 \\0 &= (3a - 10)(a - 10) \\ \therefore a &= 10 \text{ or } a = \frac{10}{3}\end{aligned}$$

Substitute into the equation (1) to solve for  $b$ :

$$\begin{aligned}\text{If } a &= 10 : b = 10 - 10 \\ &= 0 \text{ (but } b > 0) \\ \therefore &\text{ no solution}\end{aligned}$$

$$\begin{aligned}\text{If } a &= \frac{10}{3} : b = 10 - \frac{10}{3} \\ &= \frac{20}{3}\end{aligned}$$

**Step 4: Determine the second derivative  $P''(a)$** 

We check that the point  $(\frac{10}{3}; \frac{20}{3})$  is a local maximum by showing that  $P''(\frac{10}{3}) < 0$ :

$$\begin{aligned}P''(a) &= 6a - 40 \\ \therefore P''\left(\frac{10}{3}\right) &= 6\left(\frac{10}{3}\right) - 40 \\ &= 20 - 40 \\ &= -20\end{aligned}$$

**Step 5: Write the final answer**

The product is maximised when the two numbers are  $\frac{10}{3}$  and  $\frac{20}{3}$ .

**Step 6: Draw a graph**

To draw a rough sketch of the graph we need to calculate where the graph intersects with the axes and the maximum and minimum function values of the turning points:

Intercepts:

$$\begin{aligned}P(a) &= a^3 - 20a^2 + 100a \\ &= a(a - 10)^2 \\ \text{Let } P(a) &= 0 : (0; 0) \text{ and } (10; 0)\end{aligned}$$

Turning points:

$$\begin{aligned}P'(a) &= 0 \\ \therefore a &= \frac{10}{3} \text{ or } a = 10\end{aligned}$$

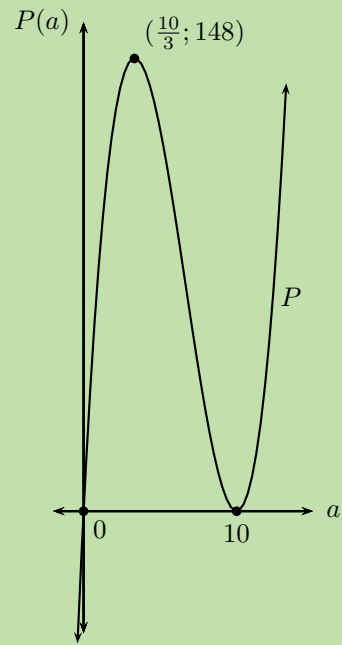
Maximum and minimum function values:

$$\begin{aligned}\text{Substitute } \left(\frac{10}{3}; \frac{20}{3}\right) : P &= ab^2 \\ &= \left(\frac{10}{3}\right) \left(\frac{20}{3}\right)^2 \\ &= \frac{4000}{27} \\ &\approx 148\end{aligned}$$

(A maximum turning point)

$$\begin{aligned}\text{Substitute } (0; 10) : P &= ab^2 \\ &= (10)(0)^2 \\ &= 0\end{aligned}$$

(A minimum turning point)

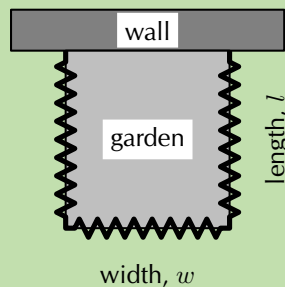


Note: the above diagram is not drawn to scale.

### Worked example 22: Optimisation problems

#### QUESTION

Michael wants to start a vegetable garden, which he decides to fence off in the shape of a rectangle from the rest of the garden. Michael has only 160 m of fencing, so he decides to use a wall as one border of the vegetable garden. Calculate the width and length of the garden that corresponds to the largest possible area that Michael can fence off.



#### SOLUTION

##### Step 1: Examine the problem and formulate the equations that are required

The important pieces of information given are related to the area and modified perimeter of the garden. We know that the area of the garden is given by the formula:

$$\text{Area} = w \times l$$

The fencing is only required for 3 sides and the three sides must add up to 160 m.

$$160 = w + l + l$$

Rearrange the formula to make  $w$  the subject of the formula:

$$w = 160 - 2l$$

Substitute the expression for  $w$  into the formula for the area of the garden. Notice that this formula now contains only one unknown variable.

$$\begin{aligned}\text{Area} &= l(160 - 2l) \\ &= 160l - 2l^2\end{aligned}$$

### Step 2: Differentiate with respect to $l$

We are interested in maximising the area of the garden, so we differentiate to get the following:

$$\frac{dA}{dl} = A' = 160 - 4l$$

### Step 3: Calculate the stationary point

To find the stationary point, we set  $A'(l) = 0$  and solve for the value(s) of  $l$  that maximises the area:

$$\begin{aligned}A'(l) &= 160 - 4l \\ 0 &= 160 - 4l \\ 4l &= 160 \\ \therefore l &= 40\end{aligned}$$

Therefore, the length of the garden is 40 m.

Substitute to solve for the width:

$$\begin{aligned}w &= 160 - 2l \\ &= 160 - 2(40) \\ &= 160 - 80 \\ &= 80\end{aligned}$$

Therefore, the width of the garden is 80 m.

### Step 4: Determine the second derivative $A''(l)$

We can check that this gives a maximum area by showing that  $A''(l) < 0$ :

$$A''(l) = -4$$

### Step 5: Write the final answer

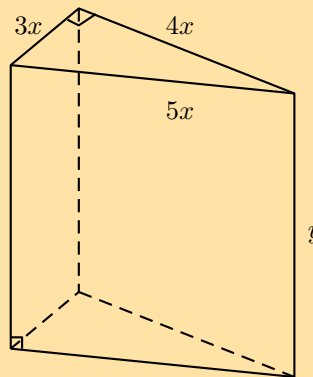
A width of 80 m and a length of 40 m will give the maximum area for the garden.

**Important note:**

The quantity that is to be minimised or maximised must be expressed in terms of only one variable. To find the optimised solution we need to determine the derivative and we only know how to differentiate with respect to one variable (more complex rules for differentiation are studied at university level).

**Exercise 6 – 11: Solving optimisation problems**

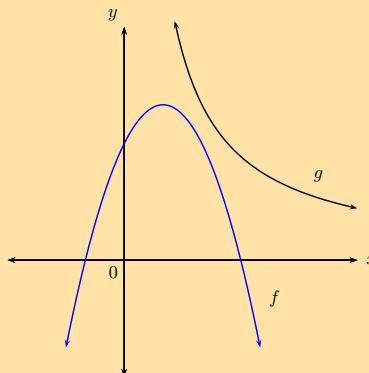
1. The sum of two positive numbers is 20. One of the numbers is multiplied by the square of the other. Find the numbers that make this product a maximum.
2. A wooden block is made as shown in the diagram. The ends are right-angled triangles having sides  $3x$ ,  $4x$  and  $5x$ . The length of the block is  $y$ . The total surface area of the block is  $3600 \text{ cm}^2$ .



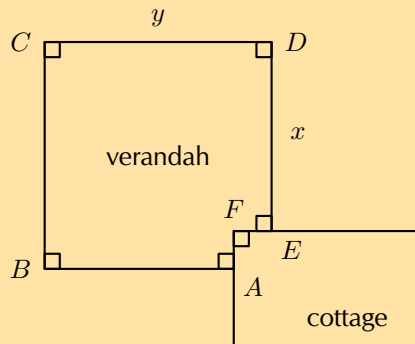
- a) Show that  $y = \frac{300 - x^2}{x}$ .
  - b) Find the value of  $x$  for which the block will have a maximum volume.  
(Volume = area of base  $\times$  height)
3. Determine the shortest vertical distance between the curves of  $f$  and  $g$  if it is given that:

$$f(x) = -x^2 + 2x + 3$$

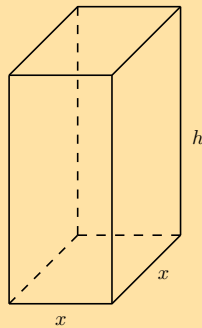
and  $g(x) = \frac{8}{x}, \quad x > 0$



4. The diagram shows the plan for a verandah which is to be built on the corner of a cottage. A railing  $ABCDE$  is to be constructed around the four edges of the verandah.



- If  $AB = DE = x$  and  $BC = CD = y$ , and the length of the railing must be 30 m, find the values of  $x$  and  $y$  for which the verandah will have a maximum area.
5. A rectangular juice container, made from cardboard, has a square base and holds  $750 \text{ cm}^3$  of juice. The container has a specially designed top that folds to close the container. The cardboard needed to fold the top of the container is twice the cardboard needed for the base, which only needs a single layer of cardboard.



- a) If the length of the sides of the base is  $x$  cm, show that the total area of the cardboard needed for one container is given by:
- $$A(\text{in square centimetres}) = \frac{3000}{x} + 3x^2$$
- b) Determine the dimensions of the container so that the area of the cardboard used is minimised.
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28YN    2a. 28YP    2b. 28YQ    3. 28YR    4. 28YS    5a. 28YT  
5b. 28YV



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It is very useful to determine how fast (the rate at which) things are changing. Mathematically we can represent change in different ways. For example we can use algebraic formulae or graphs.

Graphs give a visual representation of the rate at which the function values change as the independent (input) variable changes. This rate of change is described by the gradient of the graph and can therefore be determined by calculating the derivative.

We have learnt how to determine the average gradient of a curve and how to determine the gradient of a curve at a given point. These concepts are also referred to as the average rate of change and the instantaneous rate of change.

$$\text{Average rate of change} = \frac{f(x+h) - f(x)}{(x+h) - x}$$

$$\text{Instantaneous rate of change} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

When we mention rate of change, the instantaneous rate of change (the derivative) is implied. When average rate of change is required, it will be specifically referred to as average rate of change.

Velocity is one of the most common forms of rate of change:

$$\text{Average velocity} = \text{Average rate of change}$$

$$\begin{aligned} \text{Instantaneous velocity} &= \text{Instantaneous rate of change} \\ &= \text{Derivative} \end{aligned}$$

Velocity refers to the change in distance ( $s$ ) for a corresponding change in time ( $t$ ).

$$v(t) = \frac{ds}{dt} = s'(t)$$

Acceleration is the change in velocity for a corresponding change in time. Therefore, acceleration is the derivative of velocity

$$a(t) = v'(t)$$

This implies that acceleration is the second derivative of the distance.

$$a(t) = s''(t)$$

### QUESTION

The height (in metres) of a golf ball  $t$  seconds after it has been hit into the air, is given by  $H(t) = 20t - 5t^2$ . Determine the following:

1. The average vertical velocity of the ball during the first two seconds.
2. The vertical velocity of the ball after 1,5 s.
3. The time at which the vertical velocity is zero.
4. The vertical velocity with which the ball hits the ground.
5. The acceleration of the ball.

### SOLUTION

**Step 1: Determine the average vertical velocity during the first two seconds**

$$\begin{aligned} v_{\text{ave}} &= \frac{H(2) - H(0)}{2 - 0} \\ &= \frac{[20(2) - 5(2)^2] - [20(0) - 5(0)^2]}{2} \\ &= \frac{40 - 20}{2} \\ &= 10 \text{ m.s}^{-1} \end{aligned}$$

**Step 2: Calculate the instantaneous vertical velocity**

$$\begin{aligned} v(t) &= H'(t) \\ &= \frac{dH}{dt} \\ &= 20 - 10t \end{aligned}$$

Velocity after 1,5 s:

$$\begin{aligned} v(1,5) &= 20 - 10(1,5) \\ &= 5 \text{ m.s}^{-1} \end{aligned}$$

**Step 3: Determine the time at which the vertical velocity is zero**

$$\begin{aligned} v(t) &= 0 \\ 20 - 10t &= 0 \\ 10t &= 20 \\ t &= 2 \end{aligned}$$

Therefore, the velocity is zero after 2 s

**Step 4: Find the vertical velocity with which the ball hits the ground**

The ball hits the ground when  $H(t) = 0$

$$\begin{aligned}20t - 5t^2 &= 0 \\5t(4 - t) &= 0 \\t &= 0 \text{ or } t = 4\end{aligned}$$

The ball hits the ground after 4 s. The velocity after 4 s will be:

$$\begin{aligned}v(4) &= H'(4) \\&= 20 - 10(4) \\&= -20 \text{ m.s}^{-1}\end{aligned}$$

The ball hits the ground at a speed of  $20 \text{ m.s}^{-1}$ . Notice that the sign of the velocity is negative which means that the ball is moving downward (a positive velocity is used for upwards motion).

**Step 5: Acceleration**

$$\begin{aligned}a = v'(t) &= H''(t) \\&= -10\end{aligned}$$

$$\therefore a = -10 \text{ m.s}^{-2}$$

Just because gravity is constant does not mean we should necessarily think of acceleration as a constant. We should still consider it a function.

**Exercise 6 – 12: Rates of change**

1. A pump is connected to a water reservoir. The volume of the water is controlled by the pump and is given by the formula:

$$\begin{aligned}V(d) &= 64 + 44d - 3d^2 \\ \text{where } V &= \text{volume in kilolitres} \\ d &= \text{days}\end{aligned}$$

- Determine the rate of change of the volume of the reservoir with respect to time after 8 days.
- Is the volume of the water increasing or decreasing at the end of 8 days. Explain your answer.
- After how many days will the reservoir be empty?
- When will the amount of water be at a maximum?
- Calculate the maximum volume.
- Draw a graph of  $V(d)$ .

2. A soccer ball is kicked vertically into the air and its motion is represented by the equation:

$$D(t) = 1 + 18t - 3t^2$$

where  $D$  = distance above the ground (in metres)

$t$  = time elapsed (in seconds)

- Determine the initial height of the ball at the moment it is being kicked.
  - Find the initial velocity of the ball.
  - Determine the velocity of the ball after 1,5 s.
  - Calculate the maximum height of the ball.
  - Determine the acceleration of the ball after 1 second and explain the meaning of the answer.
  - Calculate the average velocity of the ball during the third second.
  - Determine the velocity of the ball after 3 seconds and interpret the answer.
  - How long will it take for the ball to hit the ground?
  - Determine the velocity of the ball when it hits the ground.
3. If the displacement  $s$  (in metres) of a particle at time  $t$  (in seconds) is governed by the equation  $s = \frac{1}{2}t^3 - 2t$ , find its acceleration after 2 seconds.
4. During an experiment the temperature  $T$  (in degrees Celsius) varies with time  $t$  (in hours) according to the formula:  $T(t) = 30 + 4t - \frac{1}{2}t^2$ ,  $t \in [1; 10]$ .
- Determine an expression for the rate of change of temperature with time.
  - During which time interval was the temperature dropping?
5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [28YW](#)   2a. [28YX](#)   2b. [28YY](#)   2c. [28YZ](#)   2d. [28Z2](#)   2e. [28Z3](#)  
2f. [28Z4](#)   2g. [28Z5](#)   2h. [28Z6](#)   2i. [28Z7](#)   3. [28Z8](#)   4a. [28Z9](#)  
4b. [28ZB](#)



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- The limit of a function exists and is equal to  $L$  if the values of  $f(x)$  get closer to  $L$  from both sides as  $x$  gets closer to  $a$ .

$$\lim_{x \rightarrow a} f(x) = L$$

- **Average gradient or average rate of change:**

$$\text{Average gradient} = \frac{f(x+h) - f(x)}{h}$$

- **Gradient at a point or instantaneous rate of change:**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- **Notation**

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}[f(x)] = Df(x) = D_x y$$

- **Differentiating from first principles:**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- **Rules for differentiation:**

- General rule for differentiation:

$$\frac{d}{dx}[x^n] = nx^{n-1}, \text{ where } n \in \mathbb{R} \text{ and } n \neq 0.$$

- The derivative of a constant is equal to zero.

$$\frac{d}{dx}[k] = 0$$

- The derivative of a constant multiplied by a function is equal to the constant multiplied by the derivative of the function.

$$\frac{d}{dx}[k \cdot f(x)] = k \frac{d}{dx}[f(x)]$$

- The derivative of a sum is equal to the sum of the derivatives.

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$$

- The derivative of a difference is equal to the difference of the derivatives.

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}[f(x)] - \frac{d}{dx}[g(x)]$$

- **Second derivative:**

$$f''(x) = \frac{d}{dx}[f'(x)]$$

- **Sketching graphs:**

The gradient of the curve and the tangent to the curve at stationary points is zero.  
Finding the stationary points: let  $f'(x) = 0$  and solve for  $x$ .

A stationary point can either be a local maximum, a local minimum or a point of inflection.

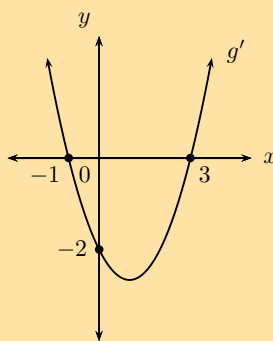
- **Optimisation problems:**

Use the given information to formulate an expression that contains only one variable.

Differentiate the expression, let the derivative equal zero and solve the equation.

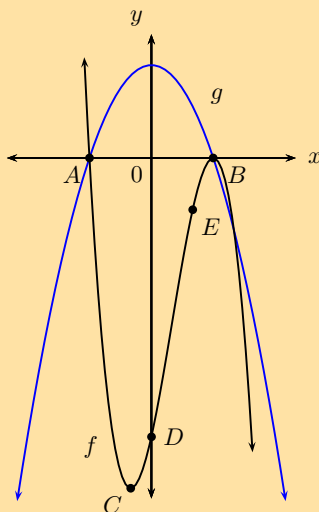
1. Determine  $f'(x)$  from first principles if  $f(x) = 2x - x^2$ .
2. Given  $f(x) = \frac{1}{x} + 3$ , find  $f'(x)$  using the definition of the derivative.
3. Calculate:  $\lim_{x \rightarrow 1} \frac{1 - x^3}{1 - x}$
4. Determine  $\frac{dy}{dx}$  if:
  - a)  $y = (x + 2)(7 - 5x)$
  - b)  $y = \frac{8x^3 + 1}{2x + 1}$
  - c)  $y = (2x)^2 - \frac{1}{3x}$
  - d)  $y = \frac{2\sqrt{x-5}}{\sqrt{x}}$
5. Given:  $f(x) = 2x^2 - x$ 
  - a) Use the definition of the derivative to calculate  $f'(x)$ .
  - b) Hence, calculate the coordinates of the point at which the gradient of the tangent to the graph of  $f$  is 7.
6. If  $g(x) = (x^{-2} + x^2)^2$ , calculate  $g'(2)$ .
7. Given:  $f(x) = 2x - 3$ 
  - a) Find  $f^{-1}(x)$ .
  - b) Solve  $f^{-1}(x) = 3f'(x)$ .
8. Find the derivative for each of the following:
  - a)  $p(t) = \frac{\sqrt[5]{t^3}}{3} + 10$
  - b)  $k(n) = \frac{(2n^2 - 5)(3n + 2)}{n^2}$
9. If  $xy - 5 = \sqrt{x^3}$ , determine  $\frac{dy}{dx}$ .
10. Given:  $y = x^3$ 
  - a) Determine  $\frac{dy}{dx}$ .
  - b) Find  $\frac{dx}{dy}$ .
  - c) Show that  $\frac{dy}{dx} \times \frac{dx}{dy} = 1$ .
11. Given:  $f(x) = x^3 - 3x^2 + 4$ 
  - a) Calculate  $f(-1)$ .
  - b) Hence, solve  $f(x) = 0$ .
  - c) Determine  $f'(x)$ .
  - d) Sketch the graph of  $f$ , showing the coordinates of the turning points and the intercepts on both axes.
  - e) Determine the coordinates of the points on the graph of  $f$  where the gradient is 9.
  - f) Draw the graph of  $f'(x)$  on the same system of axes.
  - g) Determine  $f''(x)$  and use this to make conclusions about the concavity of  $f$ .
12. Given  $f(x) = 2x^3 - 5x^2 - 4x + 3$ .
  - a) If  $f(-1) = 0$ , determine the  $x$ -intercepts of  $f$ .
  - b) Determine the coordinates of the turning points of  $f$ .

- c) Draw a sketch graph of  $f$ . Clearly indicate the coordinates of the turning points and the intercepts with the axes.
- d) For which value(s) of  $k$  will the equation  $f(x) = k$  have three real roots of which two are equal?
- e) Determine the equation of the tangent to the graph of  $f(x) = 2x^3 - 5x^2 - 4x + 3$  at the point where  $x = 1$ .
13. Given the function  $f(x) = x^3 + bx^2 + cx + d$  with  $y$ -intercept  $(0; 26)$ ,  $x$ -intercept  $(-2; 0)$  and a point of inflection at  $x = -3$ .
- Show by calculation that  $b = 9$ ,  $c = 27$  and  $d = 26$ .
  - Find the  $y$ -coordinate of the point of inflection.
  - Draw the graph of  $f$ .
  - Discuss the gradient of  $f$ .
  - Discuss the concavity of  $f$ .
14. The sketch shows the graph of  $g'(x)$ .



- Identify the stationary points of the cubic function,  $g(x)$ .
  - What is the gradient of function  $g$  where  $x = 0$ .
  - If it is further given that  $g(x)$  has only two real roots, draw a rough sketch of  $g(x)$ . Intercept values do not need to be shown.
15. Given that  $h(x)$  is a linear function with  $h(2) = 11$  and  $h'(2) = -1$ , find the equation of  $h(x)$ .
16. The graphs of  $f$  and  $g$  and the following points are given below:

$$A(-3; 0) \quad B(3; 0) \quad C(-1; -32) \quad D(0; -27) \quad E(2; y)$$



- a) Use the graphs and determine the values of  $x$  for which:
- $f(x)$  is a decreasing function.
  - $f(x) \cdot g(x) \geq 0$ .
  - $f'(x)$  and  $g(x)$  are both negative.
- b) Given  $f(x) = -x^3 + 3x^2 + 9x - 27$ , determine the equation of the tangent to  $f$  at the point  $E(2; y)$ .
- c) Find the coordinates of the point(s) where the tangent in the question above meets the graph of  $f$  again.
- d) Without any calculations, give the  $x$ -intercepts of the graph of  $f'(x)$ . Explain reasoning.
17. a) Sketch the graph of  $f(x) = x^3 - 9x^2 + 24x - 20$ , show all intercepts with the axes and turning points.
- b) Find the equation of the tangent to  $f(x)$  at  $x = 4$ .
- c) Determine the point of inflection and discuss the concavity of  $f$ .
18. Determine the minimum value of the sum of a positive number and its reciprocal.
19.  $t$  minutes after a kettle starts to boil, the height of the water in the kettle is given by  $d = 86 - \frac{1}{8}t - \frac{1}{4}t^3$ , where  $d$  is measured in millimetres.
- Calculate the height of the water level in the kettle just before it starts to boil.
  - As the water boils, the water level in the kettle decreases. Determine the rate at which the water level is decreasing when  $t = 2$  minutes.
  - How many minutes after the kettle starts to boil will the water level be decreasing at a rate of  $12\frac{1}{8}$  mm per minute?
20. The displacement of a moving object is represented by the equation:

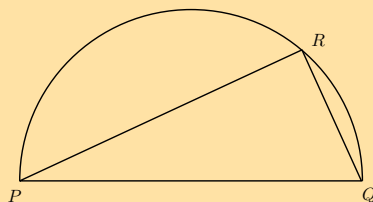
$$D(t) = \frac{4}{3}t^3 - 3t$$

where  $D$  = distance travelled in metres

$t$  = time in seconds

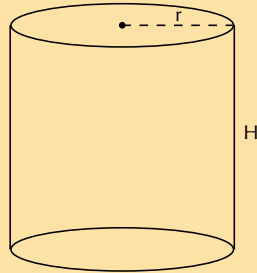
Calculate the acceleration of the object after 3 seconds.

21. In the figure  $PQ$  is the diameter of the semi-circle  $PRQ$ . The sum of the lengths of  $PR$  and  $QR$  is 10 units. Calculate the perimeter of  $\triangle PQR$  when  $\triangle PQR$  covers the maximum area in the semi-circle. Leave the answer in simplified surd form.



22. The capacity of a cylindrical water tank is 1000 litres. Let the height be  $H$  and the radius be  $r$ . The material used for the bottom of the tank is twice as thick and also twice as expensive as the material used for the curved part of the tank and the top of the tank.

Remember:  $1000 \ell = 1 \text{ m}^3$



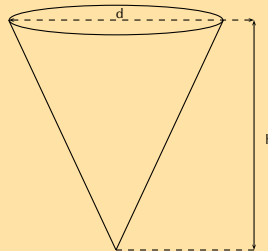
- Express  $H$  in terms of  $r$ .
- Show that the cost of the material for the tank can be expressed as:

$$C = 3\pi r^2 + \frac{2}{r}$$

- Determine the diameter of the tank that gives the minimum cost of the materials.

[IEB, 2006]

23. The diameter of an icecream cone is  $d$  and the vertical height is  $h$ . The sum of the diameter and the height of the cone is 10 cm.



- Determine the volume of the cone in terms of  $h$  and  $d$ .  
(Volume of a cone:  $V = \frac{1}{3}\pi r^2 h$ )
- Determine the radius and height of the cone for the volume to be a maximum.
- Calculate the maximum volume of the cone.

24. A water reservoir has both an inlet and an outlet pipe to regulate the depth of the water in the reservoir. The depth is given by the function:

$$D(h) = 3 + \frac{1}{2}h - \frac{1}{4}h^3$$

where  $D$  = depth in metres

$h$  = hours after 06h00

- Determine the rate at which the depth of the water is changing at 10h00.
- Is the depth of the water increasing or decreasing?
- At what time will the inflow of water be the same as the outflow?

[IEB, 2006]

25. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |                           |                           |                           |                           |                           |                           |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1. <a href="#">28ZC</a>   | 2. <a href="#">28ZD</a>   | 3. <a href="#">28ZF</a>   | 4a. <a href="#">28ZG</a>  | 4b. <a href="#">28ZH</a>  | 4c. <a href="#">28ZJ</a>  |
| 4d. <a href="#">28ZK</a>  | 5a. <a href="#">28ZM</a>  | 5b. <a href="#">28ZN</a>  | 6. <a href="#">28ZP</a>   | 7a. <a href="#">28ZQ</a>  | 7b. <a href="#">28ZR</a>  |
| 8a. <a href="#">28ZS</a>  | 8b. <a href="#">28ZT</a>  | 9. <a href="#">28ZV</a>   | 10a. <a href="#">28ZW</a> | 10b. <a href="#">28ZX</a> | 10c. <a href="#">28ZY</a> |
| 11a. <a href="#">28ZZ</a> | 11b. <a href="#">2922</a> | 11c. <a href="#">2923</a> | 11d. <a href="#">2924</a> | 11e. <a href="#">2925</a> | 11f. <a href="#">2926</a> |
| 11g. <a href="#">2927</a> | 12a. <a href="#">2928</a> | 12b. <a href="#">2929</a> | 12c. <a href="#">292B</a> | 12d. <a href="#">292C</a> | 12e. <a href="#">292D</a> |
| 13a. <a href="#">292F</a> | 13b. <a href="#">292G</a> | 13c. <a href="#">292H</a> | 13d. <a href="#">292J</a> | 13e. <a href="#">292K</a> | 14a. <a href="#">292M</a> |
| 14b. <a href="#">292N</a> | 14c. <a href="#">292P</a> | 15. <a href="#">292Q</a>  | 16a. <a href="#">292R</a> | 16b. <a href="#">292S</a> | 16c. <a href="#">292T</a> |
| 16d. <a href="#">292V</a> | 17. <a href="#">292W</a>  | 18. <a href="#">292X</a>  | 19a. <a href="#">292Y</a> | 19b. <a href="#">292Z</a> | 19c. <a href="#">2932</a> |
| 20. <a href="#">2933</a>  | 21. <a href="#">2934</a>  | 22a. <a href="#">2935</a> | 22b. <a href="#">2936</a> | 22c. <a href="#">2937</a> | 23a. <a href="#">2938</a> |
| 23b. <a href="#">2939</a> | 23c. <a href="#">293B</a> | 24a. <a href="#">293C</a> | 24b. <a href="#">293D</a> | 24c. <a href="#">293F</a> |                           |



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